

# The Notebooks

*Testing Mode · Master Question Bank (prompts)*  
UK · IB · US · Answers: see per-era PDFs / question-bank.json

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1000 skills · 3000 items · Unique prompts: 3000

# Era I — Count

Stone tablets / tally notches

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## T.I.1.001 Count in Roman numerals to XII

- (a) Convert 4 to Roman numerals (skill: Count in Roman numerals to XII).
- (b) What Arabic number is I?
- (c) A page is numbered IV. Write the next page number in Roman numerals.

## T.I.1.002 Cardinality: last number said is how many

- (a) Continue (step 2): 4, 6, 8, 10, ... Write the next two terms.
- (b) Fill ■: 4, 6, 8, 10, ■, 14, 16.
- (c) Count back from 16 in steps of 2. List the first four numbers said.

## T.I.1.003 Subitize small sets without counting

- (a) Continue (step 2): 2, 4, 6, 8, ... Write the next two terms.
- (b) Fill ■: 2, 4, 6, 8, ■, 12, 14.
- (c) Count back from 14 in steps of 2. List the first four numbers said.

## T.I.1.004 Skip-count by 2, 5, 10 exam drills

- (a) Continue (step 10): 40, 50, 60, 70, ... Write the next two terms.
- (b) Fill ■: 40, 50, 60, ■, 80, 90, 100.
- (c) Count back from 100 in steps of 10. List the first four numbers said.

## T.I.1.005 Count in steps of 3 and 4

- (a) Continue (step 3): 9, 12, 15, 18, ... Write the next two terms.
- (b) Fill ■: 9, 12, ■, 18, 21, 24, 27.
- (c) Count back from 27 in steps of 3. List the first four numbers said.

## T.I.1.006 Count back through zero into negatives

- (a) Continue (step 2): 4, 6, 8, 10, ... Write the next two terms.
- (b) Fill ■: 4, 6, 8, 10, ■, 14, 16.
- (c) Count back from 16 in steps of 2. List the first four numbers said.

## T.I.2.001 Compare using more/fewer/equal SATs stems

- (a) Partition 853 into hundreds, tens and ones.
- (b) Write 853 in expanded form.
- (c) Exchange: 15 ones equal how many tens and ones?

## T.I.2.002 Order lengths and masses with units

- (a) Partition 342 into hundreds, tens and ones.
- (b) Write 342 in expanded form.
- (c) Exchange: 14 ones equal how many tens and ones?

## T.I.2.003 Order numbers to 100 with inequality signs

- (a) Partition 256 into hundreds, tens and ones.
- (b) Write 256 in expanded form.
- (c) Exchange: 15 ones equal how many tens and ones?

## T.I.3.001 Partition with UK part-whole models

- (a) Partition 391 into hundreds, tens and ones.
- (b) Write 391 in expanded form.
- (c) Exchange: 19 ones equal how many tens and ones?

## T.I.3.002 Write numbers in expanded form (UK wording)

- (a) Partition 857 into hundreds, tens and ones.
- (b) Write 857 in expanded form.
- (c) Exchange: 15 ones equal how many tens and ones?

## T.I.3.003 Roman numerals to 100 (I–C)

- (a) Convert 74 to Roman numerals (skill: Roman numerals to 100 (I–C)).

- (b) What Arabic number is X?
- (c) A page is numbered LXXIV. Write the next page number in Roman numerals.

#### **T.I.3.004 Roman numerals to 1,000 (M)**

- (a) Convert 500 to Roman numerals (skill: Roman numerals to 1,000 (M)).
- (b) What Arabic number is XL?
- (c) A page is numbered D. Write the next page number in Roman numerals.

#### **T.I.3.005 Exchange ten ones for one ten**

- (a) Partition 189 into hundreds, tens and ones.
- (b) Write 189 in expanded form.
- (c) Exchange: 18 ones equal how many tens and ones?

#### **T.I.3.006 Flexibly partition numbers many ways**

- (a) Partition 928 into hundreds, tens and ones.
- (b) Write 928 in expanded form.
- (c) Exchange: 12 ones equal how many tens and ones?

#### **T.I.4.001 Column addition (Appendix formal method)**

- (a) Column addition:  $707 + 346$ .
- (b) Work out  $717 + 341$ .
- (c) Score 707 then 346 more. Total?

#### **T.I.4.002 Column subtraction with exchange**

- (a) Column subtraction:  $514 - 281$ .
- (b) Calculate  $511 - 283$ .
- (c) A library has 514 books; 281 are loaned. How many remain?

#### **T.I.4.003 Number bonds to 10 and 20 speed packs**

- (a) Column addition:  $277 + 628$ .
- (b) Work out  $287 + 623$ .
- (c) Score 277 then 628 more. Total?

#### **T.I.4.004 Add/subtract money in £ and p**

- (a) Column subtraction:  $623 - 348$ .
- (b) Calculate  $620 - 350$ .
- (c) A library has 623 books; 348 are loaned. How many remain?

#### **T.I.4.005 Worded multi-step add/sub (SATs style)**

- (a) Column subtraction:  $986 - 726$ .
- (b) Calculate  $983 - 728$ .
- (c) A library has 986 books; 726 are loaned. How many remain?

#### **T.I.4.006 Near doubles addition strategies**

- (a) Column addition:  $656 + 254$ .
- (b) Work out  $666 + 249$ .
- (c) Score 656 then 254 more. Total?

#### **T.I.4.007 Bridging through 10 subtraction**

- (a) Column subtraction:  $521 - 420$ .
- (b) Calculate  $518 - 422$ .
- (c) A library has 521 books; 420 are loaned. How many remain?

#### **T.I.4.008 Inverse operations check add/sub**

- (a) Column subtraction:  $808 - 358$ .
- (b) Calculate  $805 - 360$ .
- (c) A library has 808 books; 358 are loaned. How many remain?

#### **T.I.4.009 Column methods with money £.p**

- (a) Column addition:  $350 + 552$ .
- (b) Work out  $360 + 547$ .

(c) Score 350 then 552 more. Total?

#### **T.I.5.001 Repeating pattern SAT stems**

(a) Continue (step 2): 2, 4, 6, 8, ... Write the next two terms.

(b) Fill ■: 2, 4, 6, ■, 10, 12, 14.

(c) Count back from 14 in steps of 2. List the first four numbers said.

#### **T.I.5.002 Missing-number sentences with boxes**

(a) Continue (step 2): 6, 8, 10, 12, ... Write the next two terms.

(b) Fill ■: 6, 8, 10, 12, ■, 16, 18.

(c) Count back from 18 in steps of 2. List the first four numbers said.

#### **T.I.6.001 Name 2D shapes with UK vocabulary**

(a) [Name 2D shapes with UK vocabulary] Which view is the plan — front, side, or top?

(b) Cuboid  $8 \times 4 \times 2$ : how many rectangles in a correct net?

(c) Describe the front elevation of a square-based pyramid (base 8) standing on its base.

#### **T.I.6.002 Turns: quarter, half, three-quarter, full**

(a) [Turns: quarter, half, three-quarter, full] Which view is the plan — front, side, or top?

(b) Cuboid  $5 \times 6 \times 4$ : how many rectangles in a correct net?

(c) Describe the front elevation of a square-based pyramid (base 5) standing on its base.

#### **T.I.6.003 Left/right and compass directions early**

(a) [Left/right and compass directions early] Which view is the plan — front, side, or top?

(b) Cuboid  $9 \times 4 \times 4$ : how many rectangles in a correct net?

(c) Describe the front elevation of a square-based pyramid (base 9) standing on its base.

#### **T.I.6.004 Symmetry: complete mirror patterns**

(a) [Symmetry: complete mirror patterns] Which view is the plan — front, side, or top?

(b) Cuboid  $5 \times 7 \times 5$ : how many rectangles in a correct net?

(c) Describe the front elevation of a square-based pyramid (base 5) standing on its base.

#### **T.I.7.001 Tell time to the nearest minute (analogue)**

(a) Write 7:25 pm in 24-hour time.

(b) Start 7:25; duration 95 min. End time?

(c) Train 14:25 → 16:00. Journey length?

#### **T.I.7.002 12-hour clock with am/pm**

(a) Write 11:35 pm in 24-hour time.

(b) Start 11:35; duration 35 min. End time?

(c) Train 14:35 → 15:10. Journey length?

#### **T.I.7.003 24-hour clock conversion**

(a) Write 1:00 pm in 24-hour time.

(b) Start 1:00; duration 95 min. End time?

(c) Train 14:00 → 15:35. Journey length?

#### **T.I.7.004 Read simple bus/train timetables**

(a) Write 6:00 pm in 24-hour time.

(b) Start 6:00; duration 70 min. End time?

(c) Train 14:00 → 15:10. Journey length?

#### **T.I.7.005 British coins and notes identification**

(a) Total of £0.92 and £0.48?

(b) Pay £5.00 for goods costing £1.40. Change?

(c) Buy three items £0.92, £0.48, £2.16 with a £10 note. Enough? Change?

#### **T.I.7.006 Give change from £1, £5, £10, £20**

(a) Total of £2.48 and £0.68?

(b) Pay £5.00 for goods costing £3.16. Change?

(c) Buy three items £2.48, £0.68, £1.63 with a £10 note. Enough? Change?

**T.I.7.007 Temperature in °C: read and compare**

- (a) Convert 5 cm to mm.
- (b) Which is heavier: 500 g or 2 kg? (compare in g)
- (c) A jug holds 500 ml. How many litres?

**T.I.7.008 Tally charts exam pack**

- (a) Mean of 6, 11, 11, 11, 14, 18?
- (b) Range of 6, 11, 11, 11, 14, 18?
- (c) Mean of five numbers is 13. Four are 6, 11, 11, 11. Find the fifth.

**T.I.7.009 Pictograms with keys (one symbol = many)**

- (a) Mean of 4, 6, 9, 12, 13, 18?
- (b) Range of 4, 6, 9, 12, 13, 18?
- (c) Mean of five numbers is 11. Four are 4, 6, 9, 12. Find the fifth.

**T.I.7.010 Block diagrams / bar charts simple scale**

- (a) Mean of 3, 8, 9, 12, 13, 18?
- (b) Range of 3, 8, 9, 12, 13, 18?
- (c) Mean of five numbers is 11. Four are 3, 8, 9, 12. Find the fifth.

**T.I.7.011 Measure length in cm and mm**

- (a) Convert 16 cm to mm.
- (b) Which is heavier: 1600 g or 6 kg? (compare in g)
- (c) A jug holds 1600 ml. How many litres?

**T.I.7.012 Mass in g and kg comparison**

- (a) Convert 16 cm to mm.
- (b) Which is heavier: 1600 g or 7 kg? (compare in g)
- (c) A jug holds 1600 ml. How many litres?

**T.I.7.013 Capacity in ml and l**

- (a) Convert 3 cm to mm.
- (b) Which is heavier: 300 g or 8 kg? (compare in g)
- (c) A jug holds 300 ml. How many litres?

**T.I.7.014 CCSS line plots with fractional lengths**

- (a) Mean of 3, 7, 8, 9, 12, 14?
- (b) Range of 3, 7, 8, 9, 12, 14?
- (c) Mean of five numbers is 8. Four are 3, 7, 8, 9. Find the fifth.

**T.I.7.015 Elapsed time across hour boundaries**

- (a) Write 1:25 pm in 24-hour time.
- (b) Start 1:25; duration 95 min. End time?
- (c) Train 14:25 → 16:00. Journey length?

**T.I.7.016 Calendar: days in months fluency**

- (a) Write 7:05 pm in 24-hour time.
- (b) Start 7:05; duration 25 min. End time?
- (c) Train 14:05 → 14:30. Journey length?

**T.I.7.017 Compare durations in hours and minutes**

- (a) Write 10:50 pm in 24-hour time.
- (b) Start 10:50; duration 95 min. End time?
- (c) Train 14:50 → 16:25. Journey length?

**T.I.7.018 Weigh using scales reading dials**

- (a) Convert 16 cm to mm.
- (b) Which is heavier: 1600 g or 15 kg? (compare in g)
- (c) A jug holds 1600 ml. How many litres?

**T.I.7.019 Capacity practical measuring jugs**

- (a) Convert 10 cm to mm.
- (b) Which is heavier: 1000 g or 9 kg? (compare in g)
- (c) A jug holds 1000 ml. How many litres?

**T.I.7.020 Money change multi-item shop 1**

- (a) Total of £0.66 and £1.43?
- (b) Pay £5.00 for goods costing £2.09. Change?
- (c) Buy three items £0.66, £1.43, £2.59 with a £10 note. Enough? Change?

**T.I.7.021 Money change multi-item shop 2**

- (a) Total of £1.07 and £1.55?
- (b) Pay £5.00 for goods costing £2.62. Change?
- (c) Buy three items £1.07, £1.55, £1.84 with a £10 note. Enough? Change?

**T.I.7.022 Money change multi-item shop 3**

- (a) Total of £0.50 and £2.25?
- (b) Pay £5.00 for goods costing £2.75. Change?
- (c) Buy three items £0.50, £2.25, £2.98 with a £10 note. Enough? Change?

**T.I.7.023 Money change multi-item shop 4**

- (a) Total of £1.37 and £0.55?
- (b) Pay £5.00 for goods costing £1.92. Change?
- (c) Buy three items £1.37, £0.55, £1.53 with a £10 note. Enough? Change?

**T.I.7.024 Timetable reading scenario 1**

- (a) Write 10:50 pm in 24-hour time.
- (b) Start 10:50; duration 95 min. End time?
- (c) Train 14:50 → 16:25. Journey length?

**T.I.7.025 Timetable reading scenario 2**

- (a) Write 7:55 pm in 24-hour time.
- (b) Start 7:55; duration 95 min. End time?
- (c) Train 14:55 → 16:30. Journey length?

**T.I.7.026 Timetable reading scenario 3**

- (a) Write 10:30 pm in 24-hour time.
- (b) Start 10:30; duration 95 min. End time?
- (c) Train 14:30 → 16:05. Journey length?

# Era II — Operate

Gears / early mechanism

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## **T.II.1.001 Roman numerals review in large-number context**

- (a) Convert 66 to Roman numerals (skill: Roman numerals review in large-number context).
- (b) What Arabic number is LXXX?
- (c) A page is numbered LXVI. Write the next page number in Roman numerals.

## **T.II.1.002 Formal column addition multi-digit exam pack**

- (a) Column addition:  $762 + 350$ .
- (b) Work out  $772 + 345$ .
- (c) Score 762 then 350 more. Total?

## **T.II.1.003 Formal column subtraction multi-digit exam pack**

- (a) Column subtraction:  $755 - 377$ .
- (b) Calculate  $752 - 379$ .
- (c) A library has 755 books; 377 are loaned. How many remain?

## **T.II.1.004 Round to estimate before calculating (UK)**

- (a) Column addition:  $438 + 425$ .
- (b) Work out  $448 + 420$ .
- (c) Score 438 then 425 more. Total?

## **T.II.1.005 Numbers to 10 million reading/writing**

- (a) Column addition:  $574 + 380$ .
- (b) Work out  $584 + 375$ .
- (c) Score 574 then 380 more. Total?

## **T.II.1.006 Negative numbers on number lines (Y5)**

- (a) Column addition:  $301 + 505$ .
- (b) Work out  $311 + 500$ .
- (c) Score 301 then 505 more. Total?

## **T.II.1.007 Count forwards/backwards in powers of 10**

- (a) Column addition:  $691 + 174$ .
- (b) Work out  $701 + 169$ .
- (c) Score 691 then 174 more. Total?

## **T.II.1.008 Compare numbers using place-value charts**

- (a) Column addition:  $735 + 259$ .
- (b) Work out  $745 + 254$ .
- (c) Score 735 then 259 more. Total?

## **T.II.1.009 Formal column methods mixed 1**

- (a) Column addition:  $617 + 623$ .
- (b) Work out  $627 + 618$ .
- (c) Score 617 then 623 more. Total?

## **T.II.1.010 Formal column methods mixed 2**

- (a) Column addition:  $426 + 391$ .
- (b) Work out  $436 + 386$ .
- (c) Score 426 then 391 more. Total?

## **T.II.1.011 Formal column methods mixed 3**

- (a) Column addition:  $810 + 597$ .
- (b) Work out  $820 + 592$ .
- (c) Score 810 then 597 more. Total?

## **T.II.2.001 12×12 tables fluency milestone pack**

- (a) Calculate  $48 \times 28$  (formal method).

- (b) What is  $48 \times 10$ ?
- (c) A pack has 48 stickers; buy 28 packs. Total stickers?

**T.II.2.002 Long multiplication (formal written method)**

- (a) Column addition:  $573 + 391$ .
- (b) Work out  $583 + 386$ .
- (c) Score 573 then 391 more. Total?

**T.II.2.003 Grid / area-model to column bridge**

- (a) Calculate  $47 \times 25$  (formal method).
- (b) What is  $47 \times 10$ ?
- (c) A pack has 47 stickers; buy 25 packs. Total stickers?

**T.II.2.004 Multiplication worded multi-step SATs**

- (a) Calculate  $14 \times 16$  (formal method).
- (b) What is  $14 \times 10$ ?
- (c) A pack has 14 stickers; buy 16 packs. Total stickers?

**T.II.2.005 Multiply by 10, 100, 1000 place-value moves**

- (a) Calculate  $41 \times 29$  (formal method).
- (b) What is  $41 \times 10$ ?
- (c) A pack has 41 stickers; buy 29 packs. Total stickers?

**T.II.2.006 Square numbers visual arrays pack**

- (a) Calculate  $42 \times 15$  (formal method).
- (b) What is  $42 \times 10$ ?
- (c) A pack has 42 stickers; buy 15 packs. Total stickers?

**T.II.2.007 Distributive property for mental mult**

- (a) Calculate  $47 \times 22$  (formal method).
- (b) What is  $47 \times 10$ ?
- (c) A pack has 47 stickers; buy 22 packs. Total stickers?

**T.II.2.008 Multiples vs factors language traps**

- (a) Calculate  $45 \times 15$  (formal method).
- (b) What is  $45 \times 10$ ?
- (c) A pack has 45 stickers; buy 15 packs. Total stickers?

**T.II.2.009 Long multiplication accuracy set 1**

- (a) Column addition:  $287 + 317$ .
- (b) Work out  $297 + 312$ .
- (c) Score 287 then 317 more. Total?

**T.II.2.010 Long multiplication accuracy set 2**

- (a) Column addition:  $465 + 257$ .
- (b) Work out  $475 + 252$ .
- (c) Score 465 then 257 more. Total?

**T.II.2.011 Long multiplication accuracy set 3**

- (a) Column addition:  $303 + 442$ .
- (b) Work out  $313 + 437$ .
- (c) Score 303 then 442 more. Total?

**T.II.3.001 Long division formal method (one-digit divisor)**

- (a) Column addition:  $505 + 350$ .
- (b) Work out  $515 + 345$ .
- (c) Score 505 then 350 more. Total?

**T.II.3.002 Long division by two-digit divisors**

- (a) Column addition:  $284 + 583$ .
- (b) Work out  $294 + 578$ .

(c) Score 284 then 583 more. Total?

**T.II.3.003 Division with remainders as fractions/decimals**

(a) Divide  $135 \div 3$ .

(b) Using short division:  $137 \div 3$ . Give quotient and remainder.

(c) 137 sweets shared among 3 children. How many each and remainder?

**T.II.3.004 Short division (bus-stop) method fluency**

(a) Column addition:  $385 + 263$ .

(b) Work out  $395 + 258$ .

(c) Score 385 then 263 more. Total?

**T.II.3.005 Division worded multi-step SATs**

(a) Divide  $312 \div 8$ .

(b) Using short division:  $319 \div 8$ . Give quotient and remainder.

(c) 319 sweets shared among 8 children. How many each and remainder?

**T.II.3.006 Chunking division as bridge method**

(a) Column addition:  $826 + 395$ .

(b) Work out  $836 + 390$ .

(c) Score 826 then 395 more. Total?

**T.II.3.007 Worded division sharing vs grouping**

(a) Divide  $369 \div 9$ .

(b) Using short division:  $376 \div 9$ . Give quotient and remainder.

(c) 376 sweets shared among 9 children. How many each and remainder?

**T.II.3.008 Long division accuracy set 1**

(a) Column addition:  $600 + 247$ .

(b) Work out  $610 + 242$ .

(c) Score 600 then 247 more. Total?

**T.II.3.009 Long division accuracy set 2**

(a) Column addition:  $769 + 386$ .

(b) Work out  $779 + 381$ .

(c) Score 769 then 386 more. Total?

**T.II.3.010 Long division accuracy set 3**

(a) Column addition:  $381 + 269$ .

(b) Work out  $391 + 264$ .

(c) Score 381 then 269 more. Total?

**T.II.4.001 Prime numbers to 100 recognition drills**

(a) List all factors of 20.

(b)  $HCF(20,30)$ ?

(c)  $LCM(20,30)$ ?

**T.II.4.002 HCF and LCM worded problems (UK)**

(a) List all factors of 14.

(b)  $HCF(14,21)$ ?

(c)  $LCM(14,21)$ ?

**T.II.4.003 Square and cube numbers recall pack**

(a) List all factors of 16.

(b)  $HCF(16,24)$ ?

(c)  $LCM(16,24)$ ?

**T.II.4.004 Common factors and common multiples lists**

(a) List all factors of 14.

(b)  $HCF(14,21)$ ?

(c)  $LCM(14,21)$ ?

**T.II.5.001 Unit fractions of amounts (UK wording)**

- (a) Simplify  $\frac{6}{10}$ .
- (b) Find  $\frac{3}{5}$  of 55.
- (c) Calculate  $\frac{3}{5} + \frac{2}{5}$ .

**T.II.5.002 Non-unit fractions of amounts**

- (a) Simplify  $\frac{2}{4}$ .
- (b) Find  $\frac{1}{2}$  of 12.
- (c) Calculate  $\frac{1}{2} + \frac{1}{2}$ .

**T.II.5.003 Equivalent fractions wall fluency**

- (a) Simplify  $\frac{6}{16}$ .
- (b) Find  $\frac{3}{8}$  of 64.
- (c) Calculate  $\frac{3}{8} + \frac{2}{3}$ .

**T.II.5.004 Improper fractions  $\leftrightarrow$  mixed numbers fluency**

- (a) Simplify  $\frac{2}{10}$ .
- (b) Find  $\frac{1}{5}$  of 40.
- (c) Calculate  $\frac{1}{5} + \frac{4}{7}$ .

**T.II.5.005 Fractions greater than 1 on number lines**

- (a) Simplify  $\frac{6}{16}$ .
- (b) Find  $\frac{3}{8}$  of 56.
- (c) Calculate  $\frac{3}{8} + \frac{1}{6}$ .

**T.II.5.006 Fraction of a set with remainders context**

- (a) Simplify  $\frac{6}{6}$ .
- (b) Find  $\frac{3}{3}$  of 12.
- (c) Calculate  $1 + \frac{4}{3}$ .

**T.II.6.001 Add/subtract fractions same denominator exam**

- (a) Simplify  $\frac{4}{12}$ .
- (b) Find  $\frac{2}{6}$  of 42.
- (c) Calculate  $\frac{1}{3} + \frac{1}{2}$ .

**T.II.6.002 Add/subtract mixed numbers exam pack**

- (a) Simplify  $\frac{8}{16}$ .
- (b) Find  $\frac{4}{8}$  of 56.
- (c) Calculate  $\frac{1}{2} + \frac{3}{8}$ .

**T.II.6.003 Multiply fractions by wholes and fractions**

- (a) Simplify  $\frac{6}{18}$ .
- (b) Find  $\frac{3}{9}$  of 81.
- (c) Calculate  $\frac{1}{3} + \frac{1}{4}$ .

**T.II.6.004 Divide fractions by wholes (Y6)**

- (a) Simplify  $\frac{10}{12}$ .
- (b) Find  $\frac{5}{6}$  of 72.
- (c) Calculate  $\frac{5}{6} + \frac{2}{3}$ .

**T.II.6.005 Fraction  $\leftrightarrow$  decimal  $\leftrightarrow$  percent early bridge**

- (a) Simplify  $\frac{6}{12}$ .
- (b) Find  $\frac{3}{6}$  of 48.
- (c) Calculate  $\frac{1}{2} + \frac{1}{6}$ .

**T.II.6.006 Simplify fractions to lowest terms**

- (a) Simplify  $\frac{2}{12}$ .
- (b) Find  $\frac{1}{6}$  of 36.
- (c) Calculate  $\frac{1}{6} + \frac{1}{6}$ .

**T.II.6.007 Fraction of a remaining amount multi-step**

- (a) Simplify  $\frac{6}{4}$ .
- (b) Find  $\frac{3}{2}$  of 18.
- (c) Calculate  $\frac{3}{2} + \frac{2}{3}$ .

#### **T.II.6.008 Compare fractions with different denominators**

- (a) Simplify  $\frac{4}{14}$ .
- (b) Find  $\frac{2}{7}$  of 77.
- (c) Calculate  $\frac{2}{7} + \frac{3}{5}$ .

#### **T.II.7.001 Money calculations with £ and p (two decimals)**

- (a) Total of £1.66 and £1.83?
- (b) Pay £5.00 for goods costing £3.49. Change?
- (c) Buy three items £1.66, £1.83, £3.27 with a £10 note. Enough? Change?

#### **T.II.7.002 Rounding money to nearest pound/penny**

- (a) Total of £2.50 and £0.64?
- (b) Pay £5.00 for goods costing £3.14. Change?
- (c) Buy three items £2.50, £0.64, £1.49 with a £10 note. Enough? Change?

#### **T.II.7.003 Decimal ordering to 3 d.p. exam drills**

- (a) Total of £1.13 and £2.60?
- (b) Pay £5.00 for goods costing £3.73. Change?
- (c) Buy three items £1.13, £2.60, £2.28 with a £10 note. Enough? Change?

#### **T.II.7.004 Multiply/divide decimals by 10/100/1000**

- (a) Total of £2.09 and £0.44?
- (b) Pay £5.00 for goods costing £2.53. Change?
- (c) Buy three items £2.09, £0.44, £3.23 with a £10 note. Enough? Change?

#### **T.II.7.005 Thousandths place value (Y5–6)**

- (a) Total of £1.29 and £2.69?
- (b) Pay £5.00 for goods costing £3.98. Change?
- (c) Buy three items £1.29, £2.69, £3.38 with a £10 note. Enough? Change?

#### **T.II.7.006 Decimal × whole number money contexts**

- (a) Total of £2.18 and £2.37?
- (b) Pay £5.00 for goods costing £4.55. Change?
- (c) Buy three items £2.18, £2.37, £1.54 with a £10 note. Enough? Change?

#### **T.II.7.007 Decimal problems in measure contexts**

- (a) Total of £0.95 and £2.26?
- (b) Pay £5.00 for goods costing £3.21. Change?
- (c) Buy three items £0.95, £2.26, £1.47 with a £10 note. Enough? Change?

#### **T.II.8.001 Simple formulae in words then letters (Y6)**

- (a) nth term of 2, 9, 16, 23, ...
- (b) 20th term:  $a=2, d=7$ .
- (c)  $S_n$  for  $a=2, d=7$ .

#### **T.II.8.002 Find pairs of numbers satisfying an equation**

- (a) nth term of 4, 8, 12, 16, ...
- (b) 20th term:  $a=4, d=4$ .
- (c)  $S_n$  for  $a=4, d=4$ .

#### **T.II.8.003 Generate and describe linear sequences (Y6)**

- (a) nth term of 8, 12, 16, 20, ...
- (b) 20th term:  $a=8, d=4$ .
- (c)  $S_n$  for  $a=8, d=4$ .

#### **T.II.8.004 Use simple ratio language (Y6)**

- (a) nth term of 3, 7, 11, 15, ...

(b) 20th term:  $a=3$ ,  $d=4$ .

(c)  $S_n$  for  $a=3$ ,  $d=4$ .

#### **T.II.8.005 Enumerate possibilities systematically (Y6)**

(a) nth term of 9, 16, 23, 30, ...

(b) 20th term:  $a=9$ ,  $d=7$ .

(c)  $S_n$  for  $a=9$ ,  $d=7$ .

#### **T.II.9.001 Imperial ↔ metric approximate equivalences**

(a) [Imperial ↔ metric approximate equivalences] Which view is the plan — front, side, or top?

(b) Cuboid  $3 \times 6 \times 2$ : how many rectangles in a correct net?

(c) Describe the front elevation of a square-based pyramid (base 3) standing on its base.

#### **T.II.9.002 Convert between metric units fluently**

(a) [Convert between metric units fluently] Which view is the plan — front, side, or top?

(b) Cuboid  $4 \times 2 \times 4$ : how many rectangles in a correct net?

(c) Describe the front elevation of a square-based pyramid (base 4) standing on its base.

#### **T.II.9.003 Area and perimeter of rectilinear shapes**

(a) [Area and perimeter of rectilinear shapes] Which view is the plan — front, side, or top?

(b) Cuboid  $6 \times 5 \times 3$ : how many rectangles in a correct net?

(c) Describe the front elevation of a square-based pyramid (base 6) standing on its base.

#### **T.II.9.004 Volume of cubes and cuboids ( $\text{cm}^3$ )**

(a) [Volume of cubes and cuboids ( $\text{cm}^3$ )] Which view is the plan — front, side, or top?

(b) Cuboid  $3 \times 2 \times 4$ : how many rectangles in a correct net?

(c) Describe the front elevation of a square-based pyramid (base 3) standing on its base.

#### **T.II.9.005 Estimate angles; classify acute/obtuse/reflex**

(a) [Estimate angles; classify acute/obtuse/reflex] Which view is the plan — front, side, or top?

(b) Cuboid  $5 \times 2 \times 4$ : how many rectangles in a correct net?

(c) Describe the front elevation of a square-based pyramid (base 5) standing on its base.

#### **T.II.9.006 Draw and measure angles to nearest degree**

(a) [Draw and measure angles to nearest degree] Which view is the plan — front, side, or top?

(b) Cuboid  $7 \times 7 \times 5$ : how many rectangles in a correct net?

(c) Describe the front elevation of a square-based pyramid (base 7) standing on its base.

#### **T.II.9.007 Translation parallel to axes only (Y5 constraint)**

(a) Parallel lines; corresponding angle to  $46^\circ$ ?

(b) Parallel lines; co-interior with  $46^\circ$ ?

(c) Geometry construction lab for «Translation parallel to axes only (Y5 constraint)»: build the figure, measure, verify the claimed relation.

#### **T.II.9.008 Reflection in lines parallel to axes**

(a) Parallel lines; corresponding angle to  $69^\circ$ ?

(b) Parallel lines; co-interior with  $69^\circ$ ?

(c) Geometry construction lab for «Reflection in lines parallel to axes»: build the figure, measure, verify the claimed relation.

#### **T.II.9.009 Coordinates in first quadrant plot & read**

(a) [Coordinates in first quadrant plot & read] Which view is the plan — front, side, or top?

(b) Cuboid  $5 \times 5 \times 2$ : how many rectangles in a correct net?

(c) Describe the front elevation of a square-based pyramid (base 5) standing on its base.

#### **T.II.9.010 Four-quadrant coordinates intro (Y6)**

(a) [Four-quadrant coordinates intro (Y6)] Which view is the plan — front, side, or top?

(b) Cuboid  $4 \times 6 \times 5$ : how many rectangles in a correct net?

(c) Describe the front elevation of a square-based pyramid (base 4) standing on its base.

#### **T.II.9.011 Missing angles on a straight line / at a point**

(a) [Missing angles on a straight line / at a point] Which view is the plan — front, side, or top?

(b) Cuboid  $5 \times 2 \times 3$ : how many rectangles in a correct net?

(c) Describe the front elevation of a square-based pyramid (base 5) standing on its base.

#### **T.II.9.012 Properties of regular polygons (sides/angles)**

(a) [Properties of regular polygons (sides/angles)] Which view is the plan — front, side, or top?

(b) Cuboid  $5 \times 3 \times 6$ : how many rectangles in a correct net?

(c) Describe the front elevation of a square-based pyramid (base 5) standing on its base.

#### **T.II.9.013 Net of cubes and cuboids recognition**

(a) A cuboid measures  $9 \times 2 \times 6$ . How many faces does a correct net contain?

(b) Which of these cannot fold to a cube: cross of 6 squares vs row of 6 squares? Explain.

(c) Geometry construction lab for «Net of cubes and cuboids recognition»: build the figure, measure, verify the claimed relation.

#### **T.II.9.014 Identify 3D faces, edges, vertices**

(a) [Identify 3D faces, edges, vertices] Which view is the plan — front, side, or top?

(b) Cuboid  $6 \times 4 \times 3$ : how many rectangles in a correct net?

(c) Describe the front elevation of a square-based pyramid (base 6) standing on its base.

#### **T.II.9.015 Area of triangles by counting half-squares**

(a) [Area of triangles by counting half-squares] Which view is the plan — front, side, or top?

(b) Cuboid  $9 \times 5 \times 3$ : how many rectangles in a correct net?

(c) Describe the front elevation of a square-based pyramid (base 9) standing on its base.

#### **T.II.9.016 Scale factor enlarge by integer (Y6)**

(a) Lengths scale by 3. Area scale factor?

(b) A side 6 cm becomes ? after enlargement  $SF=3$ .

(c) Geometry construction lab for «Scale factor enlarge by integer (Y6)»: build the figure, measure, verify the claimed relation.

#### **T.II.9.017 Perimeter of polygons with missing sides**

(a) [Perimeter of polygons with missing sides] Which view is the plan — front, side, or top?

(b) Cuboid  $7 \times 3 \times 5$ : how many rectangles in a correct net?

(c) Describe the front elevation of a square-based pyramid (base 7) standing on its base.

#### **T.II.9.018 Draw circles with compass; label radius/diameter**

(a) [Draw circles with compass; label radius/diameter] Which view is the plan — front, side, or top?

(b) Cuboid  $6 \times 2 \times 6$ : how many rectangles in a correct net?

(c) Geometry construction lab for «Draw circles with compass; label radius/diameter»: build the figure, measure, verify the claimed relation.

#### **T.II.9.019 Time duration worded crossing midnight**

(a) [Time duration worded crossing midnight] Which view is the plan — front, side, or top?

(b) Cuboid  $3 \times 2 \times 3$ : how many rectangles in a correct net?

(c) Describe the front elevation of a square-based pyramid (base 3) standing on its base.

#### **T.II.9.020 Convert hours–minutes–seconds**

(a) [Convert hours–minutes–seconds] Which view is the plan — front, side, or top?

(b) Cuboid  $3 \times 6 \times 3$ : how many rectangles in a correct net?

(c) Describe the front elevation of a square-based pyramid (base 3) standing on its base.

#### **T.II.9.021 Find missing angle in triangle (Y6)**

(a) [Find missing angle in triangle (Y6)] Which view is the plan — front, side, or top?

(b) Cuboid  $4 \times 5 \times 4$ : how many rectangles in a correct net?

(c) Describe the front elevation of a square-based pyramid (base 4) standing on its base.

#### **T.II.9.022 Imperial-metric estimation set 1**

(a) [Imperial-metric estimation set 1] Which view is the plan — front, side, or top?

(b) Cuboid  $7 \times 5 \times 3$ : how many rectangles in a correct net?

(c) Describe the front elevation of a square-based pyramid (base 7) standing on its base.

#### **T.II.9.023 Axis-parallel transform drill 1**

(a) Parallel lines; corresponding angle to  $58^\circ$ ?

(b) Parallel lines; co-interior with  $58^\circ$ ?

(c) Geometry construction lab for «Axis-parallel transform drill 1»: build the figure, measure, verify the claimed relation.

**T.II.9.024 Imperial-metric estimation set 2**

- (a) [Imperial-metric estimation set 2] Which view is the plan — front, side, or top?
- (b) Cuboid  $9 \times 7 \times 5$ : how many rectangles in a correct net?
- (c) Describe the front elevation of a square-based pyramid (base 9) standing on its base.

**T.II.9.025 Axis-parallel transform drill 2**

- (a) Parallel lines; corresponding angle to  $50^\circ$ ?
- (b) Parallel lines; co-interior with  $50^\circ$ ?
- (c) Geometry construction lab for «Axis-parallel transform drill 2»: build the figure, measure, verify the claimed relation.

**T.II.10.001 Pie charts: interpret simple sectors (Y6)**

- (a) Mean of 3, 8, 10, 16, 16, 17?
- (b) Range of 3, 8, 10, 16, 16, 17?
- (c) Mean of five numbers is 9. Four are 3, 8, 10, 16. Find the fifth.

**T.II.10.002 Calculate the mean (Y6 statutory)**

- (a) Mean of 5, 6, 9, 11, 15, 15?
- (b) Range of 5, 6, 9, 11, 15, 15?
- (c) Mean of five numbers is 8. Four are 5, 6, 9, 11. Find the fifth.

**T.II.10.003 Line graphs: interpret dual scales carefully**

- (a) Mean of 3, 10, 14, 14, 17, 17?
- (b) Range of 3, 10, 14, 14, 17, 17?
- (c) Mean of five numbers is 13. Four are 3, 10, 14, 14. Find the fifth.

**T.II.10.004 Timetables and dual bar charts combined**

- (a) Mean of 5, 6, 6, 8, 15, 16?
- (b) Range of 5, 6, 6, 8, 15, 16?
- (c) Mean of five numbers is 8. Four are 5, 6, 6, 8. Find the fifth.

**T.II.10.005 CCSS measurement data displays pack**

- (a) Mean of 6, 9, 15, 16, 17, 18?
- (b) Range of 6, 9, 15, 16, 17, 18?
- (c) Mean of five numbers is 14. Four are 6, 9, 15, 16. Find the fifth.

**T.II.10.006 Mode and range early (UK secondary bridge)**

- (a) Mean of 5, 7, 11, 12, 15, 17?
- (b) Range of 5, 7, 11, 12, 15, 17?
- (c) Mean of five numbers is 8. Four are 5, 7, 11, 12. Find the fifth.

**T.II.10.007 Interpret misleading graphs critically**

- (a) Mean of 8, 10, 10, 12, 13, 15?
- (b) Range of 8, 10, 10, 12, 13, 15?
- (c) Mean of five numbers is 13. Four are 8, 10, 10, 12. Find the fifth.

**T.II.10.008 Dual bar charts compare categories**

- (a) Mean of 4, 6, 16, 18, 18, 18?
- (b) Range of 4, 6, 16, 18, 18, 18?
- (c) Mean of five numbers is 14. Four are 4, 6, 16, 18. Find the fifth.

**T.II.10.009 Pie chart and mean combo 1**

- (a) Mean of 5, 7, 8, 15, 16, 18?
- (b) Range of 5, 7, 8, 15, 16, 18?
- (c) Mean of five numbers is 12. Four are 5, 7, 8, 15. Find the fifth.

**T.II.10.010 Pie chart and mean combo 2**

- (a) Mean of 7, 10, 11, 12, 16, 18?
- (b) Range of 7, 10, 11, 12, 16, 18?
- (c) Mean of five numbers is 11. Four are 7, 10, 11, 12. Find the fifth.

# Era III — Relate

Graph paper / coordinate lounge

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## T.III.1.001 Standard form $A \times 10^B$ introduction (KS3)

- (a) Write 5700000 in standard form.
- (b) Write  $6.5 \times 10^5$  as an ordinary number.
- (c) Calculate  $(5.7 \times 10^6) \times (3.0 \times 10^2)$  in standard form.

## T.III.1.002 Convert ordinary $\leftrightarrow$ standard form fluency

- (a) Write 36000 in standard form.
- (b) Write  $4.0 \times 10^2$  as an ordinary number.
- (c) Calculate  $(3.6 \times 10^4) \times (2.2 \times 10^2)$  in standard form.

## T.III.1.003 Order numbers in standard form

- (a) Write 2500000 in standard form.
- (b) Write  $7.5 \times 10^2$  as an ordinary number.
- (c) Calculate  $(2.5 \times 10^6) \times (1.8 \times 10^2)$  in standard form.

## T.III.1.004 Calculate with standard form (non-calc)

- (a) Write 0.0064 in standard form.
- (b) Write  $3.1 \times 10^3$  as an ordinary number.
- (c) Calculate  $(6.4 \times 10^2) \times (2.4 \times 10^2)$  in standard form.

## T.III.1.005 Standard form with calculator (GCSE)

- (a) Write  $9.2 \times 10^{-5}$  in standard form.
- (b) Write  $3.3 \times 10^3$  as an ordinary number.
- (c) GDC lab for «Standard form with calculator (GCSE)»: record menu path and numerical output.

## T.III.1.006 Negative numbers in financial contexts

- (a) Write 24000 in standard form.
- (b) Write  $3.2 \times 10^4$  as an ordinary number.
- (c) Calculate  $(2.4 \times 10^4) \times (1.3 \times 10^2)$  in standard form.

## T.III.1.007 Recurring decimal notation (dot notation UK)

- (a) Write 3000 in standard form.
- (b) Write  $4.4 \times 10^4$  as an ordinary number.
- (c) Calculate  $(3.0 \times 10^3) \times (2.6 \times 10^2)$  in standard form.

## T.III.1.008 Convert recurring decimals to fractions

- (a) Write  $7.4 \times 10^{-5}$  in standard form.
- (b) Write  $2.9 \times 10^4$  as an ordinary number.
- (c) Calculate  $(7.4 \times 10^2) \times (2.1 \times 10^2)$  in standard form.

## T.III.1.009 Absolute value equations simple (KS3)

- (a) Write  $6.7 \times 10^{-5}$  in standard form.
- (b) Write  $4.1 \times 10^3$  as an ordinary number.
- (c) Calculate  $(6.7 \times 10^2) \times (2.6 \times 10^2)$  in standard form.

## T.III.1.010 Error intervals from rounded values intro

- (a) 190 cm to nearest 10 cm. Write the error interval for length L.
- (b)  $w=2.6$  cm,  $l=11.6$  cm each to nearest 0.1 cm. Upper bound of perimeter?
- (c) Maximise speed = distance/time given rounded measurements. Which bounds?

## T.III.1.011 Upper and lower bounds of rounded measurements

- (a) 180 cm to nearest 10 cm. Write the error interval for length L.
- (b)  $w=5.5$  cm,  $l=5.1$  cm each to nearest 0.1 cm. Upper bound of perimeter?
- (c) Maximise speed = distance/time given rounded measurements. Which bounds?

## T.III.1.012 Bounds in calculations (sum/product/quotient)

- (a) 70 cm to nearest 10 cm. Write the error interval for length L.

- (b)  $w=4.6$  cm,  $l=7.3$  cm each to nearest 0.1 cm. Upper bound of perimeter?
- (c) Maximise speed = distance/time given rounded measurements. Which bounds?

#### **T.III.1.013 Limits of accuracy wording exam pack**

- (a) 40 cm to nearest 10 cm. Write the error interval for length L.
- (b)  $w=4.5$  cm,  $l=9.8$  cm each to nearest 0.1 cm. Upper bound of perimeter?
- (c) Maximise speed = distance/time given rounded measurements. Which bounds?

#### **T.III.1.014 Bounds to 1 d.p. from truncated measurements**

- (a) 90 cm to nearest 10 cm. Write the error interval for length L.
- (b)  $w=6.3$  cm,  $l=10.8$  cm each to nearest 0.1 cm. Upper bound of perimeter?
- (c) Maximise speed = distance/time given rounded measurements. Which bounds?

#### **T.III.1.015 Error intervals for continuous measures**

- (a) 150 cm to nearest 10 cm. Write the error interval for length L.
- (b)  $w=6.1$  cm,  $l=8.0$  cm each to nearest 0.1 cm. Upper bound of perimeter?
- (c) Maximise speed = distance/time given rounded measurements. Which bounds?

#### **T.III.1.016 Bounds: upper and lower for truncated data**

- (a) 60 cm to nearest 10 cm. Write the error interval for length L.
- (b)  $w=5.9$  cm,  $l=6.3$  cm each to nearest 0.1 cm. Upper bound of perimeter?
- (c) Maximise speed = distance/time given rounded measurements. Which bounds?

#### **T.III.1.017 Calculator vs non-calculator standard form**

- (a) Write  $3.8e-05$  in standard form.
- (b) Write  $6.0 \times 10^2$  as an ordinary number.
- (c) GDC lab for «Calculator vs non-calculator standard form»: record menu path and numerical output.

#### **T.III.1.018 Standard form applied science context 1**

- (a) Write 14000 in standard form.
- (b) Write  $8.1 \times 10^4$  as an ordinary number.
- (c) Calculate  $(1.4 \times 10^4) \times (2.2 \times 10^2)$  in standard form.

#### **T.III.1.019 Standard form applied science context 2**

- (a) Write 260000 in standard form.
- (b) Write  $7.1 \times 10^2$  as an ordinary number.
- (c) Calculate  $(2.6 \times 10^5) \times (2.7 \times 10^2)$  in standard form.

#### **T.III.1.020 Standard form applied science context 3**

- (a) Write 5500 in standard form.
- (b) Write  $5.8 \times 10^5$  as an ordinary number.
- (c) Calculate  $(5.5 \times 10^3) \times (1.1 \times 10^2)$  in standard form.

#### **T.III.2.001 Compound measure: speed (m/s and km/h)**

- (a) Share £52 in the ratio 5:8. First share?
- (b) Ratio 5:8; first quantity 20. Second?
- (c) Map scale 1:25000. Map distance 8 cm. Real distance in km?

#### **T.III.2.002 Compound measure: density (mass÷volume)**

- (a) Share £130 in the ratio 5:8. First share?
- (b) Ratio 5:8; first quantity 50. Second?
- (c) Map scale 1:25000. Map distance 8 cm. Real distance in km?

#### **T.III.2.003 Compound measure: pressure**

- (a) Share £64 in the ratio 3:5. First share?
- (b) Ratio 3:5; first quantity 24. Second?
- (c) Map scale 1:25000. Map distance 6 cm. Real distance in km?

#### **T.III.2.004 Rates of pay and unit pricing**

- (a) Share £56 in the ratio 3:5. First share?
- (b) Ratio 3:5; first quantity 21. Second?

(c) Map scale 1:25000. Map distance 6 cm. Real distance in km?

**T.III.2.005 Currency conversion with exchange rates**

(a) Share £65 in the ratio 5:8. First share?

(b) Ratio 5:8; first quantity 25. Second?

(c) Map scale 1:25000. Map distance 8 cm. Real distance in km?

**T.III.2.006 Map scales as ratios (1:n)**

(a) Share £20 in the ratio 2:3. First share?

(b) Ratio 2:3; first quantity 8. Second?

(c) Map scale 1:25000. Map distance 5 cm. Real distance in km?

**T.III.2.007 Sharing in a ratio three-part**

(a) Share £130 in the ratio 5:8. First share?

(b) Ratio 5:8; first quantity 50. Second?

(c) Map scale 1:25000. Map distance 8 cm. Real distance in km?

**T.III.2.008 Ratio given difference or total variants**

(a) Share £88 in the ratio 3:5. First share?

(b) Ratio 3:5; first quantity 33. Second?

(c) Map scale 1:25000. Map distance 6 cm. Real distance in km?

**T.III.2.009 Direct proportion graphs through origin**

(a) Share £20 in the ratio 2:3. First share?

(b) Ratio 2:3; first quantity 8. Second?

(c) Map scale 1:25000. Map distance 5 cm. Real distance in km?

**T.III.2.010 Inverse proportion intro ( $xy=k$ )**

(a) Share £112 in the ratio 3:5. First share?

(b) Ratio 3:5; first quantity 42. Second?

(c) Map scale 1:25000. Map distance 6 cm. Real distance in km?

**T.III.2.011 Speed–distance–time with unit changes**

(a) Share £143 in the ratio 4:7. First share?

(b) Ratio 4:7; first quantity 52. Second?

(c) Map scale 1:25000. Map distance 7 cm. Real distance in km?

**T.III.2.012 Population density and area ratios**

(a) Share £50 in the ratio 2:3. First share?

(b) Ratio 2:3; first quantity 20. Second?

(c) Map scale 1:25000. Map distance 5 cm. Real distance in km?

**T.III.2.013 Recipe scaling and mixture ratios**

(a) Share £78 in the ratio 5:8. First share?

(b) Ratio 5:8; first quantity 30. Second?

(c) Map scale 1:25000. Map distance 8 cm. Real distance in km?

**T.III.2.014 CCSS RP: tape diagrams for percent/ratio**

(a) Share £55 in the ratio 4:7. First share?

(b) Ratio 4:7; first quantity 20. Second?

(c) Map scale 1:25000. Map distance 7 cm. Real distance in km?

**T.III.2.015 CCSS RP: percent as ratio of part to whole**

(a) Share £32 in the ratio 3:5. First share?

(b) Ratio 3:5; first quantity 12. Second?

(c) Map scale 1:25000. Map distance 6 cm. Real distance in km?

**T.III.2.016 Mixing problems with ratios**

(a) Share £35 in the ratio 2:3. First share?

(b) Ratio 2:3; first quantity 14. Second?

(c) Map scale 1:25000. Map distance 5 cm. Real distance in km?

### **T.III.2.017 Scale drawings measure and interpret**

- (a) Share £121 in the ratio 4:7. First share?
- (b) Ratio 4:7; first quantity 44. Second?
- (c) Geometry construction lab for «Scale drawings measure and interpret»: build the figure, measure, verify the claimed relation.

### **T.III.2.018 Proportion word problems multi-step**

- (a) Share £182 in the ratio 5:8. First share?
- (b) Ratio 5:8; first quantity 70. Second?
- (c) Map scale 1:25000. Map distance 8 cm. Real distance in km?

### **T.III.2.019 Average speed for multi-leg journeys**

- (a) Share £78 in the ratio 5:8. First share?
- (b) Ratio 5:8; first quantity 30. Second?
- (c) Map scale 1:25000. Map distance 8 cm. Real distance in km?

### **T.III.2.020 Convert compound units step by step**

- (a) Share £121 in the ratio 4:7. First share?
- (b) Ratio 4:7; first quantity 44. Second?
- (c) Map scale 1:25000. Map distance 7 cm. Real distance in km?

### **T.III.2.021 Compound measures mixed drill 1**

- (a) Share £112 in the ratio 3:5. First share?
- (b) Ratio 3:5; first quantity 42. Second?
- (c) Map scale 1:25000. Map distance 6 cm. Real distance in km?

### **T.III.2.022 Compound measures mixed drill 2**

- (a) Share £72 in the ratio 3:5. First share?
- (b) Ratio 3:5; first quantity 27. Second?
- (c) Map scale 1:25000. Map distance 6 cm. Real distance in km?

### **T.III.2.023 Compound measures mixed drill 3**

- (a) Share £112 in the ratio 3:5. First share?
- (b) Ratio 3:5; first quantity 42. Second?
- (c) Map scale 1:25000. Map distance 6 cm. Real distance in km?

### **T.III.2.024 Compound measures mixed drill 4**

- (a) Share £32 in the ratio 3:5. First share?
- (b) Ratio 3:5; first quantity 12. Second?
- (c) Map scale 1:25000. Map distance 6 cm. Real distance in km?

### **T.III.2.025 Compound measures mixed drill 5**

- (a) Share £169 in the ratio 5:8. First share?
- (b) Ratio 5:8; first quantity 65. Second?
- (c) Map scale 1:25000. Map distance 8 cm. Real distance in km?

### **T.III.3.001 Find original value after % change (reverse %)**

- (a) Increase £400 by 8%.
- (b) After a 8% increase a price is £432. Find the original.
- (c) A quantity rises by 8% then by another 8%. Overall % increase?

### **T.III.3.002 Simple interest exam pack ( $I = PRT/100$ )**

- (a) Increase £400 by 12%.
- (b) After a 12% increase a price is £448. Find the original.
- (c) A quantity rises by 12% then by another 12%. Overall % increase?

### **T.III.3.003 Compound interest (non-calc appreciation)**

- (a) Increase £400 by 8%.
- (b) After a 8% increase a price is £432. Find the original.
- (c) A quantity rises by 8% then by another 8%. Overall % increase?

### **T.III.3.004 Compound interest depreciation / decay**

- (a) Increase £120 by 10%.
- (b) After a 10% increase a price is £132. Find the original.
- (c) A quantity rises by 10% then by another 10%. Overall % increase?

#### **T.III.3.005 VAT and sales tax multi-step**

- (a) Increase £250 by 10%.
- (b) After a 10% increase a price is £275. Find the original.
- (c) A quantity rises by 10% then by another 10%. Overall % increase?

#### **T.III.3.006 Hire purchase and interest comparisons**

- (a) Increase £80 by 15%.
- (b) After a 15% increase a price is £92. Find the original.
- (c) A quantity rises by 15% then by another 15%. Overall % increase?

#### **T.III.3.007 Successive percentage changes (not additive)**

- (a) Increase £80 by 5%.
- (b) After a 5% increase a price is £84. Find the original.
- (c) A quantity rises by 5% then by another 5%. Overall % increase?

#### **T.III.3.008 Express one quantity as % of another**

- (a) Increase £200 by 5%.
- (b) After a 5% increase a price is £210. Find the original.
- (c) A quantity rises by 5% then by another 5%. Overall % increase?

#### **T.III.3.009 Percentage profit and loss**

- (a) Increase £200 by 12%.
- (b) After a 12% increase a price is £224. Find the original.
- (c) A quantity rises by 12% then by another 12%. Overall % increase?

#### **T.III.3.010 Repeated percentage growth GDC workflow**

- (a) Increase £400 by 8%.
- (b) After a 8% increase a price is £432. Find the original.
- (c) GDC lab for «Repeated percentage growth GDC workflow»: record menu path and numerical output.

#### **T.III.3.011 Bank statements and balance tracking**

- (a) Increase £120 by 12%.
- (b) After a 12% increase a price is £134. Find the original.
- (c) A quantity rises by 12% then by another 12%. Overall % increase?

#### **T.III.3.012 Best-buy percentage vs absolute saving**

- (a) Increase £80 by 12%.
- (b) After a 12% increase a price is £89. Find the original.
- (c) A quantity rises by 12% then by another 12%. Overall % increase?

#### **T.III.3.013 Percentage multipliers one-step fluency**

- (a) Increase £250 by 20%.
- (b) After a 20% increase a price is £300. Find the original.
- (c) A quantity rises by 20% then by another 20%. Overall % increase?

#### **T.III.3.014 Find % when given two related quantities**

- (a) Increase £400 by 20%.
- (b) After a 20% increase a price is £480. Find the original.
- (c) A quantity rises by 20% then by another 20%. Overall % increase?

#### **T.III.3.015 Inflation and real-terms value (simple)**

- (a) Increase £80 by 10%.
- (b) After a 10% increase a price is £88. Find the original.
- (c) A quantity rises by 10% then by another 10%. Overall % increase?

#### **T.III.3.016 Express changes using percentage multipliers chain**

- (a) Increase £80 by 8%.

- (b) After a 8% increase a price is £86. Find the original.
- (c) A quantity rises by 8% then by another 8%. Overall % increase?

#### **T.III.3.017 Financial % multi-step scenario 1**

- (a) Increase £250 by 5%.
- (b) After a 5% increase a price is £262. Find the original.
- (c) A quantity rises by 5% then by another 5%. Overall % increase?

#### **T.III.3.018 Financial % multi-step scenario 2**

- (a) Increase £80 by 5%.
- (b) After a 5% increase a price is £84. Find the original.
- (c) A quantity rises by 5% then by another 5%. Overall % increase?

#### **T.III.3.019 Financial % multi-step scenario 3**

- (a) Increase £250 by 5%.
- (b) After a 5% increase a price is £262. Find the original.
- (c) A quantity rises by 5% then by another 5%. Overall % increase?

#### **T.III.3.020 Financial % multi-step scenario 4**

- (a) Increase £250 by 8%.
- (b) After a 8% increase a price is £270. Find the original.
- (c) A quantity rises by 8% then by another 8%. Overall % increase?

#### **T.III.3.021 Financial % multi-step scenario 5**

- (a) Increase £80 by 15%.
- (b) After a 15% increase a price is £92. Find the original.
- (c) A quantity rises by 15% then by another 15%. Overall % increase?

#### **T.III.4.001 Laws of indices exam pack (integer)**

- (a) Simplify  $\sqrt{50}$ .
- (b) Rationalise  $1/\sqrt{2}$ .
- (c) Expand and simplify  $(\sqrt{2}+2)(\sqrt{2}-2)$ .

#### **T.III.4.002 Fractional indices as roots**

- (a) Simplify  $\sqrt{20}$ .
- (b) Rationalise  $1/\sqrt{3}$ .
- (c) Expand and simplify  $(\sqrt{3}+2)(\sqrt{3}-2)$ .

#### **T.III.4.003 Negative and fractional indices combined**

- (a) Simplify  $\sqrt{72}$ .
- (b) Rationalise  $1/\sqrt{2}$ .
- (c) Expand and simplify  $(\sqrt{2}+2)(\sqrt{2}-2)$ .

#### **T.III.4.004 Surds: simplify $\sqrt{ab}$ and $\sqrt{a/b}$**

- (a) Simplify  $\sqrt{50}$ .
- (b) Rationalise  $1/\sqrt{2}$ .
- (c) Expand and simplify  $(\sqrt{2}+2)(\sqrt{2}-2)$ .

#### **T.III.4.005 Surds: expand and simplify $(a+\sqrt{b})(c+\sqrt{d})$**

- (a) Simplify  $\sqrt{18}$ .
- (b) Rationalise  $1/\sqrt{3}$ .
- (c) Expand and simplify  $(\sqrt{3}+2)(\sqrt{3}-2)$ .

#### **T.III.4.006 Rationalise denominators (monomial and binomial)**

- (a) Simplify  $\sqrt{27}$ .
- (b) Rationalise  $1/\sqrt{7}$ .
- (c) Expand and simplify  $(\sqrt{7}+2)(\sqrt{7}-2)$ .

#### **T.III.4.007 Scientific notation vs UK standard form wording**

- (a) Write 0.0079 in standard form.
- (b) Write  $3.4 \times 10^5$  as an ordinary number.

(c) Calculate  $(7.9 \times 10^2) \times (1.6 \times 10^2)$  in standard form.

**T.III.4.008 Estimate powers and roots without calc**

(a) Simplify  $\sqrt{32}$ .

(b) Rationalise  $1/\sqrt{3}$ .

(c) Expand and simplify  $(\sqrt{3}+2)(\sqrt{3}-2)$ .

**T.III.4.009 Zero and negative indices meaning pack**

(a) Simplify  $\sqrt{27}$ .

(b) Rationalise  $1/\sqrt{7}$ .

(c) Expand and simplify  $(\sqrt{7}+2)(\sqrt{7}-2)$ .

**T.III.4.010 Work in standard form then convert back**

(a) Write 270000 in standard form.

(b) Write  $3.0 \times 10^4$  as an ordinary number.

(c) Calculate  $(2.7 \times 10^5) \times (2.5 \times 10^2)$  in standard form.

**T.III.4.011 Evaluate with fractional/negative indices non-calc**

(a) Simplify  $\sqrt{8}$ .

(b) Rationalise  $1/\sqrt{7}$ .

(c) Expand and simplify  $(\sqrt{7}+2)(\sqrt{7}-2)$ .

**T.III.5.001 Expand single and double brackets exam pack**

(a) Expand  $2(x+4)$ .

(b) Expand  $(x+5)(x+5)$ .

(c) Factorise  $x^2 + 10x + 25$ .

**T.III.5.002 Factorise linear expressions fully**

(a) Expand  $4(x+5)$ .

(b) Expand  $(x+3)(x+3)$ .

(c) Factorise  $x^2 + 6x + 9$ .

**T.III.5.003 Factorise quadratics a=1 exam pack**

(a) Expand  $4(x+4)$ .

(b) Expand  $(x+4)(x+3)$ .

(c) Factorise  $x^2 + 7x + 12$ .

**T.III.5.004 Factorise difference of two squares**

(a) Simplify  $\sqrt{8}$ .

(b) Rationalise  $1/\sqrt{3}$ .

(c) Expand and simplify  $(\sqrt{3}+2)(\sqrt{3}-2)$ .

**T.III.5.005 Algebraic fractions simplify (linear)**

(a) Expand  $2(x+3)$ .

(b) Expand  $(x+2)(x+4)$ .

(c) Factorise  $x^2 + 6x + 8$ .

**T.III.5.006 Write expressions from worded contexts**

(a) Expand  $3(x+2)$ .

(b) Expand  $(x+2)(x+3)$ .

(c) Factorise  $x^2 + 5x + 6$ .

**T.III.5.007 Identities vs equations recognition**

(a) Expand  $3(x+2)$ .

(b) Expand  $(x+1)(x+3)$ .

(c) Factorise  $x^2 + 4x + 3$ .

**T.III.5.008 Collect like terms with multiple variables**

(a) Expand  $5(x+1)$ .

(b) Expand  $(x+3)(x+6)$ .

(c) Factorise  $x^2 + 9x + 18$ .

**T.III.5.009 Substitution including negatives and powers**

- (a) Expand  $5(x+1)$ .
- (b) Expand  $(x+1)(x+2)$ .
- (c) Factorise  $x^2 + 3x + 2$ .

**T.III.5.010 Expand three binomials (Higher)**

- (a) Expand  $5(x+4)$ .
- (b) Expand  $(x+1)(x+2)$ .
- (c) Factorise  $x^2 + 3x + 2$ .

**T.III.5.011 Factorise by grouping (four terms)**

- (a) Expand  $6(x+1)$ .
- (b) Expand  $(x+4)(x+2)$ .
- (c) Factorise  $x^2 + 6x + 8$ .

**T.III.6.001 Solve linear equations with fractions**

- (a) Solve  $6x + 8 = 38$ .
- (b) Solve  $6(x - 8) = -12$ .
- (c) List integers  $n$  satisfying  $5 < n \leq 9$ .

**T.III.6.002 Form and solve equations from geometry**

- (a) Solve  $4x + 7 = 31$ .
- (b) Solve  $4(x - 7) = 0$ .
- (c) List integers  $n$  satisfying  $6 < n \leq 10$ .

**T.III.6.003 Inequalities on number lines (UK open/closed)**

- (a) Solve  $9x + 5 = 32$ .
- (b) Solve  $9(x - 5) = -9$ .
- (c) List integers  $n$  satisfying  $3 < n \leq 7$ .

**T.III.6.004 Solve linear inequalities and list integers**

- (a) Solve  $7x + 4 = 67$ .
- (b) Solve  $7(x - 4) = 42$ .
- (c) List integers  $n$  satisfying  $9 < n \leq 13$ .

**T.III.6.005 Simultaneous equations elimination (KS3/4)**

- (a) Solve  $2x + 3 = 13$ .
- (b) Solve  $2(x - 3) = 6$ .
- (c) List integers  $n$  satisfying  $5 < n \leq 9$ .

**T.III.6.006 Simultaneous equations substitution**

- (a) Solve  $4x + 2 = 22$ .
- (b) Solve  $4(x - 2) = 16$ .
- (c) List integers  $n$  satisfying  $5 < n \leq 9$ .

**T.III.6.007 Graphical solution of simultaneous equations**

- (a) Solve  $5x + 1 = 46$ .
- (b) Solve  $5(x - 1) = 45$ .
- (c) Desmos lab for «Graphical solution of simultaneous equations»: vary one parameter, capture before/after, write one observation.

**T.III.6.008 Equations with unknown on both sides**

- (a) Solve  $8x + 2 = 42$ .
- (b) Solve  $8(x - 2) = 32$ .
- (c) List integers  $n$  satisfying  $5 < n \leq 9$ .

**T.III.6.009 Inequalities with negative coefficients**

- (a) Solve  $2x + 6 = 24$ .
- (b) Solve  $2(x - 6) = 8$ .
- (c) List integers  $n$  satisfying  $9 < n \leq 13$ .

**T.III.6.010 Quadratic inequalities intro via graphs**

- (a) Solve  $4x + 6 = 10$ .
- (b) Solve  $4(x - 6) = -16$ .
- (c) Desmos lab for «Quadratic inequalities intro via graphs»: vary one parameter, capture before/after, write one observation.

#### **T.III.6.011 Trial and improvement (legacy dialect)**

- (a) Solve  $3x + 6 = 18$ .
- (b) Solve  $3(x - 6) = -3$ .
- (c) List integers  $n$  satisfying  $4 < n \leq 8$ .

#### **T.III.7.001 Gradient from coordinates (UK wording)**

- (a) Expand  $2(x+5)$ .
- (b) Expand  $(x+3)(x+4)$ .
- (c) Factorise  $x^2 + 7x + 12$ .

#### **T.III.7.002 Equation of a line $y=mx+c$ fluency**

- (a) Expand  $2(x+1)$ .
- (b) Expand  $(x+5)(x+6)$ .
- (c) Factorise  $x^2 + 11x + 30$ .

#### **T.III.7.003 Parallel and perpendicular gradients**

- (a) Expand  $6(x+6)$ .
- (b) Expand  $(x+2)(x+1)$ .
- (c) Factorise  $x^2 + 3x + 2$ .

#### **T.III.7.004 Midpoint and length of a line segment**

- (a) Expand  $5(x+2)$ .
- (b) Expand  $(x+4)(x+5)$ .
- (c) Factorise  $x^2 + 9x + 20$ .

#### **T.III.7.005 Real graphs: conversion and distance–time**

- (a) Expand  $4(x+1)$ .
- (b) Expand  $(x+2)(x+3)$ .
- (c) Graph lab for «Real graphs: conversion and distance–time»: plot, mark key points, state the trend.

#### **T.III.7.006 Speed from distance–time graph gradients**

- (a) Expand  $5(x+6)$ .
- (b) Expand  $(x+5)(x+5)$ .
- (c) Graph lab for «Speed from distance–time graph gradients»: plot, mark key points, state the trend.

#### **T.III.7.007 Instantaneous rate of change on a curve (KS4 tease)**

- (a) Expand  $4(x+5)$ .
- (b) Expand  $(x+1)(x+4)$ .
- (c) Desmos lab for «Instantaneous rate of change on a curve (KS4 tease)»: vary one parameter, capture before/after, write one observation.

#### **T.III.7.008 Find equation of parallel line through a point**

- (a) Expand  $4(x+5)$ .
- (b) Expand  $(x+3)(x+6)$ .
- (c) Factorise  $x^2 + 9x + 18$ .

#### **T.III.7.009 Interpret intercepts in context**

- (a) Expand  $2(x+5)$ .
- (b) Expand  $(x+4)(x+1)$ .
- (c) Graph lab for «Interpret intercepts in context»: plot, mark key points, state the trend.

#### **T.III.7.010 Solve linear equations graphically on Desmos**

- (a) Expand  $5(x+3)$ .
- (b) Expand  $(x+5)(x+4)$ .
- (c) Desmos lab for «Solve linear equations graphically on Desmos»: vary one parameter, capture before/after, write one observation.

#### **T.III.7.011 Find intercepts algebraically and graphically**

- (a) Expand  $5(x+5)$ .

(b) Expand  $(x+1)(x+2)$ .

(c) Desmos lab for «Find intercepts algebraically and graphically»: vary one parameter, capture before/after, write one observation.

#### **T.III.8.001 Plans and elevations of 3D shapes**

(a) [Plans and elevations of 3D shapes] Which view is the plan — front, side, or top?

(b) Cuboid  $7 \times 3 \times 5$ : how many rectangles in a correct net?

(c) Geometry construction lab for «Plans and elevations of 3D shapes»: build the figure, measure, verify the claimed relation.

#### **T.III.8.002 Nets of pyramids, prisms, cylinders**

(a) A cuboid measures  $8 \times 7 \times 6$ . How many faces does a correct net contain?

(b) Which of these cannot fold to a cube: cross of 6 squares vs row of 6 squares? Explain.

(c) Surface area of cuboid  $8 \times 7 \times 6$ ?

#### **T.III.8.003 Bearings intro: three-figure bearings**

(a) Write  $255^\circ$  from north as a three-figure bearing.

(b) Bearing of B from A is 255. Bearing of A from B?

(c) Geometry construction lab for «Bearings intro: three-figure bearings»: build the figure, measure, verify the claimed relation.

#### **T.III.8.004 Construct triangles given SSS/SAS/ASA**

(a) [Construct triangles given SSS/SAS/ASA] Which view is the plan — front, side, or top?

(b) Cuboid  $3 \times 3 \times 3$ : how many rectangles in a correct net?

(c) Geometry construction lab for «Construct triangles given SSS/SAS/ASA»: build the figure, measure, verify the claimed relation.

#### **T.III.8.005 Loci: set of points equidistant**

(a) [Loci: set of points equidistant] Which view is the plan — front, side, or top?

(b) Cuboid  $4 \times 7 \times 5$ : how many rectangles in a correct net?

(c) Geometry construction lab for «Loci: set of points equidistant»: build the figure, measure, verify the claimed relation.

#### **T.III.8.006 Loci: regions satisfying inequalities**

(a) [Loci: regions satisfying inequalities] Which view is the plan — front, side, or top?

(b) Cuboid  $5 \times 3 \times 5$ : how many rectangles in a correct net?

(c) Geometry construction lab for «Loci: regions satisfying inequalities»: build the figure, measure, verify the claimed relation.

#### **T.III.8.007 Pythagoras exam worded (2D)**

(a) Legs 6 cm, 8 cm. Hypotenuse?

(b) Hypotenuse 10 m, leg 6 m. Other leg?

(c) Ladder 10 m; base 6 m from wall. Reach height?

#### **T.III.8.008 Pythagoras in 3D (space diagonal)**

(a) Legs 6 cm, 8 cm. Hypotenuse?

(b) Hypotenuse 10 m, leg 6 m. Other leg?

(c) Ladder 10 m; base 6 m from wall. Reach height?

#### **T.III.8.009 Circle area/circumference exact answers in $\pi$**

(a) [Circle area/circumference exact answers in  $\pi$ ] Which view is the plan — front, side, or top?

(b) Cuboid  $9 \times 4 \times 2$ : how many rectangles in a correct net?

(c) Describe the front elevation of a square-based pyramid (base 9) standing on its base.

#### **T.III.8.010 Arc length and sector area (GCSE)**

(a) [Arc length and sector area (GCSE)] Which view is the plan — front, side, or top?

(b) Cuboid  $3 \times 7 \times 4$ : how many rectangles in a correct net?

(c) Describe the front elevation of a square-based pyramid (base 3) standing on its base.

#### **T.III.8.011 Volume/surface area formula selection pack**

(a) [Volume/surface area formula selection pack] Which view is the plan — front, side, or top?

(b) Cuboid  $9 \times 6 \times 6$ : how many rectangles in a correct net?

(c) Describe the front elevation of a square-based pyramid (base 9) standing on its base.

#### **T.III.8.012 Similar shapes length/area/volume scale factors**

(a) Lengths scale by 3. Area scale factor?

(b) A side 6 cm becomes ? after enlargement  $SF=3$ .

(c) Volumes scale by ? if lengths scale by 3.

#### **T.III.8.013 Congruence conditions SSS SAS ASA RHS**

(a) Name three congruence conditions for triangles.

(b) Does SSA always prove congruence? Why/why not?

(c) In «Congruence conditions SSS SAS ASA RHS»: what does CPCTC allow you to conclude after triangles are congruent?

#### **T.III.8.014 Enlargement with fractional and negative SF**

(a) Lengths scale by 5. Area scale factor?

(b) A side 6 cm becomes ? after enlargement  $SF=5$ .

(c) Geometry construction lab for «Enlargement with fractional and negative SF»: build the figure, measure, verify the claimed relation.

#### **T.III.8.015 Describe transformations fully (GCSE language)**

(a) [Describe transformations fully (GCSE language)] Which view is the plan — front, side, or top?

(b) Cuboid  $8 \times 4 \times 6$ : how many rectangles in a correct net?

(c) Describe the front elevation of a square-based pyramid (base 8) standing on its base.

#### **T.III.8.016 Interior/exterior angles of polygons exam**

(a) [Interior/exterior angles of polygons exam] Which view is the plan — front, side, or top?

(b) Cuboid  $8 \times 6 \times 6$ : how many rectangles in a correct net?

(c) Describe the front elevation of a square-based pyramid (base 8) standing on its base.

#### **T.III.8.017 Tessellation angle conditions**

(a) [Tessellation angle conditions] Which view is the plan — front, side, or top?

(b) Cuboid  $7 \times 2 \times 4$ : how many rectangles in a correct net?

(c) Geometry construction lab for «Tessellation angle conditions»: build the figure, measure, verify the claimed relation.

#### **T.III.8.018 Surface area of cylinders exact and decimal**

(a) [Surface area of cylinders exact and decimal] Which view is the plan — front, side, or top?

(b) Cuboid  $6 \times 5 \times 2$ : how many rectangles in a correct net?

(c) Describe the front elevation of a square-based pyramid (base 6) standing on its base.

#### **T.III.8.019 Exact trig values preview: 30–60–90 links**

(a) Exact value of  $\sin 30^\circ$ .

(b) Exact value of  $\cos 45^\circ$ .

(c) Without calculator, evaluate  $2 \sin 30^\circ \cos 30^\circ$ .

#### **T.III.8.020 Isometric drawing of 3D shapes**

(a) [Isometric drawing of 3D shapes] Which view is the plan — front, side, or top?

(b) Cuboid  $8 \times 2 \times 4$ : how many rectangles in a correct net?

(c) Geometry construction lab for «Isometric drawing of 3D shapes»: build the figure, measure, verify the claimed relation.

#### **T.III.8.021 Exact answers leave in terms of $\pi$ pack**

(a) [Exact answers leave in terms of  $\pi$  pack] Which view is the plan — front, side, or top?

(b) Cuboid  $9 \times 4 \times 2$ : how many rectangles in a correct net?

(c) Describe the front elevation of a square-based pyramid (base 9) standing on its base.

#### **T.III.8.022 Surface area of spheres leave in $\pi$**

(a) [Surface area of spheres leave in  $\pi$ ] Which view is the plan — front, side, or top?

(b) Cuboid  $6 \times 5 \times 5$ : how many rectangles in a correct net?

(c) Describe the front elevation of a square-based pyramid (base 6) standing on its base.

#### **T.III.8.023 Voronoi diagrams nearest-site partitioning**

(a) [Voronoi diagrams nearest-site partitioning] Which view is the plan — front, side, or top?

(b) Cuboid  $5 \times 2 \times 4$ : how many rectangles in a correct net?

(c) Geometry construction lab for «Voronoi diagrams nearest-site partitioning»: build the figure, measure, verify the claimed relation.

#### **T.III.8.024 Voronoi applications: services catchment**

(a) [Voronoi applications: services catchment] Which view is the plan — front, side, or top?

(b) Cuboid  $4 \times 2 \times 3$ : how many rectangles in a correct net?

(c) Describe the front elevation of a square-based pyramid (base 4) standing on its base.

### **T.III.8.025 Graph theory: walks trails paths cycles (light)**

- (a) [Graph theory: walks trails paths cycles (light)] Which view is the plan — front, side, or top?
- (b) Cuboid  $3 \times 7 \times 3$ : how many rectangles in a correct net?
- (c) Describe the front elevation of a square-based pyramid (base 3) standing on its base.

### **T.III.8.026 Shortest path / Chinese Postman light**

- (a) [Shortest path / Chinese Postman light] Which view is the plan — front, side, or top?
- (b) Cuboid  $8 \times 4 \times 6$ : how many rectangles in a correct net?
- (c) Describe the front elevation of a square-based pyramid (base 8) standing on its base.

### **T.III.8.027 Plans elevations practice figure 1**

- (a) [Plans elevations practice figure 1] Which view is the plan — front, side, or top?
- (b) Cuboid  $3 \times 2 \times 6$ : how many rectangles in a correct net?
- (c) Geometry construction lab for «Plans elevations practice figure 1»: build the figure, measure, verify the claimed relation.

### **T.III.8.028 Plans elevations practice figure 2**

- (a) [Plans elevations practice figure 2] Which view is the plan — front, side, or top?
- (b) Cuboid  $9 \times 4 \times 3$ : how many rectangles in a correct net?
- (c) Geometry construction lab for «Plans elevations practice figure 2»: build the figure, measure, verify the claimed relation.

### **T.III.8.029 Plans elevations practice figure 3**

- (a) [Plans elevations practice figure 3] Which view is the plan — front, side, or top?
- (b) Cuboid  $3 \times 6 \times 5$ : how many rectangles in a correct net?
- (c) Geometry construction lab for «Plans elevations practice figure 3»: build the figure, measure, verify the claimed relation.

### **T.III.9.001 Venn diagrams for probability (two sets)**

- (a) Bag: 8 red, 4 blue.  $P(\text{red})$ ?
- (b) Without replacement,  $P(\text{both red})$ ?
- (c) With replacement,  $P(\text{red then blue})$ ?

### **T.III.9.002 Venn diagrams three sets**

- (a) Bag: 3 red, 2 blue.  $P(\text{red})$ ?
- (b) Without replacement,  $P(\text{both red})$ ?
- (c) With replacement,  $P(\text{red then blue})$ ?

### **T.III.9.003 Frequency trees for probability**

- (a) Mean of 4, 10, 11, 11, 15, 18?
- (b) Range of 4, 10, 11, 11, 15, 18?
- (c) Mean of five numbers is 11. Four are 4, 10, 11, 11. Find the fifth.

### **T.III.9.004 Histograms with unequal class widths**

- (a) Class width 5, frequency 15. Frequency density?
- (b) A histogram bar has width 5 and FD 3.0. Frequency?
- (c) Why must unequal-width histograms use frequency density, not raw frequency for bar height?

### **T.III.9.005 Cumulative frequency curves and medians**

- (a) Mean of 4, 5, 5, 11, 12, 13?
- (b) Range of 4, 5, 5, 11, 12, 13?
- (c) Graph lab for «Cumulative frequency curves and medians»: plot, mark key points, state the trend.

### **T.III.9.006 Box plots from CF and compare distributions**

- (a) Mean of 3, 4, 5, 9, 9, 14?
- (b) Range of 3, 4, 5, 9, 9, 14?
- (c) Mean of five numbers is 12. Four are 3, 4, 5, 9. Find the fifth.

### **T.III.9.007 Product rule for counting (KS4 Higher)**

- (a) Mean of 5, 7, 10, 15, 15, 17?
- (b) Range of 5, 7, 10, 15, 15, 17?
- (c) Mean of five numbers is 12. Four are 5, 7, 10, 15. Find the fifth.

### **T.III.9.008 Sample space diagrams for two events**

- (a) Mean of 4, 7, 9, 10, 12, 13?
- (b) Range of 4, 7, 9, 10, 12, 13?
- (c) Mean of five numbers is 9. Four are 4, 7, 9, 10. Find the fifth.

**T.III.9.009 Relative frequency vs theoretical probability**

- (a) Mean of 5, 6, 6, 8, 15, 15?
- (b) Range of 5, 6, 6, 8, 15, 15?
- (c) Mean of five numbers is 8. Four are 5, 6, 6, 8. Find the fifth.

**T.III.9.010 Capture–recapture estimation (UK contexts)**

- (a) Mean of 4, 6, 12, 12, 12, 12?
- (b) Range of 4, 6, 12, 12, 12, 12?
- (c) Mean of five numbers is 11. Four are 4, 6, 12, 12. Find the fifth.

**T.III.9.011 Stratified sampling calculations**

- (a) Mean of 4, 4, 12, 17, 18, 18?
- (b) Range of 4, 4, 12, 17, 18, 18?
- (c) Mean of five numbers is 14. Four are 4, 4, 12, 17. Find the fifth.

**T.III.9.012 Two-way tables probability conditional**

- (a) Mean of 3, 4, 5, 8, 11, 14?
- (b) Range of 3, 4, 5, 8, 11, 14?
- (c) Mean of five numbers is 14. Four are 3, 4, 5, 8. Find the fifth.

**T.III.9.013 Pie chart construction from data**

- (a) Mean of 5, 10, 11, 14, 15, 16?
- (b) Range of 5, 10, 11, 14, 15, 16?
- (c) Mean of five numbers is 11. Four are 5, 10, 11, 14. Find the fifth.

**T.III.9.014 Moving averages (time series intro)**

- (a) Mean of 5, 5, 9, 12, 15, 16?
- (b) Range of 5, 5, 9, 12, 15, 16?
- (c) Mean of five numbers is 10. Four are 5, 5, 9, 12. Find the fifth.

**T.III.9.015 CCSS SP: chance device simulations middle school**

- (a) Mean of 3, 3, 6, 7, 13, 14?
- (b) Range of 3, 3, 6, 7, 13, 14?
- (c) Mean of five numbers is 10. Four are 3, 3, 6, 7. Find the fifth.

**T.III.9.016 CCSS SP: bivariate categorical associations**

- (a) Mean of 3, 5, 14, 16, 17, 18?
- (b) Range of 3, 5, 14, 16, 17, 18?
- (c) Mean of five numbers is 13. Four are 3, 5, 14, 16. Find the fifth.

**T.III.9.017 Mean from frequency table (estimated)**

- (a) Mean of 3, 8, 9, 12, 15, 16?
- (b) Range of 3, 8, 9, 12, 15, 16?
- (c) Mean of five numbers is 8. Four are 3, 8, 9, 12. Find the fifth.

**T.III.9.018 Modal class and median class from grouped data**

- (a) Mean of 3, 6, 6, 13, 14, 17?
- (b) Range of 3, 6, 6, 13, 14, 17?
- (c) Mean of five numbers is 9. Four are 3, 6, 6, 13. Find the fifth.

**T.III.9.019 Probability scale 0 to 1 language**

- (a) AP/IB style: Which theorem or definition is central to «Probability scale 0 to 1 language»?
- (b) Write a justification sentence frame for «Probability scale 0 to 1 language»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.
- (c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Probability scale 0 to 1 language», then interpret.

**T.III.9.020 Expectation from theoretical probability**

- (a) Mean of 4, 5, 6, 8, 10, 11?

- (b) Range of 4, 5, 6, 8, 10, 11?
- (c) Mean of five numbers is 13. Four are 4, 5, 6, 8. Find the fifth.

#### **T.III.9.021 AP Stats: describing distributions SOCS**

- (a) Mean of 9, 12, 13, 14, 17, 18?
- (b) Range of 9, 12, 13, 14, 17, 18?
- (c) Mean of five numbers is 14. Four are 9, 12, 13, 14. Find the fifth.

#### **T.III.9.022 AP Stats: comparing dual boxplots FRQ**

- (a) AP/IB style: Which theorem or definition is central to «AP Stats: comparing dual boxplots FRQ»?
- (b) Write a justification sentence frame for «AP Stats: comparing dual boxplots FRQ»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.
- (c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «AP Stats: comparing dual boxplots FRQ», then interpret.

#### **T.III.9.023 IB AI: boxplots and outliers 1.5×IQR rule**

- (a) Mean of 3, 4, 12, 16, 17, 18?
- (b) Range of 3, 4, 12, 16, 17, 18?
- (c) Mean of five numbers is 9. Four are 3, 4, 12, 16. Find the fifth.

#### **T.III.9.024 Misleading scales and truncated axes critique**

- (a) Mean of 3, 5, 6, 9, 12, 17?
- (b) Range of 3, 5, 6, 9, 12, 17?
- (c) Mean of five numbers is 10. Four are 3, 5, 6, 9. Find the fifth.

#### **T.III.9.025 Cumulative frequency ↔ box plot pipeline**

- (a) Mean of 3, 6, 7, 11, 11, 12?
- (b) Range of 3, 6, 7, 11, 11, 12?
- (c) Graph lab for «Cumulative frequency ↔ box plot pipeline»: plot, mark key points, state the trend.

#### **T.III.9.026 Experimental probability long-run estimate**

- (a) Mean of 3, 4, 6, 7, 10, 15?
- (b) Range of 3, 4, 6, 7, 10, 15?
- (c) Mean of five numbers is 14. Four are 3, 4, 6, 7. Find the fifth.

#### **T.III.9.027 Questionnaire bias and leading questions**

- (a) Mean of 3, 10, 11, 13, 17, 17?
- (b) Range of 3, 10, 11, 13, 17, 17?
- (c) Mean of five numbers is 8. Four are 3, 10, 11, 13. Find the fifth.

#### **T.III.9.028 Frequency polygon from histogram**

- (a) Class width 10, frequency 24. Frequency density?
- (b) A histogram bar has width 10 and FD 2.4. Frequency?
- (c) Graph lab for «Frequency polygon from histogram»: plot, mark key points, state the trend.

#### **T.III.9.029 Pie chart angle calculation exam**

- (a) Mean of 6, 7, 7, 11, 13, 18?
- (b) Range of 6, 7, 7, 11, 13, 18?
- (c) Mean of five numbers is 9. Four are 6, 7, 7, 11. Find the fifth.

#### **T.III.9.030 Combine probabilities mutually exclusive lists**

- (a) Mean of 3, 6, 7, 14, 17, 18?
- (b) Range of 3, 6, 7, 14, 17, 18?
- (c) Mean of five numbers is 11. Four are 3, 6, 7, 14. Find the fifth.

#### **T.III.9.031 Sampling variability dotplot intuition**

- (a) Mean of 7, 7, 16, 16, 17, 17?
- (b) Range of 7, 7, 16, 16, 17, 17?
- (c) Mean of five numbers is 12. Four are 7, 7, 16, 16. Find the fifth.

#### **T.III.9.032 Venn probability exam set 1**

- (a) Bag: 7 red, 2 blue. P(red)?
- (b) Without replacement, P(both red)?

(c) With replacement,  $P(\text{red then blue})$ ?

**T.III.9.033 Venn probability exam set 2**

(a) Bag: 3 red, 2 blue.  $P(\text{red})$ ?

(b) Without replacement,  $P(\text{both red})$ ?

(c) With replacement,  $P(\text{red then blue})$ ?

**T.III.9.034 Venn probability exam set 3**

(a) Bag: 8 red, 2 blue.  $P(\text{red})$ ?

(b) Without replacement,  $P(\text{both red})$ ?

(c) With replacement,  $P(\text{red then blue})$ ?

**T.III.9.035 Histogram unequal width frequency density set**

(a) Class width 20, frequency 22. Frequency density?

(b) A histogram bar has width 20 and FD 1.1. Frequency?

(c) Why must unequal-width histograms use frequency density, not raw frequency for bar height?

**T.III.1.021 Bounds of a calculated perimeter/area**

(a) 60 cm to nearest 10 cm. Write the error interval for length L.

(b)  $w=4.5$  cm,  $l=9.6$  cm each to nearest 0.1 cm. Upper bound of perimeter?

(c) Maximise speed = distance/time given rounded measurements. Which bounds?

**T.III.3.022 Reverse percentage after successive changes**

(a) Increase £200 by 15%.

(b) After a 15% increase a price is £229. Find the original.

(c) A quantity rises by 15% then by another 15%. Overall % increase?

# Era IV — Solve

Charcoal algebra folio

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## T.IV.5.001 Surds GCSE Higher full toolkit pack

- (a) Simplify  $\sqrt{8}$ .
- (b) Rationalise  $1/\sqrt{2}$ .
- (c) Expand and simplify  $(\sqrt{2}+2)(\sqrt{2}-2)$ .

## T.IV.5.002 Rationalise two-term denominators under time

- (a) Simplify  $\sqrt{20}$ .
- (b) Rationalise  $1/\sqrt{2}$ .
- (c) Expand and simplify  $(\sqrt{2}+2)(\sqrt{2}-2)$ .

## T.IV.5.003 Surds in exact geometric answers

- (a) Simplify  $\sqrt{27}$ .
- (b) Rationalise  $1/\sqrt{3}$ .
- (c) Expand and simplify  $(\sqrt{3}+2)(\sqrt{3}-2)$ .

## T.IV.5.004 Fractional indices $\leftrightarrow$ radical form conversions

- (a) Simplify  $\sqrt{12}$ .
- (b) Rationalise  $1/\sqrt{3}$ .
- (c) Expand and simplify  $(\sqrt{3}+2)(\sqrt{3}-2)$ .

## T.IV.5.005 Solve equations with fractional indices

- (a) Simplify  $\sqrt[3]{72}$ .
- (b) Rationalise  $1/\sqrt{2}$ .
- (c) Expand and simplify  $(\sqrt{2}+2)(\sqrt{2}-2)$ .

## T.IV.5.006 Nested radicals simplify (strange)

- (a) Simplify  $\sqrt{20}$ .
- (b) Rationalise  $1/\sqrt{5}$ .
- (c) Expand and simplify  $(\sqrt{5}+2)(\sqrt{5}-2)$ .

## T.IV.5.007 Simplify expressions with mixed surds and indices

- (a) Simplify  $\sqrt{50}$ .
- (b) Rationalise  $1/\sqrt{7}$ .
- (c) Expand and simplify  $(\sqrt{7}+2)(\sqrt{7}-2)$ .

## T.IV.5.008 Rationalise with conjugate under radicals

- (a) Simplify  $\sqrt{20}$ .
- (b) Rationalise  $1/\sqrt{7}$ .
- (c) Expand and simplify  $(\sqrt{7}+2)(\sqrt{7}-2)$ .

## T.IV.5.009 Surds non-calc paper segment 1

- (a) Simplify  $\sqrt{20}$ .
- (b) Rationalise  $1/\sqrt{7}$ .
- (c) Expand and simplify  $(\sqrt{7}+2)(\sqrt{7}-2)$ .

## T.IV.5.010 Surds non-calc paper segment 2

- (a) Simplify  $\sqrt{32}$ .
- (b) Rationalise  $1/\sqrt{7}$ .
- (c) Expand and simplify  $(\sqrt{7}+2)(\sqrt{7}-2)$ .

## T.IV.5.011 Surds non-calc paper segment 3

- (a) Simplify  $\sqrt{18}$ .
- (b) Rationalise  $1/\sqrt{3}$ .
- (c) Expand and simplify  $(\sqrt{3}+2)(\sqrt{3}-2)$ .

## T.IV.7.001 Completing the square exam method pack

- (a) Factorise  $x^2 - 8x + 15$ .

(b) Solve  $x^2 - 8x + 15 = 0$ .

(c) Complete the square:  $x^2 - 10x + 7 = (x-p)^2 + q$ . Find p,q.

**T.IV.7.002 Turning points via completing the square**

(a) Factorise  $x^2 - 11x + 28$ .

(b) Solve  $x^2 - 11x + 28 = 0$ .

(c) Complete the square:  $x^2 - 8x + 7 = (x-p)^2 + q$ . Find p,q.

**T.IV.7.003 Sketch quadratics from completed square**

(a) Factorise  $x^2 - 10x + 24$ .

(b) Solve  $x^2 - 10x + 24 = 0$ .

(c) Desmos lab for «Sketch quadratics from completed square»: vary one parameter, capture before/after, write one observation.

**T.IV.7.004 Quadratic formula exact surd answers**

(a) Simplify  $\sqrt{32}$ .

(b) Rationalise  $1/\sqrt{5}$ .

(c) Expand and simplify  $(\sqrt{5}+2)(\sqrt{5}-2)$ .

**T.IV.7.005 Discriminant: number of roots exam stems**

(a) Factorise  $x^2 - 8x + 15$ .

(b) Solve  $x^2 - 8x + 15 = 0$ .

(c) Complete the square:  $x^2 - 6x + 7 = (x-p)^2 + q$ . Find p,q.

**T.IV.7.006 Form quadratic equations from roots/context**

(a) Factorise  $x^2 - 6x + 8$ .

(b) Solve  $x^2 - 6x + 8 = 0$ .

(c) Complete the square:  $x^2 - 4x + 7 = (x-p)^2 + q$ . Find p,q.

**T.IV.7.007 Quadratic simultaneous with linear**

(a) Factorise  $x^2 - 12x + 35$ .

(b) Solve  $x^2 - 12x + 35 = 0$ .

(c) Complete the square:  $x^2 - 10x + 7 = (x-p)^2 + q$ . Find p,q.

**T.IV.7.008 Iteration: rearrange to  $x=g(x)$  form**

(a) Factorise  $x^2 - 7x + 6$ .

(b) Solve  $x^2 - 7x + 6 = 0$ .

(c) Complete the square:  $x^2 - 2x + 7 = (x-p)^2 + q$ . Find p,q.

**T.IV.7.009 Iteration: show a root lies in an interval**

(a) Factorise  $x^2 - 5x + 4$ .

(b) Solve  $x^2 - 5x + 4 = 0$ .

(c) Complete the square:  $x^2 - 2x + 7 = (x-p)^2 + q$ . Find p,q.

**T.IV.7.010 Iteration: perform and interpret convergence**

(a) Factorise  $x^2 - 3x + 2$ .

(b) Solve  $x^2 - 3x + 2 = 0$ .

(c) GDC lab for «Iteration: perform and interpret convergence»: record menu path and numerical output.

**T.IV.7.011 Quadratic inequalities critical values method**

(a) Factorise  $x^2 - 10x + 24$ .

(b) Solve  $x^2 - 10x + 24 = 0$ .

(c) Complete the square:  $x^2 - 8x + 7 = (x-p)^2 + q$ . Find p,q.

**T.IV.7.012 Projectile worded GCSE/IB pack**

(a) Factorise  $x^2 - 5x + 6$ .

(b) Solve  $x^2 - 5x + 6 = 0$ .

(c) Complete the square:  $x^2 - 4x + 7 = (x-p)^2 + q$ . Find p,q.

**T.IV.7.013 Hidden quadratics (exp/trig substitutions)**

(a) Factorise  $x^2 - 3x + 2$ .

(b) Solve  $x^2 - 3x + 2 = 0$ .

(c) Complete the square:  $x^2 - 2x + 7 = (x-p)^2 + q$ . Find p,q.

#### **T.IV.7.014 Quadratic model fitting with GDC**

(a) Factorise  $x^2 - 8x + 7$ .

(b) Solve  $x^2 - 8x + 7 = 0$ .

(c) GDC lab for «Quadratic model fitting with GDC»: record menu path and numerical output.

#### **T.IV.7.015 Sign-change decimal search before iteration**

(a) Factorise  $x^2 - 9x + 18$ .

(b) Solve  $x^2 - 9x + 18 = 0$ .

(c) Complete the square:  $x^2 - 6x + 7 = (x-p)^2 + q$ . Find p,q.

#### **T.IV.7.016 Intersecting chord/quadratic geometry link**

(a) Factorise  $x^2 - 10x + 24$ .

(b) Solve  $x^2 - 10x + 24 = 0$ .

(c) Geometry construction lab for «Intersecting chord/quadratic geometry link»: build the figure, measure, verify the claimed relation.

#### **T.IV.7.017 Completing square exam battery 1**

(a) Factorise  $x^2 - 4x + 3$ .

(b) Solve  $x^2 - 4x + 3 = 0$ .

(c) Complete the square:  $x^2 - 2x + 7 = (x-p)^2 + q$ . Find p,q.

#### **T.IV.7.018 Completing square exam battery 2**

(a) Factorise  $x^2 - 6x + 5$ .

(b) Solve  $x^2 - 6x + 5 = 0$ .

(c) Complete the square:  $x^2 - 2x + 7 = (x-p)^2 + q$ . Find p,q.

#### **T.IV.7.019 Completing square exam battery 3**

(a) Factorise  $x^2 - 9x + 14$ .

(b) Solve  $x^2 - 9x + 14 = 0$ .

(c) Complete the square:  $x^2 - 4x + 7 = (x-p)^2 + q$ . Find p,q.

#### **T.IV.7.020 Completing square exam battery 4**

(a) Factorise  $x^2 - 7x + 10$ .

(b) Solve  $x^2 - 7x + 10 = 0$ .

(c) Complete the square:  $x^2 - 10x + 7 = (x-p)^2 + q$ . Find p,q.

#### **T.IV.7.021 Iteration numerical table 1**

(a) Factorise  $x^2 - 9x + 20$ .

(b) Solve  $x^2 - 9x + 20 = 0$ .

(c) GDC lab for «Iteration numerical table 1»: record menu path and numerical output.

#### **T.IV.7.022 Iteration numerical table 2**

(a) Factorise  $x^2 - 11x + 28$ .

(b) Solve  $x^2 - 11x + 28 = 0$ .

(c) GDC lab for «Iteration numerical table 2»: record menu path and numerical output.

#### **T.IV.7.023 Iteration numerical table 3**

(a) Factorise  $x^2 - 7x + 10$ .

(b) Solve  $x^2 - 7x + 10 = 0$ .

(c) GDC lab for «Iteration numerical table 3»: record menu path and numerical output.

#### **T.IV.1.001 Inequalities regions on graphs (linear)**

(a) Solve  $7x + 7 = 56$ .

(b) Solve  $7(x - 7) = 7$ .

(c) Desmos lab for «Inequalities regions on graphs (linear)»: vary one parameter, capture before/after, write one observation.

#### **T.IV.1.002 Set notation for solution sets (UK)**

(a) Solve  $7x + 7 = 28$ .

(b) Solve  $7(x - 7) = -21$ .

(c) List integers n satisfying  $3 < n \leq 7$ .

**T.IV.1.003 Exact vs approximate solutions language**

- (a) Simplify  $\sqrt{12}$ .
- (b) Rationalise  $1/\sqrt{3}$ .
- (c) Expand and simplify  $(\sqrt{3}+2)(\sqrt{3}-2)$ .

**T.IV.2.001 Function notation  $f(x)$  exam fluency**

- (a) Expand  $6(x+6)$ .
- (b) Expand  $(x+1)(x+6)$ .
- (c) Factorise  $x^2 + 7x + 6$ .

**T.IV.2.002 Domain and range from graphs and equations**

- (a) Expand  $7(x+3)$ .
- (b) Expand  $(x+3)(x+6)$ .
- (c) Desmos lab for «Domain and range from graphs and equations»: vary one parameter, capture before/after, write one observation.

**T.IV.2.003 Composite functions fog and gof**

- (a) Expand  $7(x+4)$ .
- (b) Expand  $(x+5)(x+1)$ .
- (c) Factorise  $x^2 + 6x + 5$ .

**T.IV.2.004 Inverse functions algebraically**

- (a) Expand  $2(x+5)$ .
- (b) Expand  $(x+1)(x+1)$ .
- (c) Factorise  $x^2 + 2x + 1$ .

**T.IV.2.005 Piecewise functions interpret and graph**

- (a) Expand  $4(x+2)$ .
- (b) Expand  $(x+2)(x+4)$ .
- (c) Desmos lab for «Piecewise functions interpret and graph»: vary one parameter, capture before/after, write one observation.

**T.IV.2.006 Transformations of graphs  $y=f(x)$  exam pack**

- (a) Expand  $6(x+1)$ .
- (b) Expand  $(x+5)(x+6)$ .
- (c) Desmos lab for «Transformations of graphs  $y=f(x)$  exam pack»: vary one parameter, capture before/after, write one observation.

**T.IV.2.007 Modulus function graphs and equations**

- (a) Expand  $4(x+5)$ .
- (b) Expand  $(x+1)(x+3)$ .
- (c) Desmos lab for «Modulus function graphs and equations»: vary one parameter, capture before/after, write one observation.

**T.IV.2.008 Vertical line test and function definition exam**

- (a) Expand  $5(x+5)$ .
- (b) Expand  $(x+2)(x+4)$ .
- (c) Desmos lab for «Vertical line test and function definition exam»: vary one parameter, capture before/after, write one observation.

**T.IV.2.009 1–1 and onto language for inverses**

- (a) Expand  $6(x+1)$ .
- (b) Expand  $(x+4)(x+3)$ .
- (c) Factorise  $x^2 + 7x + 12$ .

**T.IV.2.010 Parametric definition of functions intro**

- (a) Expand  $5(x+1)$ .
- (b) Expand  $(x+5)(x+2)$ .
- (c) Desmos lab for «Parametric definition of functions intro»: vary one parameter, capture before/after, write one observation.

**T.IV.2.011 Self-inverse functions recognition**

- (a) Expand  $4(x+1)$ .
- (b) Expand  $(x+3)(x+3)$ .
- (c) Factorise  $x^2 + 6x + 9$ .

**T.IV.3.001 Perpendicular bisector equation**

- (a) [Perpendicular bisector equation] Which view is the plan — front, side, or top?
- (b) Cuboid  $9 \times 3 \times 2$ : how many rectangles in a correct net?
- (c) Describe the front elevation of a square-based pyramid (base 9) standing on its base.

#### **T.IV.3.002 Distance from point to line formula use**

- (a) [Distance from point to line formula use] Which view is the plan — front, side, or top?
- (b) Cuboid  $9 \times 3 \times 5$ : how many rectangles in a correct net?
- (c) Describe the front elevation of a square-based pyramid (base 9) standing on its base.

#### **T.IV.3.003 Equation of circle centre origin + tangent (GCSE H)**

- (a) [Equation of circle centre origin + tangent (GCSE H)] Which view is the plan — front, side, or top?
- (b) Cuboid  $4 \times 6 \times 3$ : how many rectangles in a correct net?
- (c) Geometry construction lab for «Equation of circle centre origin + tangent (GCSE H)»: build the figure, measure, verify the claimed relation.

#### **T.IV.3.004 Equation of circle general; find centre/radius**

- (a) [Equation of circle general; find centre/radius] Which view is the plan — front, side, or top?
- (b) Cuboid  $3 \times 3 \times 3$ : how many rectangles in a correct net?
- (c) Describe the front elevation of a square-based pyramid (base 3) standing on its base.

#### **T.IV.4.001 Simultaneous three equations intro**

- (a)  $\det([[4,3],[2,5]])$ ?
- (b)  $[[4,3],[2,5]] \times \mathbf{I}$ ?
- (c) If  $\det \neq 0$ , inverse of  $[[4,3],[2,5]]$ ?

#### **T.IV.4.002 Non-linear simultaneous exam pack**

- (a)  $\det([[1,1],[4,3]])$ ?
- (b)  $[[1,1],[4,3]] \times \mathbf{I}$ ?
- (c) If  $\det \neq 0$ , inverse of  $[[1,1],[4,3]]$ ?

#### **T.IV.4.003 Matrices $2 \times 2$ multiply and inverse (GCSE+)**

- (a)  $\det([[3,1],[5,3]])$ ?
- (b)  $[[3,1],[5,3]] \times \mathbf{I}$ ?
- (c) If  $\det \neq 0$ , inverse of  $[[3,1],[5,3]]$ ?

#### **T.IV.4.004 Matrix transformations of the plane intro**

- (a)  $\det([[3,1],[1,2]])$ ?
- (b)  $[[3,1],[1,2]] \times \mathbf{I}$ ?
- (c) Geometry construction lab for «Matrix transformations of the plane intro»: build the figure, measure, verify the claimed relation.

#### **T.IV.4.005 Graphical linear programming light (regions)**

- (a)  $\det([[1,4],[1,3]])$ ?
- (b)  $[[1,4],[1,3]] \times \mathbf{I}$ ?
- (c) Desmos lab for «Graphical linear programming light (regions)»: vary one parameter, capture before/after, write one observation.

#### **T.IV.6.001 Factor theorem exam use**

- (a) Expand  $7(x+3)$ .
- (b) Expand  $(x+1)(x+1)$ .
- (c) Factorise  $x^2 + 2x + 1$ .

#### **T.IV.6.002 Remainder theorem exam use**

- (a) Expand  $2(x+4)$ .
- (b) Expand  $(x+5)(x+1)$ .
- (c) Factorise  $x^2 + 6x + 5$ .

#### **T.IV.6.003 Algebraic division of cubics**

- (a) Expand  $7(x+2)$ .
- (b) Expand  $(x+1)(x+4)$ .
- (c) Factorise  $x^2 + 5x + 4$ .

#### **T.IV.6.004 Sketch cubic and quartic from factorised form**

- (a) Expand  $3(x+3)$ .

(b) Expand  $(x+5)(x+4)$ .

(c) Desmos lab for «Sketch cubic and quartic from factorised form»: vary one parameter, capture before/after, write one observation.

#### **T.IV.6.005 Expand and factorise mixed higher exam set**

(a) Expand  $7(x+1)$ .

(b) Expand  $(x+5)(x+4)$ .

(c) Factorise  $x^2 + 9x + 20$ .

#### **T.IV.6.006 Binomial expansion positive integer n**

(a) Expand  $3(x+5)$ .

(b) Expand  $(x+4)(x+1)$ .

(c) Factorise  $x^2 + 5x + 4$ .

#### **T.IV.6.007 Binomial approximation $(1+x)^n$ for small x**

(a) Expand  $4(x+4)$ .

(b) Expand  $(x+4)(x+4)$ .

(c) Factorise  $x^2 + 8x + 16$ .

#### **T.IV.6.008 Sketch from leading term and roots only**

(a) Expand  $5(x+4)$ .

(b) Expand  $(x+3)(x+4)$ .

(c) Desmos lab for «Sketch from leading term and roots only»: vary one parameter, capture before/after, write one observation.

#### **T.IV.8.001 Argand diagram exam plotting**

(a) Expand  $3(x+6)$ .

(b) Expand  $(x+1)(x+4)$ .

(c) Geometry construction lab for «Argand diagram exam plotting»: build the figure, measure, verify the claimed relation.

#### **T.IV.8.002 Modulus-argument form conversions**

(a) Expand  $4(x+6)$ .

(b) Expand  $(x+1)(x+5)$ .

(c) Factorise  $x^2 + 6x + 5$ .

#### **T.IV.8.003 De Moivre exam applications**

(a) Expand  $5(x+4)$ .

(b) Expand  $(x+5)(x+4)$ .

(c) Factorise  $x^2 + 9x + 20$ .

#### **T.IV.8.004 Complex loci $|z-a|=r$ and $\arg(z-a)=\theta$**

(a) Expand  $7(x+4)$ .

(b) Expand  $(x+3)(x+1)$ .

(c) Factorise  $x^2 + 4x + 3$ .

#### **T.IV.8.005 Solve equations in complex numbers**

(a) Expand  $6(x+6)$ .

(b) Expand  $(x+2)(x+3)$ .

(c) Factorise  $x^2 + 5x + 6$ .

#### **T.IV.9.001 Even/odd functions recognition**

(a) Expand  $2(x+4)$ .

(b) Expand  $(x+4)(x+6)$ .

(c) Factorise  $x^2 + 10x + 24$ .

#### **T.IV.9.002 Absolute value inequalities**

(a) Expand  $4(x+4)$ .

(b) Expand  $(x+5)(x+2)$ .

(c) Factorise  $x^2 + 7x + 10$ .

#### **T.IV.9.003 Asymptotes of rational functions exam**

(a) Expand  $5(x+2)$ .

(b) Expand  $(x+4)(x+2)$ .

(c) Desmos lab for «Asymptotes of rational functions exam»: vary one parameter, capture before/after, write one observation.

#### **T.IV.9.004 Combined transformations order matters**

(a) Expand  $2(x+5)$ .

(b) Expand  $(x+2)(x+4)$ .

(c) Desmos lab for «Combined transformations order matters»: vary one parameter, capture before/after, write one observation.

#### **T.IV.9.005 Step functions and floor in modelling (light)**

(a) Expand  $3(x+3)$ .

(b) Expand  $(x+2)(x+5)$ .

(c) Desmos lab for «Step functions and floor in modelling (light)»: vary one parameter, capture before/after, write one observation.

#### **T.IV.9.006 Oblique asymptotes by division**

(a) Expand  $6(x+2)$ .

(b) Expand  $(x+3)(x+4)$ .

(c) Desmos lab for «Oblique asymptotes by division»: vary one parameter, capture before/after, write one observation.

#### **T.IV.10.001 Polynomial inequalities on number line**

(a) Solve  $4x + 7 = 43$ .

(b) Solve  $4(x - 7) = 12$ .

(c) List integers  $n$  satisfying  $9 < n \leq 13$ .

#### **T.IV.11.001 Solve rational equations check extraneous**

(a) Solve  $8x + 7 = 31$ .

(b) Solve  $8(x - 7) = -24$ .

(c) List integers  $n$  satisfying  $3 < n \leq 7$ .

#### **T.IV.11.002 Radical equations isolate and check**

(a) Simplify  $\sqrt{50}$ .

(b) Rationalise  $1/\sqrt{2}$ .

(c) Expand and simplify  $(\sqrt{2}+2)(\sqrt{2}-2)$ .

#### **T.IV.11.003 Graphical solution of radical/rational eqs**

(a) Simplify  $\sqrt{75}$ .

(b) Rationalise  $1/\sqrt{3}$ .

(c) Desmos lab for «Graphical solution of radical/rational eqs»: vary one parameter, capture before/after, write one observation.

#### **T.IV.12.001 Compound interest continuous ( $A=Pe^{rt}$ )**

(a) Increase £120 by 5%.

(b) After a 5% increase a price is £126. Find the original.

(c) A quantity rises by 5% then by another 5%. Overall % increase?

#### **T.IV.12.002 GCSE growth and decay worded pack**

(a) Increase £120 by 10%.

(b) After a 10% increase a price is £132. Find the original.

(c) A quantity rises by 10% then by another 10%. Overall % increase?

#### **T.IV.12.003 IB GDC exponential regression workflow**

(a) Increase £120 by 12%.

(b) After a 12% increase a price is £134. Find the original.

(c) GDC lab for «IB GDC exponential regression workflow»: record menu path and numerical output.

#### **T.IV.12.004 Log scales interpretation**

(a) Increase £120 by 10%.

(b) After a 10% increase a price is £132. Find the original.

(c) Graph lab for «Log scales interpretation»: plot, mark key points, state the trend.

#### **T.IV.12.005 Solve exp/log equations exam mixed pack**

(a) Increase £250 by 20%.

(b) After a 20% increase a price is £300. Find the original.

(c) A quantity rises by 20% then by another 20%. Overall % increase?

**T.IV.12.006 Change of base in exam calculations**

- (a) Increase £250 by 15%.
- (b) After a 15% increase a price is £287. Find the original.
- (c) GDC lab for «Change of base in exam calculations»: record menu path and numerical output.

**T.IV.12.007 Half-life multi-step worded**

- (a) Increase £200 by 25%.
- (b) After a 25% increase a price is £250. Find the original.
- (c) A quantity rises by 25% then by another 25%. Overall % increase?

**T.IV.12.008 Modelling cycle: formulate–solve–interpret–evaluate**

- (a) Increase £80 by 12%.
- (b) After a 12% increase a price is £89. Find the original.
- (c) A quantity rises by 12% then by another 12%. Overall % increase?

**T.IV.12.009 Compare linear vs exponential models**

- (a) Increase £250 by 20%.
- (b) After a 20% increase a price is £300. Find the original.
- (c) GDC lab for «Compare linear vs exponential models»: record menu path and numerical output.

**T.IV.12.010 Logarithmic linearisation of data**

- (a) Increase £120 by 15%.
- (b) After a 15% increase a price is £138. Find the original.
- (c) GDC lab for «Logarithmic linearisation of data»: record menu path and numerical output.

**T.IV.12.011 Newton's law of cooling discrete steps**

- (a) Increase £400 by 5%.
- (b) After a 5% increase a price is £420. Find the original.
- (c) A quantity rises by 5% then by another 5%. Overall % increase?

**T.IV.12.012 Effective annual rate vs nominal**

- (a) Increase £250 by 20%.
- (b) After a 20% increase a price is £300. Find the original.
- (c) A quantity rises by 20% then by another 20%. Overall % increase?

**T.IV.12.013 Exp/log modelling context 1**

- (a) Increase £400 by 12%.
- (b) After a 12% increase a price is £448. Find the original.
- (c) GDC lab for «Exp/log modelling context 1»: record menu path and numerical output.

**T.IV.12.014 Exp/log modelling context 2**

- (a) Increase £250 by 5%.
- (b) After a 5% increase a price is £262. Find the original.
- (c) GDC lab for «Exp/log modelling context 2»: record menu path and numerical output.

**T.IV.12.015 Exp/log modelling context 3**

- (a) Increase £250 by 25%.
- (b) After a 25% increase a price is £312. Find the original.
- (c) GDC lab for «Exp/log modelling context 3»: record menu path and numerical output.

**T.IV.13.001 nth term of linear sequences (GCSE)**

- (a) nth term of 7, 13, 19, 25, ...
- (b) 20th term:  $a=7, d=6$ .
- (c)  $S_n$  for  $a=7, d=6$ .

**T.IV.13.002 Quadratic sequences nth term**

- (a) nth term of 2, 4, 6, 8, ...
- (b) 20th term:  $a=2, d=2$ .
- (c)  $S_n$  for  $a=2, d=2$ .

**T.IV.13.003 Geometric sequences GCSE/IB worded**

(a) nth term of 5, 11, 17, 23, ...

(b) 20th term:  $a=5$ ,  $d=6$ .

(c)  $S_n$  for  $a=5$ ,  $d=6$ .

#### T.IV.13.004 Arithmetic series exam applications

(a) nth term of 9, 15, 21, 27, ...

(b) 20th term:  $a=9$ ,  $d=6$ .

(c)  $S_n$  for  $a=9$ ,  $d=6$ .

#### T.IV.13.005 Infinite geometric series $|r|<1$ condition

(a) nth term of 4, 10, 16, 22, ...

(b) 20th term:  $a=4$ ,  $d=6$ .

(c)  $S_n$  for  $a=4$ ,  $d=6$ .

#### T.IV.13.006 Sigma notation exam fluency

(a) nth term of 9, 14, 19, 24, ...

(b) 20th term:  $a=9$ ,  $d=5$ .

(c)  $S_n$  for  $a=9$ ,  $d=5$ .

#### T.IV.13.007 Recurrence relations $x_n=ax_{n-1}+b$

(a) nth term of 7, 12, 17, 22, ...

(b) 20th term:  $a=7$ ,  $d=5$ .

(c)  $S_n$  for  $a=7$ ,  $d=5$ .

#### T.IV.13.008 Fibonacci-type sequences properties

(a) nth term of 6, 12, 18, 24, ...

(b) 20th term:  $a=6$ ,  $d=6$ .

(c)  $S_n$  for  $a=6$ ,  $d=6$ .

#### T.IV.13.009 Sequences modelling IB AI contexts

(a) nth term of 7, 11, 15, 19, ...

(b) 20th term:  $a=7$ ,  $d=4$ .

(c) GDC lab for «Sequences modelling IB AI contexts»: record menu path and numerical output.

#### T.IV.13.010 Prove a sequence is arithmetic/geometric

(a) nth term of 4, 11, 18, 25, ...

(b) 20th term:  $a=4$ ,  $d=7$ .

(c)  $S_n$  for  $a=4$ ,  $d=7$ .

#### T.IV.13.011 Financial series: annuities simple

(a) nth term of 5, 7, 9, 11, ...

(b) 20th term:  $a=5$ ,  $d=2$ .

(c)  $S_n$  for  $a=5$ ,  $d=2$ .

#### T.IV.13.012 IB AA HL induction on sequences preview

(a) List the three structural steps of mathematical induction.

(b) Inductive step only: assume  $1+\dots+k=k(k+1)/2$ ; prove for  $k+1$ .

(c) Name one write-up error that loses IB AA HL induction marks.

#### T.IV.13.013 Compare arithmetic vs geometric growth stories

(a) nth term of 6, 13, 20, 27, ...

(b) 20th term:  $a=6$ ,  $d=7$ .

(c)  $S_n$  for  $a=6$ ,  $d=7$ .

#### T.IV.13.014 Sequences GCSE mixed paper 1

(a) nth term of 2, 4, 6, 8, ...

(b) 20th term:  $a=2$ ,  $d=2$ .

(c)  $S_n$  for  $a=2$ ,  $d=2$ .

#### T.IV.13.015 Sequences GCSE mixed paper 2

(a) nth term of 3, 5, 7, 9, ...

(b) 20th term:  $a=3$ ,  $d=2$ .

(c)  $S_n$  for  $a=3$ ,  $d=2$ .

#### **T.IV.13.016 Sequences GCSE mixed paper 3**

(a) nth term of 2, 6, 10, 14, ...

(b) 20th term:  $a=2$ ,  $d=4$ .

(c)  $S_n$  for  $a=2$ ,  $d=4$ .

#### **T.IV.13.017 Sequences GCSE mixed paper 4**

(a) nth term of 3, 10, 17, 24, ...

(b) 20th term:  $a=3$ ,  $d=7$ .

(c)  $S_n$  for  $a=3$ ,  $d=7$ .

#### **T.IV.14.001 Product rule and permutations exam pack**

(a) Bag: 5 red, 2 blue.  $P(\text{red})$ ?

(b) Without replacement,  $P(\text{both red})$ ?

(c) With replacement,  $P(\text{red then blue})$ ?

#### **T.IV.14.002 Combinations in probability contexts**

(a) Bag: 5 red, 4 blue.  $P(\text{red})$ ?

(b) Without replacement,  $P(\text{both red})$ ?

(c) With replacement,  $P(\text{red then blue})$ ?

#### **T.IV.14.003 Conditional probability tree diagrams (GCSE H)**

(a) Bag: 4 red, 5 blue.  $P(\text{red})$ ?

(b) Without replacement,  $P(\text{both red})$ ?

(c) With replacement,  $P(\text{red then blue})$ ?

#### **T.IV.14.004 Independent vs mutually exclusive exam traps**

(a) Bag: 7 red, 6 blue.  $P(\text{red})$ ?

(b) Without replacement,  $P(\text{both red})$ ?

(c) With replacement,  $P(\text{red then blue})$ ?

#### **T.IV.14.005 Venn $\leftrightarrow$ tree $\leftrightarrow$ two-way translation**

(a) Bag: 6 red, 2 blue.  $P(\text{red})$ ?

(b) Without replacement,  $P(\text{both red})$ ?

(c) With replacement,  $P(\text{red then blue})$ ?

#### **T.IV.14.006 Expected value decision problems (S-MD)**

(a) Bag: 7 red, 4 blue.  $P(\text{red})$ ?

(b) Without replacement,  $P(\text{both red})$ ?

(c) With replacement,  $P(\text{red then blue})$ ?

#### **T.IV.14.007 Bayes with medical/test contexts**

(a) Bag: 8 red, 4 blue.  $P(\text{red})$ ?

(b) Without replacement,  $P(\text{both red})$ ?

(c) With replacement,  $P(\text{red then blue})$ ?

#### **T.IV.14.008 Inclusive exclusive counting with Venn**

(a) Bag: 4 red, 3 blue.  $P(\text{red})$ ?

(b) Without replacement,  $P(\text{both red})$ ?

(c) With replacement,  $P(\text{red then blue})$ ?

#### **T.IV.14.009 Conditional probability given complementary info**

(a) Bag: 7 red, 6 blue.  $P(\text{red})$ ?

(b) Without replacement,  $P(\text{both red})$ ?

(c) With replacement,  $P(\text{red then blue})$ ?

#### **T.IV.14.010 At least one complement strategy pack**

(a) Bag: 4 red, 4 blue.  $P(\text{red})$ ?

(b) Without replacement,  $P(\text{both red})$ ?

(c) With replacement,  $P(\text{red then blue})$ ?

**T.IV.14.011 Law of large numbers simulation demo**

(a) Bag: 8 red, 6 blue.  $P(\text{red})$ ?

(b) Without replacement,  $P(\text{both red})$ ?

(c) With replacement,  $P(\text{red then blue})$ ?

**T.IV.14.012 Rare event rule intuition**

(a) Bag: 8 red, 2 blue.  $P(\text{red})$ ?

(b) Without replacement,  $P(\text{both red})$ ?

(c) With replacement,  $P(\text{red then blue})$ ?

**T.IV.14.013 Expected value of games fair/unfair**

(a) Bag: 7 red, 7 blue.  $P(\text{red})$ ?

(b) Without replacement,  $P(\text{both red})$ ?

(c) With replacement,  $P(\text{red then blue})$ ?

**T.IV.14.014 Conditional probability medical screening FRQ**

(a) AP/IB style: Which theorem or definition is central to «Conditional probability medical screening FRQ»?

(b) Write a justification sentence frame for «Conditional probability medical screening FRQ»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.

(c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Conditional probability medical screening FRQ», then interpret.

**T.IV.14.015 mutually exclusive addition rule drills**

(a) Bag: 7 red, 4 blue.  $P(\text{red})$ ?

(b) Without replacement,  $P(\text{both red})$ ?

(c) With replacement,  $P(\text{red then blue})$ ?

**T.IV.14.016 Tree diagrams with three stages**

(a) Bag: 8 red, 5 blue.  $P(\text{red})$ ?

(b) Without replacement,  $P(\text{both red})$ ?

(c) With replacement,  $P(\text{red then blue})$ ?

**T.IV.14.017 Conditional probability FRQ-style 1**

(a) AP/IB style: Which theorem or definition is central to «Conditional probability FRQ-style 1»?

(b) Write a justification sentence frame for «Conditional probability FRQ-style 1»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.

(c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Conditional probability FRQ-style 1», then interpret.

**T.IV.14.018 Conditional probability FRQ-style 2**

(a) AP/IB style: Which theorem or definition is central to «Conditional probability FRQ-style 2»?

(b) Write a justification sentence frame for «Conditional probability FRQ-style 2»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.

(c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Conditional probability FRQ-style 2», then interpret.

**T.IV.14.019 Conditional probability FRQ-style 3**

(a) AP/IB style: Which theorem or definition is central to «Conditional probability FRQ-style 3»?

(b) Write a justification sentence frame for «Conditional probability FRQ-style 3»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.

(c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Conditional probability FRQ-style 3», then interpret.

**T.IV.15.001 Normal distribution empirical rule exam**

(a)  $X \sim N(70, 10^2)$ . Approx  $P(60 < X < 80)$  (empirical rule)?

(b) z-score of  $x=90$  ( $\mu=70$ ,  $\sigma=10$ )?

(c) GDC: outline finding  $P(X > 80)$  for  $X \sim N(70, 10^2)$ .

**T.IV.15.002 z-score calculations and interpretation**

(a)  $X \sim N(100, 15^2)$ . Approx  $P(85 < X < 115)$  (empirical rule)?

(b) z-score of  $x=130$  ( $\mu=100$ ,  $\sigma=15$ )?

(c) GDC: outline finding  $P(X > 115)$  for  $X \sim N(100, 15^2)$ .

**T.IV.15.003 Binomial probability GDC workflow**

(a)  $X \sim N(70, 10^2)$ . Approx  $P(60 < X < 80)$  (empirical rule)?

(b) z-score of  $x=90$  ( $\mu=70$ ,  $\sigma=10$ )?

(c) GDC lab for «Binomial probability GDC workflow»: record menu path and numerical output.

**T.IV.15.004 Scatter plots: describe strength/direction/form**

- (a)  $X \sim N(70, 10^2)$ . Approx  $P(60 < X < 80)$  (empirical rule)?
- (b) z-score of  $x=90$  ( $\mu=70$ ,  $\sigma=10$ )?
- (c) GDC: outline finding  $P(X > 80)$  for  $X \sim N(70, 10^2)$ .

**T.IV.15.005 Correlation vs causation exam stems**

- (a)  $X \sim N(70, 10^2)$ . Approx  $P(60 < X < 80)$  (empirical rule)?
- (b) z-score of  $x=90$  ( $\mu=70$ ,  $\sigma=10$ )?
- (c) GDC: outline finding  $P(X > 80)$  for  $X \sim N(70, 10^2)$ .

**T.IV.15.006 Least-squares residual interpretation**

- (a)  $X \sim N(50, 5^2)$ . Approx  $P(45 < X < 55)$  (empirical rule)?
- (b) z-score of  $x=60$  ( $\mu=50$ ,  $\sigma=5$ )?
- (c) GDC lab for «Least-squares residual interpretation»: record menu path and numerical output.

**T.IV.15.007 Outliers effect on mean/median/regression**

- (a)  $X \sim N(100, 15^2)$ . Approx  $P(85 < X < 115)$  (empirical rule)?
- (b) z-score of  $x=130$  ( $\mu=100$ ,  $\sigma=15$ )?
- (c) GDC: outline finding  $P(X > 115)$  for  $X \sim N(100, 15^2)$ .

**T.IV.15.008 Comparing distributions with multiple measures**

- (a)  $X \sim N(50, 5^2)$ . Approx  $P(45 < X < 55)$  (empirical rule)?
- (b) z-score of  $x=60$  ( $\mu=50$ ,  $\sigma=5$ )?
- (c) GDC: outline finding  $P(X > 55)$  for  $X \sim N(50, 5^2)$ .

**T.IV.15.009 Percentiles and cumulative relative frequency**

- (a)  $X \sim N(100, 15^2)$ . Approx  $P(85 < X < 115)$  (empirical rule)?
- (b) z-score of  $x=130$  ( $\mu=100$ ,  $\sigma=15$ )?
- (c) GDC: outline finding  $P(X > 115)$  for  $X \sim N(100, 15^2)$ .

**T.IV.15.010 Skew interpretation from histograms/boxplots**

- (a)  $X \sim N(50, 5^2)$ . Approx  $P(45 < X < 55)$  (empirical rule)?
- (b) z-score of  $x=60$  ( $\mu=50$ ,  $\sigma=5$ )?
- (c) GDC: outline finding  $P(X > 55)$  for  $X \sim N(50, 5^2)$ .

**T.IV.15.011 Binomial vs geometric settings recognition**

- (a)  $X \sim N(70, 10^2)$ . Approx  $P(60 < X < 80)$  (empirical rule)?
- (b) z-score of  $x=90$  ( $\mu=70$ ,  $\sigma=10$ )?
- (c) GDC: outline finding  $P(X > 80)$  for  $X \sim N(70, 10^2)$ .

**T.IV.15.012 Geometric distribution probability**

- (a)  $X \sim N(100, 15^2)$ . Approx  $P(85 < X < 115)$  (empirical rule)?
- (b) z-score of  $x=130$  ( $\mu=100$ ,  $\sigma=15$ )?
- (c) GDC: outline finding  $P(X > 115)$  for  $X \sim N(100, 15^2)$ .

**T.IV.15.013 Normal probability GDC invNorm/normalcdf**

- (a)  $X \sim N(100, 15^2)$ . Approx  $P(85 < X < 115)$  (empirical rule)?
- (b) z-score of  $x=130$  ( $\mu=100$ ,  $\sigma=15$ )?
- (c) GDC lab for «Normal probability GDC invNorm/normalcdf»: record menu path and numerical output.

**T.IV.15.014 Assessing normality: plots and rules**

- (a)  $X \sim N(50, 5^2)$ . Approx  $P(45 < X < 55)$  (empirical rule)?
- (b) z-score of  $x=60$  ( $\mu=50$ ,  $\sigma=5$ )?
- (c) GDC lab for «Assessing normality: plots and rules»: record menu path and numerical output.

**T.IV.15.015 Transforming data: effect on mean/SD**

- (a)  $X \sim N(70, 10^2)$ . Approx  $P(60 < X < 80)$  (empirical rule)?
- (b) z-score of  $x=90$  ( $\mu=70$ ,  $\sigma=10$ )?
- (c) GDC: outline finding  $P(X > 80)$  for  $X \sim N(70, 10^2)$ .

**T.IV.15.016 Residual plots: pattern means non-linear**

- (a)  $X \sim N(50, 5^2)$ . Approx  $P(45 < X < 55)$  (empirical rule)?
- (b) z-score of  $x=60$  ( $\mu=50$ ,  $\sigma=5$ )?
- (c) GDC lab for «Residual plots: pattern means non-linear»: record menu path and numerical output.

#### **T.IV.15.017 Combining random variables mean/variance**

- (a)  $X \sim N(100, 15^2)$ . Approx  $P(85 < X < 115)$  (empirical rule)?
- (b) z-score of  $x=130$  ( $\mu=100$ ,  $\sigma=15$ )?
- (c) GDC: outline finding  $P(X > 115)$  for  $X \sim N(100, 15^2)$ .

#### **T.IV.15.018 Linear regression prediction vs extrapolation**

- (a)  $X \sim N(50, 5^2)$ . Approx  $P(45 < X < 55)$  (empirical rule)?
- (b) z-score of  $x=60$  ( $\mu=50$ ,  $\sigma=5$ )?
- (c) GDC: outline finding  $P(X > 55)$  for  $X \sim N(50, 5^2)$ .

#### **T.IV.15.019 Discrete RV probability distribution tables**

- (a)  $X \sim N(100, 15^2)$ . Approx  $P(85 < X < 115)$  (empirical rule)?
- (b) z-score of  $x=130$  ( $\mu=100$ ,  $\sigma=15$ )?
- (c) GDC: outline finding  $P(X > 115)$  for  $X \sim N(100, 15^2)$ .

#### **T.IV.15.020 Inverse normal cut-points for percentiles**

- (a)  $X \sim N(50, 5^2)$ . Approx  $P(45 < X < 55)$  (empirical rule)?
- (b) z-score of  $x=60$  ( $\mu=50$ ,  $\sigma=5$ )?
- (c) GDC lab for «Inverse normal cut-points for percentiles»: record menu path and numerical output.

#### **T.IV.15.021 Effect of coding data (subtract/divide)**

- (a)  $X \sim N(100, 15^2)$ . Approx  $P(85 < X < 115)$  (empirical rule)?
- (b) z-score of  $x=130$  ( $\mu=100$ ,  $\sigma=15$ )?
- (c) GDC: outline finding  $P(X > 115)$  for  $X \sim N(100, 15^2)$ .

#### **T.IV.5.012 Surds in exact trig non-calc hybrids**

- (a) Simplify  $\sqrt{8}$ .
- (b) Rationalise  $1/\sqrt{2}$ .
- (c) Expand and simplify  $(\sqrt{2}+2)(\sqrt{2}-2)$ .

#### **T.IV.5.013 Index equations leading to quadratics**

- (a) Simplify  $\sqrt{18}$ .
- (b) Rationalise  $1/\sqrt{2}$ .
- (c) Expand and simplify  $(\sqrt{2}+2)(\sqrt{2}-2)$ .

#### **T.IV.7.024 Completing square with $a \neq 1$ fluency**

- (a) Factorise  $x^2 - 8x + 15$ .
- (b) Solve  $x^2 - 8x + 15 = 0$ .
- (c) Complete the square:  $x^2 - 6x + 7 = (x-p)^2 + q$ . Find p,q.

#### **T.IV.7.025 Iteration: cobweb diagram interpretation**

- (a) Factorise  $x^2 - 9x + 20$ .
- (b) Solve  $x^2 - 9x + 20 = 0$ .
- (c) Desmos lab for «Iteration: cobweb diagram interpretation»: vary one parameter, capture before/after, write one observation.

#### **T.IV.7.026 Quadratic inequalities on number line shading**

- (a) Factorise  $x^2 - 8x + 15$ .
- (b) Solve  $x^2 - 8x + 15 = 0$ .
- (c) Complete the square:  $x^2 - 10x + 7 = (x-p)^2 + q$ . Find p,q.

#### **T.IV.2.012 Restrict domain to make inverse**

- (a) Expand  $3(x+2)$ .
- (b) Expand  $(x+4)(x+5)$ .
- (c) Desmos lab for «Restrict domain to make inverse»: vary one parameter, capture before/after, write one observation.

#### **T.IV.2.013 Function arithmetic (sum product quotient)**

- (a) Expand  $4(x+4)$ .

(b) Expand  $(x+1)(x+5)$ .

(c) Desmos lab for «Function arithmetic (sum product quotient)»: vary one parameter, capture before/after, write one observation.

#### **T.IV.6.009 Factorise cubics fully after one root**

(a) Expand  $5(x+5)$ .

(b) Expand  $(x+1)(x+1)$ .

(c) Factorise  $x^2 + 2x + 1$ .

#### **T.IV.6.010 Polynomial division with remainders interpreted**

(a) Expand  $7(x+2)$ .

(b) Expand  $(x+2)(x+6)$ .

(c) Factorise  $x^2 + 8x + 12$ .

#### **T.IV.12.016 Compound interest vs simple interest compare**

(a) Increase £250 by 15%.

(b) After a 15% increase a price is £287. Find the original.

(c) A quantity rises by 15% then by another 15%. Overall % increase?

#### **T.IV.12.017 Solve for time in exponential models**

(a) Increase £400 by 20%.

(b) After a 20% increase a price is £480. Find the original.

(c) A quantity rises by 20% then by another 20%. Overall % increase?

#### **T.IV.13.018 nth term from practical sequences contexts**

(a) nth term of 5, 11, 17, 23, ...

(b) 20th term:  $a=5$ ,  $d=6$ .

(c)  $S_n$  for  $a=5$ ,  $d=6$ .

#### **T.IV.13.019 Sum of first n naturals / odds identities**

(a) nth term of 6, 12, 18, 24, ...

(b) 20th term:  $a=6$ ,  $d=6$ .

(c)  $S_n$  for  $a=6$ ,  $d=6$ .

#### **T.IV.14.020 Conditional probability given restricted sample space**

(a) Bag: 6 red, 4 blue.  $P(\text{red})$ ?

(b) Without replacement,  $P(\text{both red})$ ?

(c) With replacement,  $P(\text{red then blue})$ ?

#### **T.IV.14.021 Counting with restrictions (no two adjacent)**

(a) Bag: 4 red, 4 blue.  $P(\text{red})$ ?

(b) Without replacement,  $P(\text{both red})$ ?

(c) With replacement,  $P(\text{red then blue})$ ?

#### **T.IV.15.022 Empirical rule vs Chebyshev light contrast**

(a)  $X \sim N(50, 5^2)$ . Approx  $P(45 < X < 55)$  (empirical rule)?

(b) z-score of  $x=60$  ( $\mu=50$ ,  $\sigma=5$ )?

(c) GDC: outline finding  $P(X > 55)$  for  $X \sim N(50, 5^2)$ .

#### **T.IV.15.023 Binomial mean np and SD $\sqrt{npq}$ fluency**

(a)  $X \sim N(70, 10^2)$ . Approx  $P(60 < X < 80)$  (empirical rule)?

(b) z-score of  $x=90$  ( $\mu=70$ ,  $\sigma=10$ )?

(c) GDC: outline finding  $P(X > 80)$  for  $X \sim N(70, 10^2)$ .

#### **T.IV.4.006 Determinant condition for unique solution $2 \times 2$**

(a)  $\det\left(\begin{bmatrix} 4 & 5 \\ 4 & 5 \end{bmatrix}\right)$ ?

(b)  $\begin{bmatrix} 4 & 5 \\ 4 & 5 \end{bmatrix} \times I$ ?

(c) If  $\det \neq 0$ , inverse of  $\begin{bmatrix} 4 & 5 \\ 4 & 5 \end{bmatrix}$ ?

#### **T.IV.8.006 Powers of i cycle fluency**

(a) Expand  $6(x+3)$ .

(b) Expand  $(x+5)(x+6)$ .

(c) Factorise  $x^2 + 11x + 30$ .

**T.IV.9.007 Graph  $y=1/f(x)$  from  $y=f(x)$  features**

(a) Expand  $2(x+4)$ .

(b) Expand  $(x+1)(x+1)$ .

(c) Desmos lab for «Graph  $y=1/f(x)$  from  $y=f(x)$  features»: vary one parameter, capture before/after, write one observation.

**T.IV.3.005 Find intersection of two circles algebraically**

(a) [Find intersection of two circles algebraically] Which view is the plan — front, side, or top?

(b) Cuboid  $5 \times 3 \times 4$ : how many rectangles in a correct net?

(c) Describe the front elevation of a square-based pyramid (base 5) standing on its base.

**T.IV.1.004 Inequalities with fractions clear carefully**

(a) Solve  $2x + 6 = 8$ .

(b) Solve  $2(x - 6) = -8$ .

(c) List integers  $n$  satisfying  $1 < n \leq 5$ .

**T.IV.11.004 Domain of radical and rational combined**

(a) Simplify  $\sqrt{27}$ .

(b) Rationalise  $1/\sqrt{7}$ .

(c) Expand and simplify  $(\sqrt{7}+2)(\sqrt{7}-2)$ .

**T.IV.12.018 Doubling time from continuous and discrete models**

(a) Increase £200 by 5%.

(b) After a 5% increase a price is £210. Find the original.

(c) A quantity rises by 5% then by another 5%. Overall % increase?

**T.IV.13.020 Geometric series in repeating decimals link**

(a)  $n$ th term of 4, 7, 10, 13, ...

(b) 20th term:  $a=4$ ,  $d=3$ .

(c)  $S_n$  for  $a=4$ ,  $d=3$ .

**T.IV.7.027 Worded optimisation without calculus (complete square)**

(a) Factorise  $x^2 - 4x + 3$ .

(b) Solve  $x^2 - 4x + 3 = 0$ .

(c) Complete the square:  $x^2 - 2x + 7 = (x-p)^2 + q$ . Find  $p, q$ .

**T.IV.14.022 Probability with without replacement trees**

(a) Bag: 8 red, 5 blue.  $P(\text{red})$ ?

(b) Without replacement,  $P(\text{both red})$ ?

(c) With replacement,  $P(\text{red then blue})$ ?

**T.IV.15.024 Lurking variables in observational studies**

(a)  $X \sim N(50, 5^2)$ . Approx  $P(45 < X < 55)$  (empirical rule)?

(b)  $z$ -score of  $x=60$  ( $\mu=50$ ,  $\sigma=5$ )?

(c) GDC: outline finding  $P(X > 55)$  for  $X \sim N(50, 5^2)$ .

**T.IV.13.021 GCSE Higher special sequences (triangular)**

(a)  $n$ th term of 6, 9, 12, 15, ...

(b) 20th term:  $a=6$ ,  $d=3$ .

(c)  $S_n$  for  $a=6$ ,  $d=3$ .

# Era V — Prove

Cream construction folio

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## **T.V.1.001 GCSE Higher geometric proof write-ups**

- (a) List the three structural steps of mathematical induction.
- (b) Inductive step only: assume  $1 + \dots + k = k(k+1)/2$ ; prove for  $k+1$ .
- (c) Name one write-up error that loses IB AA HL induction marks.

## **T.V.1.002 Proof by induction AA HL staple pack**

- (a) List the three structural steps of mathematical induction.
- (b) Inductive step only: assume  $1 + \dots + k = k(k+1)/2$ ; prove for  $k+1$ .
- (c) Name one write-up error that loses IB AA HL induction marks.

## **T.V.1.003 Induction: inequalities and sequences**

- (a) List the three structural steps of mathematical induction.
- (b) Inductive step only: assume  $1 + \dots + k = k(k+1)/2$ ; prove for  $k+1$ .
- (c) Name one write-up error that loses IB AA HL induction marks.

## **T.V.1.004 Counterexample hunting exam style**

- (a) List the three structural steps of mathematical induction.
- (b) Inductive step only: assume  $1 + \dots + k = k(k+1)/2$ ; prove for  $k+1$ .
- (c) Name one write-up error that loses IB AA HL induction marks.

## **T.V.1.005 Two-column congruence proof drills**

- (a) List the three structural steps of mathematical induction.
- (b) Inductive step only: assume  $1 + \dots + k = k(k+1)/2$ ; prove for  $k+1$ .
- (c) Name one write-up error that loses IB AA HL induction marks.

## **T.V.1.006 IB exploration mini-write structure**

- (a) List the three structural steps of mathematical induction.
- (b) Inductive step only: assume  $1 + \dots + k = k(k+1)/2$ ; prove for  $k+1$ .
- (c) Name one write-up error that loses IB AA HL induction marks.

## **T.V.1.007 Indirect proof (contradiction) geometry classic**

- (a) List the three structural steps of mathematical induction.
- (b) Inductive step only: assume  $1 + \dots + k = k(k+1)/2$ ; prove for  $k+1$ .
- (c) Name one write-up error that loses IB AA HL induction marks.

## **T.V.2.001 Parallel line angle chase exam pack**

- (a) Parallel lines; corresponding angle to  $46^\circ$ ?
- (b) Parallel lines; co-interior with  $46^\circ$ ?
- (c) Find alternate angle to  $46^\circ$  between parallels.

## **T.V.2.002 Prove corresponding/alternate equal**

- (a) Parallel lines; corresponding angle to  $43^\circ$ ?
- (b) Parallel lines; co-interior with  $43^\circ$ ?
- (c) Find alternate angle to  $43^\circ$  between parallels.

## **T.V.3.001 Congruence proof multi-step GCSE/US**

- (a) Name three congruence conditions for triangles.
- (b) Does SSA always prove congruence? Why/why not?
- (c) In «Congruence proof multi-step GCSE/US»: what does CPCTC allow you to conclude after triangles are congruent?

## **T.V.3.002 Isosceles triangle theorem proofs**

- (a) [Isosceles triangle theorem proofs] Which view is the plan — front, side, or top?
- (b) Cuboid  $3 \times 3 \times 6$ : how many rectangles in a correct net?
- (c) Describe the front elevation of a square-based pyramid (base 3) standing on its base.

## **T.V.3.003 Points of concurrency justification pack**

- (a) AP/IB style: Which theorem or definition is central to «Points of concurrency justification pack»?

- (b) Write a justification sentence frame for «Points of concurrency justification pack»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.
- (c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Points of concurrency justification pack», then interpret.

#### **T.V.3.004 Exterior angle of triangle proof applications**

- (a) [Exterior angle of triangle proof applications] Which view is the plan — front, side, or top?
- (b) Cuboid  $7 \times 5 \times 5$ : how many rectangles in a correct net?
- (c) Geometry construction lab for «Exterior angle of triangle proof applications»: build the figure, measure, verify the claimed relation.

#### **T.V.3.005 CPCTC in multi-triangle proofs**

- (a) [CPCTC in multi-triangle proofs] Which view is the plan — front, side, or top?
- (b) Cuboid  $4 \times 2 \times 3$ : how many rectangles in a correct net?
- (c) Describe the front elevation of a square-based pyramid (base 4) standing on its base.

#### **T.V.4.001 Interior angle sum proof for polygons**

- (a) [Interior angle sum proof for polygons] Which view is the plan — front, side, or top?
- (b) Cuboid  $4 \times 2 \times 5$ : how many rectangles in a correct net?
- (c) Describe the front elevation of a square-based pyramid (base 4) standing on its base.

#### **T.V.4.002 Regular polygon exterior angle proofs**

- (a) [Regular polygon exterior angle proofs] Which view is the plan — front, side, or top?
- (b) Cuboid  $8 \times 5 \times 6$ : how many rectangles in a correct net?
- (c) Describe the front elevation of a square-based pyramid (base 8) standing on its base.

#### **T.V.4.003 Tessellation and interior angle conditions proof**

- (a) [Tessellation and interior angle conditions proof] Which view is the plan — front, side, or top?
- (b) Cuboid  $3 \times 2 \times 4$ : how many rectangles in a correct net?
- (c) Geometry construction lab for «Tessellation and interior angle conditions proof»: build the figure, measure, verify the claimed relation.

#### **T.V.5.001 Similar triangles exam multi-step**

- (a) Lengths scale by 2. Area scale factor?
- (b) A side 6 cm becomes ? after enlargement  $SF=2$ .
- (c) Volumes scale by ? if lengths scale by 2.

#### **T.V.5.002 AA/SSS/SAS similarity justification**

- (a) AP/IB style: Which theorem or definition is central to «AA/SSS/SAS similarity justification»?
- (b) Write a justification sentence frame for «AA/SSS/SAS similarity justification»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.
- (c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «AA/SSS/SAS similarity justification», then interpret.

#### **T.V.5.003 Area ratios of similar shapes exam**

- (a) Lengths scale by 2. Area scale factor?
- (b) A side 6 cm becomes ? after enlargement  $SF=2$ .
- (c) Volumes scale by ? if lengths scale by 2.

#### **T.V.5.004 Intercept theorems (basic proportionality)**

- (a) [Intercept theorems (basic proportionality)] Which view is the plan — front, side, or top?
- (b) Cuboid  $5 \times 5 \times 6$ : how many rectangles in a correct net?
- (c) Describe the front elevation of a square-based pyramid (base 5) standing on its base.

#### **T.V.5.005 Shadow and mirror similar-triangle measurement**

- (a) Lengths scale by 5. Area scale factor?
- (b) A side 6 cm becomes ? after enlargement  $SF=5$ .
- (c) Geometry construction lab for «Shadow and mirror similar-triangle measurement»: build the figure, measure, verify the claimed relation.

#### **T.V.5.006 Parallel lines $\Rightarrow$ similar triangles patterns**

- (a) Parallel lines; corresponding angle to  $63^\circ$ ?
- (b) Parallel lines; co-interior with  $63^\circ$ ?
- (c) Geometry construction lab for «Parallel lines  $\Rightarrow$  similar triangles patterns»: build the figure, measure, verify the claimed relation.

#### **T.V.6.001 Exact trig values $0^\circ, 30^\circ, 45^\circ, 60^\circ, 90^\circ$ pack**

- (a) Exact value of  $\sin 30^\circ$ .
- (b) Exact value of  $\cos 45^\circ$ .

(c) Without calculator, evaluate  $2 \sin 30^\circ \cos 30^\circ$ .

**T.V.6.002 Exact value multi-step without calculator**

(a) Exact value of  $\sin 60^\circ$ .

(b) Exact value of  $\cos 45^\circ$ .

(c) Without calculator, evaluate  $2 \sin 60^\circ \cos 60^\circ$ .

**T.V.6.003 Angle of elevation multi-step exam**

(a) From a point 32 m from a tower, angle of elevation to the top is  $\theta$ . Write  $\tan \theta$  in terms of height  $h$ .

(b) Angle of elevation  $30^\circ$ , distance 32 m. Exact height of tower?

(c) Observer eye height 1.5 m; angle of elevation to a 32 m tree top from 32 m away. Set up the equation for the angle  $\theta$ .

**T.V.6.004 Angle of depression with bearings mix**

(a) Write  $75^\circ$  from north as a three-figure bearing.

(b) Bearing of B from A is 075. Bearing of A from B?

(c) Geometry construction lab for «Angle of depression with bearings mix»: build the figure, measure, verify the claimed relation.

**T.V.6.005 SOHCAHTOA worded GCSE Foundation/Higher**

(a) Exact value of  $\sin 45^\circ$ .

(b) Exact value of  $\cos 60^\circ$ .

(c) Without calculator, evaluate  $2 \sin 45^\circ \cos 45^\circ$ .

**T.V.6.006 Find sides with reciprocal ratios**

(a) Exact value of  $\sin 60^\circ$ .

(b) Exact value of  $\cos 30^\circ$ .

(c) Without calculator, evaluate  $2 \sin 60^\circ \cos 60^\circ$ .

**T.V.6.007 Trig in isosceles / split base techniques**

(a) Exact value of  $\sin 45^\circ$ .

(b) Exact value of  $\cos 60^\circ$ .

(c) Without calculator, evaluate  $2 \sin 45^\circ \cos 45^\circ$ .

**T.V.6.008 3D trigonometry: angle between line and plane**

(a) Exact value of  $\sin 60^\circ$ .

(b) Exact value of  $\cos 60^\circ$ .

(c) Geometry construction lab for «3D trigonometry: angle between line and plane»: build the figure, measure, verify the claimed relation.

**T.V.6.009 3D trigonometry: angle between two planes**

(a) Exact value of  $\sin 60^\circ$ .

(b) Exact value of  $\cos 60^\circ$ .

(c) Geometry construction lab for «3D trigonometry: angle between two planes»: build the figure, measure, verify the claimed relation.

**T.V.6.010 3D Pythagoras + trig combined**

(a) Legs 7 cm, 24 cm. Hypotenuse?

(b) Hypotenuse 25 m, leg 7 m. Other leg?

(c) Ladder 25 m; base 7 m from wall. Reach height?

**T.V.6.011 Right-trig calculator vs exact decision**

(a) Exact value of  $\sin 60^\circ$ .

(b) Exact value of  $\cos 30^\circ$ .

(c) Without calculator, evaluate  $2 \sin 60^\circ \cos 60^\circ$ .

**T.V.6.012 Inverse trig range awareness (principal values)**

(a) Exact value of  $\sin 30^\circ$ .

(b) Exact value of  $\cos 45^\circ$ .

(c) Without calculator, evaluate  $2 \sin 30^\circ \cos 30^\circ$ .

**T.V.6.013 Clinometer / practical elevation contexts**

(a) From a point 27 m from a tower, angle of elevation to the top is  $\theta$ . Write  $\tan \theta$  in terms of height  $h$ .

(b) Angle of elevation  $30^\circ$ , distance 27 m. Exact height of tower?

(c) Geometry construction lab for «Clinometer / practical elevation contexts»: build the figure, measure, verify the claimed relation.

**T.V.6.014 Trig ratios as slopes and unit rates**

- (a) Exact value of  $\sin 60^\circ$ .
- (b) Exact value of  $\cos 30^\circ$ .
- (c) Without calculator, evaluate  $2 \sin 60^\circ \cos 60^\circ$ .

**T.V.6.015 Exact tan 30/45/60 without calculator**

- (a) Exact value of  $\sin 30^\circ$ .
- (b) Exact value of  $\cos 30^\circ$ .
- (c) Without calculator, evaluate  $2 \sin 30^\circ \cos 30^\circ$ .

**T.V.6.016 Two-triangle elevation problems**

- (a) From a point 21 m from a tower, angle of elevation to the top is  $\theta$ . Write  $\tan \theta$  in terms of height  $h$ .
- (b) Angle of elevation  $30^\circ$ , distance 21 m. Exact height of tower?
- (c) Geometry construction lab for «Two-triangle elevation problems»: build the figure, measure, verify the claimed relation.

**T.V.6.017 Bearing and distance from trig in right  $\Delta$** 

- (a) Write  $200^\circ$  from north as a three-figure bearing.
- (b) Bearing of B from A is  $200^\circ$ . Bearing of A from B?
- (c) Geometry construction lab for «Bearing and distance from trig in right  $\Delta$ »: build the figure, measure, verify the claimed relation.

**T.V.6.018 Trigonometric ratios of complementary angles**

- (a) Exact value of  $\sin 60^\circ$ .
- (b) Exact value of  $\cos 60^\circ$ .
- (c) Without calculator, evaluate  $2 \sin 60^\circ \cos 60^\circ$ .

**T.V.6.019 Exact values for related angles  $120^\circ, 135^\circ, 150^\circ$** 

- (a) Exact value of  $\sin 45^\circ$ .
- (b) Exact value of  $\cos 60^\circ$ .
- (c) Unit-circle lab for «Exact values for related angles  $120^\circ, 135^\circ, 150^\circ$ »: sample three angles; record ( $\cos, \sin$ ).

**T.V.6.020 Slope angle of a roof / ramp accessibility**

- (a) Exact value of  $\sin 30^\circ$ .
- (b) Exact value of  $\cos 60^\circ$ .
- (c) Without calculator, evaluate  $2 \sin 30^\circ \cos 30^\circ$ .

**T.V.6.021 Trigonometry non-calc paper strategies**

- (a) Exact value of  $\sin 60^\circ$ .
- (b) Exact value of  $\cos 60^\circ$ .
- (c) Without calculator, evaluate  $2 \sin 60^\circ \cos 60^\circ$ .

**T.V.6.022 Double triangle ladder against wall classic**

- (a) Exact value of  $\sin 30^\circ$ .
- (b) Exact value of  $\cos 60^\circ$ .
- (c) Geometry construction lab for «Double triangle ladder against wall classic»: build the figure, measure, verify the claimed relation.

**T.V.6.023 Trigonometry in non-right by dropping altitude**

- (a) Exact value of  $\sin 45^\circ$ .
- (b) Exact value of  $\cos 60^\circ$ .
- (c) Geometry construction lab for «Trigonometry in non-right by dropping altitude»: build the figure, measure, verify the claimed relation.

**T.V.6.024 Inclination of a line from  $\tan \theta = m$** 

- (a) Exact value of  $\sin 45^\circ$ .
- (b) Exact value of  $\cos 30^\circ$ .
- (c) Without calculator, evaluate  $2 \sin 45^\circ \cos 45^\circ$ .

**T.V.6.025 Exact values application set 1**

- (a) Exact value of  $\sin 30^\circ$ .
- (b) Exact value of  $\cos 45^\circ$ .
- (c) Without calculator, evaluate  $2 \sin 30^\circ \cos 30^\circ$ .

**T.V.6.026 Exact values application set 2**

- (a) Exact value of  $\sin 30^\circ$ .
- (b) Exact value of  $\cos 60^\circ$ .
- (c) Without calculator, evaluate  $2 \sin 30^\circ \cos 30^\circ$ .

#### **T.V.6.027 Exact values application set 3**

- (a) Exact value of  $\sin 30^\circ$ .
- (b) Exact value of  $\cos 30^\circ$ .
- (c) Without calculator, evaluate  $2 \sin 30^\circ \cos 30^\circ$ .

#### **T.V.6.028 Exact values application set 4**

- (a) Exact value of  $\sin 30^\circ$ .
- (b) Exact value of  $\cos 60^\circ$ .
- (c) Without calculator, evaluate  $2 \sin 30^\circ \cos 30^\circ$ .

#### **T.V.6.029 3D trig optional challenge 1**

- (a) Exact value of  $\sin 45^\circ$ .
- (b) Exact value of  $\cos 60^\circ$ .
- (c) Geometry construction lab for «3D trig optional challenge 1»: build the figure, measure, verify the claimed relation.

#### **T.V.6.030 3D trig optional challenge 2**

- (a) Exact value of  $\sin 60^\circ$ .
- (b) Exact value of  $\cos 60^\circ$ .
- (c) Geometry construction lab for «3D trig optional challenge 2»: build the figure, measure, verify the claimed relation.

#### **T.V.6.031 3D trig optional challenge 3**

- (a) Exact value of  $\sin 60^\circ$ .
- (b) Exact value of  $\cos 60^\circ$ .
- (c) Geometry construction lab for «3D trig optional challenge 3»: build the figure, measure, verify the claimed relation.

#### **T.V.7.001 Bearings: three-figure notation drills**

- (a) Write  $145^\circ$  from north as a three-figure bearing.
- (b) Bearing of B from A is  $145$ . Bearing of A from B?
- (c) Geometry construction lab for «Bearings: three-figure notation drills»: build the figure, measure, verify the claimed relation.

#### **T.V.7.002 Bearings: reverse bearings (+180° rule)**

- (a) Write  $35^\circ$  from north as a three-figure bearing.
- (b) Bearing of B from A is  $035$ . Bearing of A from B?
- (c) Geometry construction lab for «Bearings: reverse bearings (+180° rule)»: build the figure, measure, verify the claimed relation.

#### **T.V.7.003 Bearings: multi-leg route problems**

- (a) Write  $310^\circ$  from north as a three-figure bearing.
- (b) Bearing of B from A is  $310$ . Bearing of A from B?
- (c) Geometry construction lab for «Bearings: multi-leg route problems»: build the figure, measure, verify the claimed relation.

#### **T.V.7.004 Bearings: exam worded navigation Qs**

- (a) Write  $145^\circ$  from north as a three-figure bearing.
- (b) Bearing of B from A is  $145$ . Bearing of A from B?
- (c) Geometry construction lab for «Bearings: exam worded navigation Qs»: build the figure, measure, verify the claimed relation.

#### **T.V.7.005 Bearings with cosine/sine rule**

- (a) Write  $35^\circ$  from north as a three-figure bearing.
- (b) Bearing of B from A is  $035$ . Bearing of A from B?
- (c) Geometry construction lab for «Bearings with cosine/sine rule»: build the figure, measure, verify the claimed relation.

#### **T.V.7.006 Law of sines ambiguous case SSA deep pack**

- (a) Write  $255^\circ$  from north as a three-figure bearing.
- (b) Bearing of B from A is  $255$ . Bearing of A from B?
- (c) Ship sails  $255$  for  $12$  km then  $345$  for  $5$  km. Name the rule to find straight-line return distance.

#### **T.V.7.007 Law of cosines rearrange for angles**

- (a) Write  $310^\circ$  from north as a three-figure bearing.

- (b) Bearing of B from A is 310. Bearing of A from B?
- (c) Ship sails 310 for 12 km then 040 for 5 km. Name the rule to find straight-line return distance.

#### **T.V.7.008 Area $\frac{1}{2}ab \sin C$ exam pack**

- (a) Write  $255^\circ$  from north as a three-figure bearing.
- (b) Bearing of B from A is 255. Bearing of A from B?
- (c) Ship sails 255 for 12 km then 345 for 5 km. Name the rule to find straight-line return distance.

#### **T.V.7.009 Heron vs sine-area method choice**

- (a) Write  $255^\circ$  from north as a three-figure bearing.
- (b) Bearing of B from A is 255. Bearing of A from B?
- (c) Ship sails 255 for 12 km then 345 for 5 km. Name the rule to find straight-line return distance.

#### **T.V.7.010 Surveying triangulation exam contexts**

- (a) Write  $48^\circ$  from north as a three-figure bearing.
- (b) Bearing of B from A is 048. Bearing of A from B?
- (c) Ship sails 048 for 12 km then 138 for 5 km. Name the rule to find straight-line return distance.

#### **T.V.7.011 Solve any triangle mixed timed set**

- (a) Write  $110^\circ$  from north as a three-figure bearing.
- (b) Bearing of B from A is 110. Bearing of A from B?
- (c) Ship sails 110 for 12 km then 200 for 5 km. Name the rule to find straight-line return distance.

#### **T.V.7.012 Area of triangle given SAS multi-step**

- (a) Write  $48^\circ$  from north as a three-figure bearing.
- (b) Bearing of B from A is 048. Bearing of A from B?
- (c) Ship sails 048 for 12 km then 138 for 5 km. Name the rule to find straight-line return distance.

#### **T.V.7.013 Obtuse triangles cosine rule careful signs**

- (a) Write  $255^\circ$  from north as a three-figure bearing.
- (b) Bearing of B from A is 255. Bearing of A from B?
- (c) Ship sails 255 for 12 km then 345 for 5 km. Name the rule to find straight-line return distance.

#### **T.V.7.014 Navigation: wind/current vector triangle**

- (a)  $a=(6, -1)$ ,  $b=(5, 5)$ . Find  $a+b$  as a column vector.
- (b)  $|a|$  for  $a=(6, -1)$ ?
- (c) Geometry construction lab for «Navigation: wind/current vector triangle»: build the figure, measure, verify the claimed relation.

#### **T.V.7.015 Ambiguous case: height vs given side decision tree**

- (a) Write  $62^\circ$  from north as a three-figure bearing.
- (b) Bearing of B from A is 062. Bearing of A from B?
- (c) Geometry construction lab for «Ambiguous case: height vs given side decision tree»: build the figure, measure, verify the claimed relation.

#### **T.V.7.016 Largest angle opposite largest side justification**

- (a) AP/IB style: Which theorem or definition is central to «Largest angle opposite largest side justification»?
- (b) Write a justification sentence frame for «Largest angle opposite largest side justification»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.
- (c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Largest angle opposite largest side justification», then interpret.

#### **T.V.7.017 Flight path bearings with ground speed**

- (a) Write  $110^\circ$  from north as a three-figure bearing.
- (b) Bearing of B from A is 110. Bearing of A from B?
- (c) Geometry construction lab for «Flight path bearings with ground speed»: build the figure, measure, verify the claimed relation.

#### **T.V.7.018 Bearings: airport runway / orienteering contexts**

- (a) Write  $255^\circ$  from north as a three-figure bearing.
- (b) Bearing of B from A is 255. Bearing of A from B?
- (c) Geometry construction lab for «Bearings: airport runway / orienteering contexts»: build the figure, measure, verify the claimed relation.

#### **T.V.7.019 Bearings navigation paper 1**

- (a) Write  $75^\circ$  from north as a three-figure bearing.
- (b) Bearing of B from A is 075. Bearing of A from B?

(c) Geometry construction lab for «Bearings navigation paper 1»: build the figure, measure, verify the claimed relation.

#### **T.V.7.020 Bearings navigation paper 2**

(a) Write  $62^\circ$  from north as a three-figure bearing.

(b) Bearing of B from A is 062. Bearing of A from B?

(c) Geometry construction lab for «Bearings navigation paper 2»: build the figure, measure, verify the claimed relation.

#### **T.V.7.021 Bearings navigation paper 3**

(a) Write  $200^\circ$  from north as a three-figure bearing.

(b) Bearing of B from A is 200. Bearing of A from B?

(c) Geometry construction lab for «Bearings navigation paper 3»: build the figure, measure, verify the claimed relation.

#### **T.V.7.022 Bearings navigation paper 4**

(a) Write  $310^\circ$  from north as a three-figure bearing.

(b) Bearing of B from A is 310. Bearing of A from B?

(c) Geometry construction lab for «Bearings navigation paper 4»: build the figure, measure, verify the claimed relation.

#### **T.V.8.001 Circle theorems: angle in a semicircle**

(a) Angle at centre  $40^\circ$ . Angle at circumference (same arc)?

(b) Triangle in a semicircle; one acute angle  $32^\circ$ . Other acute angle?

(c) Cyclic quadrilateral;  $\angle A = 40^\circ$ . Find  $\angle C$ .

#### **T.V.8.002 Circle theorems: centre angle is twice**

(a) Angle at centre  $80^\circ$ . Angle at circumference (same arc)?

(b) Triangle in a semicircle; one acute angle  $32^\circ$ . Other acute angle?

(c) Cyclic quadrilateral;  $\angle A = 80^\circ$ . Find  $\angle C$ .

#### **T.V.8.003 Circle theorems: angles in same segment**

(a) Angle at centre  $72^\circ$ . Angle at circumference (same arc)?

(b) Triangle in a semicircle; one acute angle  $32^\circ$ . Other acute angle?

(c) Cyclic quadrilateral;  $\angle A = 72^\circ$ . Find  $\angle C$ .

#### **T.V.8.004 Circle theorems: opposite angles cyclic quad**

(a) Angle at centre  $80^\circ$ . Angle at circumference (same arc)?

(b) Triangle in a semicircle; one acute angle  $32^\circ$ . Other acute angle?

(c) Cyclic quadrilateral;  $\angle A = 80^\circ$ . Find  $\angle C$ .

#### **T.V.8.005 Circle theorems: alternate segment**

(a) Angle at centre  $72^\circ$ . Angle at circumference (same arc)?

(b) Triangle in a semicircle; one acute angle  $32^\circ$ . Other acute angle?

(c) Cyclic quadrilateral;  $\angle A = 72^\circ$ . Find  $\angle C$ .

#### **T.V.8.006 Circle theorems: tangent-chord angle**

(a) Angle at centre  $40^\circ$ . Angle at circumference (same arc)?

(b) Triangle in a semicircle; one acute angle  $32^\circ$ . Other acute angle?

(c) Cyclic quadrilateral;  $\angle A = 40^\circ$ . Find  $\angle C$ .

#### **T.V.8.007 Circle theorems mixed proof pack**

(a) Angle at centre  $64^\circ$ . Angle at circumference (same arc)?

(b) Triangle in a semicircle; one acute angle  $32^\circ$ . Other acute angle?

(c) Geometry construction lab for «Circle theorems mixed proof pack»: build the figure, measure, verify the claimed relation.

#### **T.V.8.008 Equation of circle centre origin + tangent**

(a) Angle at centre  $80^\circ$ . Angle at circumference (same arc)?

(b) Triangle in a semicircle; one acute angle  $32^\circ$ . Other acute angle?

(c) Cyclic quadrilateral;  $\angle A = 80^\circ$ . Find  $\angle C$ .

#### **T.V.8.009 Tangent from external point length**

(a) Angle at centre  $100^\circ$ . Angle at circumference (same arc)?

(b) Triangle in a semicircle; one acute angle  $32^\circ$ . Other acute angle?

(c) Cyclic quadrilateral;  $\angle A = 100^\circ$ . Find  $\angle C$ .

**T.V.8.010 Intersecting chords length theorem**

- (a) Angle at centre  $40^\circ$ . Angle at circumference (same arc)?
- (b) Triangle in a semicircle; one acute angle  $32^\circ$ . Other acute angle?
- (c) Cyclic quadrilateral;  $\angle A = 40^\circ$ . Find  $\angle C$ .

**T.V.8.011 Sector/segment area exact exam**

- (a) Angle at centre  $64^\circ$ . Angle at circumference (same arc)?
- (b) Triangle in a semicircle; one acute angle  $32^\circ$ . Other acute angle?
- (c) Cyclic quadrilateral;  $\angle A = 64^\circ$ . Find  $\angle C$ .

**T.V.8.012 Radians arc/sector IB/AP bridge**

- (a) Angle at centre  $50^\circ$ . Angle at circumference (same arc)?
- (b) Triangle in a semicircle; one acute angle  $32^\circ$ . Other acute angle?
- (c) Cyclic quadrilateral;  $\angle A = 50^\circ$ . Find  $\angle C$ .

**T.V.8.013 Proof: tangent  $\perp$  radius**

- (a) Angle at centre  $100^\circ$ . Angle at circumference (same arc)?
- (b) Triangle in a semicircle; one acute angle  $32^\circ$ . Other acute angle?
- (c) Geometry construction lab for «Proof: tangent  $\perp$  radius»: build the figure, measure, verify the claimed relation.

**T.V.8.014 Arc length vs sector vs segment decision**

- (a) Angle at centre  $64^\circ$ . Angle at circumference (same arc)?
- (b) Triangle in a semicircle; one acute angle  $32^\circ$ . Other acute angle?
- (c) Cyclic quadrilateral;  $\angle A = 64^\circ$ . Find  $\angle C$ .

**T.V.8.015 Trig in circle: chord length via sine**

- (a) Angle at centre  $72^\circ$ . Angle at circumference (same arc)?
- (b) Triangle in a semicircle; one acute angle  $32^\circ$ . Other acute angle?
- (c) Geometry construction lab for «Trig in circle: chord length via sine»: build the figure, measure, verify the claimed relation.

**T.V.8.016 Circle theorems + trig hybrid problems**

- (a) Angle at centre  $100^\circ$ . Angle at circumference (same arc)?
- (b) Triangle in a semicircle; one acute angle  $32^\circ$ . Other acute angle?
- (c) Geometry construction lab for «Circle theorems + trig hybrid problems»: build the figure, measure, verify the claimed relation.

**T.V.8.017 Circle theorems: radius to tangent proof write**

- (a) Angle at centre  $64^\circ$ . Angle at circumference (same arc)?
- (b) Triangle in a semicircle; one acute angle  $32^\circ$ . Other acute angle?
- (c) Geometry construction lab for «Circle theorems: radius to tangent proof write»: build the figure, measure, verify the claimed relation.

**T.V.8.018 Circle theorems mixed figure 1**

- (a) Angle at centre  $100^\circ$ . Angle at circumference (same arc)?
- (b) Triangle in a semicircle; one acute angle  $32^\circ$ . Other acute angle?
- (c) Geometry construction lab for «Circle theorems mixed figure 1»: build the figure, measure, verify the claimed relation.

**T.V.8.019 Circle theorems mixed figure 2**

- (a) Angle at centre  $50^\circ$ . Angle at circumference (same arc)?
- (b) Triangle in a semicircle; one acute angle  $32^\circ$ . Other acute angle?
- (c) Geometry construction lab for «Circle theorems mixed figure 2»: build the figure, measure, verify the claimed relation.

**T.V.8.020 Circle theorems mixed figure 3**

- (a) Angle at centre  $50^\circ$ . Angle at circumference (same arc)?
- (b) Triangle in a semicircle; one acute angle  $32^\circ$ . Other acute angle?
- (c) Geometry construction lab for «Circle theorems mixed figure 3»: build the figure, measure, verify the claimed relation.

**T.V.8.021 Circle theorems mixed figure 4**

- (a) Angle at centre  $80^\circ$ . Angle at circumference (same arc)?
- (b) Triangle in a semicircle; one acute angle  $32^\circ$ . Other acute angle?
- (c) Geometry construction lab for «Circle theorems mixed figure 4»: build the figure, measure, verify the claimed relation.

**T.V.8.022 Circle theorems mixed figure 5**

- (a) Angle at centre  $80^\circ$ . Angle at circumference (same arc)?
- (b) Triangle in a semicircle; one acute angle  $32^\circ$ . Other acute angle?
- (c) Geometry construction lab for «Circle theorems mixed figure 5»: build the figure, measure, verify the claimed relation.

#### **T.V.9.001 Unit circle: verticality of sine**

- (a) Unit circle: coordinates at  $135^\circ$ ?
- (b) Reference angle for  $135^\circ$ ?
- (c) Unit-circle lab for «Unit circle: verticality of sine»: sample three angles; record (cos,sin).

#### **T.V.9.002 Unit circle: horizontality of cosine**

- (a) Unit circle: coordinates at  $0^\circ$ ?
- (b) Reference angle for  $0^\circ$ ?
- (c) Unit-circle lab for «Unit circle: horizontality of cosine»: sample three angles; record (cos,sin).

#### **T.V.9.003 Unit circle: tangent as slope of radius**

- (a) Unit circle: coordinates at  $135^\circ$ ?
- (b) Reference angle for  $135^\circ$ ?
- (c) Unit-circle lab for «Unit circle: tangent as slope of radius»: sample three angles; record (cos,sin).

#### **T.V.9.004 Unit circle special angles exact coordinates**

- (a) Unit circle: coordinates at  $120^\circ$ ?
- (b) Reference angle for  $120^\circ$ ?
- (c) Unit-circle lab for «Unit circle special angles exact coordinates»: sample three angles; record (cos,sin).

#### **T.V.9.005 Reference angles across four quadrants drills**

- (a) Unit circle: coordinates at  $150^\circ$ ?
- (b) Reference angle for  $150^\circ$ ?
- (c) Unit-circle lab for «Reference angles across four quadrants drills»: sample three angles; record (cos,sin).

#### **T.V.9.006 ASTC / CAST diagram fluency (UK)**

- (a) Unit circle: coordinates at  $60^\circ$ ?
- (b) Reference angle for  $60^\circ$ ?
- (c) Unit-circle lab for «ASTC / CAST diagram fluency (UK)»: sample three angles; record (cos,sin).

#### **T.V.9.007 Coterminal angles in degrees and radians**

- (a) Unit circle: coordinates at  $60^\circ$ ?
- (b) Reference angle for  $60^\circ$ ?
- (c) Using CAST/ASTC, sign of  $\cos 60^\circ$ ?

#### **T.V.9.008 Radians as arc length on unit circle**

- (a) Unit circle: coordinates at  $135^\circ$ ?
- (b) Reference angle for  $135^\circ$ ?
- (c) Unit-circle lab for «Radians as arc length on unit circle»: sample three angles; record (cos,sin).

#### **T.V.9.009 Graph sine: build from unit circle animation**

- (a) Unit circle: coordinates at  $0^\circ$ ?
- (b) Reference angle for  $0^\circ$ ?
- (c) Unit-circle lab for «Graph sine: build from unit circle animation»: sample three angles; record (cos,sin).

#### **T.V.9.010 Graph cosine: build from unit circle animation**

- (a) Unit circle: coordinates at  $120^\circ$ ?
- (b) Reference angle for  $120^\circ$ ?
- (c) Unit-circle lab for «Graph cosine: build from unit circle animation»: sample three angles; record (cos,sin).

#### **T.V.9.011 Amplitude/period/phase exam transformations**

- (a) Unit circle: coordinates at  $120^\circ$ ?
- (b) Reference angle for  $120^\circ$ ?
- (c) Desmos lab for «Amplitude/period/phase exam transformations»: vary one parameter, capture before/after, write one observation.

#### **T.V.9.012 Desmos: match equation to trig graph challenges**

- (a) Unit circle: coordinates at  $135^\circ$ ?

- (b) Reference angle for  $135^\circ$ ?
- (c) Desmos lab for «Desmos: match equation to trig graph challenges»: vary one parameter, capture before/after, write one observation.

#### **T.V.9.013 Periodic modelling: tides and ferris wheels**

- (a) Unit circle: coordinates at  $60^\circ$ ?
- (b) Reference angle for  $60^\circ$ ?
- (c) Desmos lab for «Periodic modelling: tides and ferris wheels»: vary one parameter, capture before/after, write one observation.

#### **T.V.9.014 Graph tangent with asymptotes exam**

- (a) Unit circle: coordinates at  $60^\circ$ ?
- (b) Reference angle for  $60^\circ$ ?
- (c) Desmos lab for «Graph tangent with asymptotes exam»: vary one parameter, capture before/after, write one observation.

#### **T.V.9.015 Inverse trig graphs and ranges**

- (a) Unit circle: coordinates at  $60^\circ$ ?
- (b) Reference angle for  $60^\circ$ ?
- (c) Desmos lab for «Inverse trig graphs and ranges»: vary one parameter, capture before/after, write one observation.

#### **T.V.9.016 Composition $\arcsin(\sin \theta)$ pitfalls**

- (a) Unit circle: coordinates at  $150^\circ$ ?
- (b) Reference angle for  $150^\circ$ ?
- (c) Using CAST/ASTC, sign of  $\cos 150^\circ$ ?

#### **T.V.9.017 Desmos: explore a $\sin(bx+c)+d$ parameters live**

- (a) Unit circle: coordinates at  $30^\circ$ ?
- (b) Reference angle for  $30^\circ$ ?
- (c) Desmos lab for «Desmos: explore a  $\sin(bx+c)+d$  parameters live»: vary one parameter, capture before/after, write one observation.

#### **T.V.9.018 Unit circle: project to axes interactively**

- (a) Unit circle: coordinates at  $90^\circ$ ?
- (b) Reference angle for  $90^\circ$ ?
- (c) Unit-circle lab for «Unit circle: project to axes interactively»: sample three angles; record  $(\cos, \sin)$ .

#### **T.V.9.019 Negative angles on unit circle**

- (a) Unit circle: coordinates at  $0^\circ$ ?
- (b) Reference angle for  $0^\circ$ ?
- (c) Unit-circle lab for «Negative angles on unit circle»: sample three angles; record  $(\cos, \sin)$ .

#### **T.V.9.020 Co-terminal and reference mixed mega-drill**

- (a) Unit circle: coordinates at  $180^\circ$ ?
- (b) Reference angle for  $180^\circ$ ?
- (c) Unit-circle lab for «Co-terminal and reference mixed mega-drill»: sample three angles; record  $(\cos, \sin)$ .

#### **T.V.9.021 Desmos: compare $\sin x$ vs $\sin(2x)$ vs $2\sin x$**

- (a) Unit circle: coordinates at  $0^\circ$ ?
- (b) Reference angle for  $0^\circ$ ?
- (c) Desmos lab for «Desmos: compare  $\sin x$  vs  $\sin(2x)$  vs  $2\sin x$ »: vary one parameter, capture before/after, write one observation.

#### **T.V.9.022 Unit circle: quiz coordinates at $\pi/6$ multiples**

- (a) Unit circle: coordinates at  $180^\circ$ ?
- (b) Reference angle for  $180^\circ$ ?
- (c) Unit-circle lab for «Unit circle: quiz coordinates at  $\pi/6$  multiples»: sample three angles; record  $(\cos, \sin)$ .

#### **T.V.9.023 Convert DMS $\leftrightarrow$ decimal degrees exam**

- (a) Unit circle: coordinates at  $30^\circ$ ?
- (b) Reference angle for  $30^\circ$ ?
- (c) Using CAST/ASTC, sign of  $\cos 30^\circ$ ?

#### **T.V.9.024 Angular speed and linear speed link**

- (a) Unit circle: coordinates at  $60^\circ$ ?
- (b) Reference angle for  $60^\circ$ ?

(c) Using CAST/ASTC, sign of  $\cos 60^\circ$ ?

#### **T.V.9.025 Graph cosec/sec/cot from sine/cosine**

(a) Unit circle: coordinates at  $120^\circ$ ?

(b) Reference angle for  $120^\circ$ ?

(c) Desmos lab for «Graph cosec/sec/cot from sine/cosine»: vary one parameter, capture before/after, write one observation.

#### **T.V.9.026 Phase shift left/right common student errors**

(a) Unit circle: coordinates at  $120^\circ$ ?

(b) Reference angle for  $120^\circ$ ?

(c) Desmos lab for «Phase shift left/right common student errors»: vary one parameter, capture before/after, write one observation.

#### **T.V.9.027 Interactive: drag angle see sin/cos update**

(a) Unit circle: coordinates at  $135^\circ$ ?

(b) Reference angle for  $135^\circ$ ?

(c) Unit-circle lab for «Interactive: drag angle see sin/cos update»: sample three angles; record (cos,sin).

#### **T.V.9.028 Fourier intuition: sum of sines preview**

(a) Unit circle: coordinates at  $180^\circ$ ?

(b) Reference angle for  $180^\circ$ ?

(c) Desmos lab for «Fourier intuition: sum of sines preview»: vary one parameter, capture before/after, write one observation.

#### **T.V.9.029 Desmos: find period from two zeros**

(a) Unit circle: coordinates at  $90^\circ$ ?

(b) Reference angle for  $90^\circ$ ?

(c) Desmos lab for «Desmos: find period from two zeros»: vary one parameter, capture before/after, write one observation.

#### **T.V.9.030 Unit circle symmetry: odd/even visual proof**

(a) Unit circle: coordinates at  $180^\circ$ ?

(b) Reference angle for  $180^\circ$ ?

(c) Unit-circle lab for «Unit circle symmetry: odd/even visual proof»: sample three angles; record (cos,sin).

#### **T.V.9.031 Model daylight hours with cosine**

(a) Unit circle: coordinates at  $30^\circ$ ?

(b) Reference angle for  $30^\circ$ ?

(c) Desmos lab for «Model daylight hours with cosine»: vary one parameter, capture before/after, write one observation.

#### **T.V.9.032 Radians exclusively: no degree crutches**

(a) Unit circle: coordinates at  $30^\circ$ ?

(b) Reference angle for  $30^\circ$ ?

(c) Unit-circle lab for «Radians exclusively: no degree crutches»: sample three angles; record (cos,sin).

#### **T.V.9.033 Interactive geometry: isolate sin vs cos projections**

(a) Unit circle: coordinates at  $120^\circ$ ?

(b) Reference angle for  $120^\circ$ ?

(c) Unit-circle lab for «Interactive geometry: isolate sin vs cos projections»: sample three angles; record (cos,sin).

#### **T.V.9.034 Desmos: solve $\sin x = \cos x$ graphically and exactly**

(a) Unit circle: coordinates at  $90^\circ$ ?

(b) Reference angle for  $90^\circ$ ?

(c) Desmos lab for «Desmos: solve  $\sin x = \cos x$  graphically and exactly»: vary one parameter, capture before/after, write one observation.

#### **T.V.9.035 Amplitude from max–min; midline average**

(a) Unit circle: coordinates at  $45^\circ$ ?

(b) Reference angle for  $45^\circ$ ?

(c) Desmos lab for «Amplitude from max–min; midline average»: vary one parameter, capture before/after, write one observation.

#### **T.V.9.036 Unit circle: memory palace for special angles**

(a) Unit circle: coordinates at  $150^\circ$ ?

(b) Reference angle for  $150^\circ$ ?

(c) Unit-circle lab for «Unit circle: memory palace for special angles»: sample three angles; record (cos,sin).

**T.V.9.037 Desmos: Fourier square-wave tease with odd harmonics**

- (a) Unit circle: coordinates at  $135^\circ$ ?
- (b) Reference angle for  $135^\circ$ ?
- (c) Desmos lab for «Desmos: Fourier square-wave tease with odd harmonics»: vary one parameter, capture before/after, write one observation.

**T.V.9.038 Unit circle fluency round 1**

- (a) Unit circle: coordinates at  $45^\circ$ ?
- (b) Reference angle for  $45^\circ$ ?
- (c) Unit-circle lab for «Unit circle fluency round 1»: sample three angles; record (cos,sin).

**T.V.9.039 Unit circle fluency round 2**

- (a) Unit circle: coordinates at  $90^\circ$ ?
- (b) Reference angle for  $90^\circ$ ?
- (c) Unit-circle lab for «Unit circle fluency round 2»: sample three angles; record (cos,sin).

**T.V.9.040 Unit circle fluency round 3**

- (a) Unit circle: coordinates at  $150^\circ$ ?
- (b) Reference angle for  $150^\circ$ ?
- (c) Unit-circle lab for «Unit circle fluency round 3»: sample three angles; record (cos,sin).

**T.V.9.041 Unit circle fluency round 4**

- (a) Unit circle: coordinates at  $60^\circ$ ?
- (b) Reference angle for  $60^\circ$ ?
- (c) Unit-circle lab for «Unit circle fluency round 4»: sample three angles; record (cos,sin).

**T.V.9.042 Unit circle fluency round 5**

- (a) Unit circle: coordinates at  $90^\circ$ ?
- (b) Reference angle for  $90^\circ$ ?
- (c) Unit-circle lab for «Unit circle fluency round 5»: sample three angles; record (cos,sin).

**T.V.9.043 Unit circle fluency round 6**

- (a) Unit circle: coordinates at  $60^\circ$ ?
- (b) Reference angle for  $60^\circ$ ?
- (c) Unit-circle lab for «Unit circle fluency round 6»: sample three angles; record (cos,sin).

**T.V.9.044 Unit circle fluency round 7**

- (a) Unit circle: coordinates at  $150^\circ$ ?
- (b) Reference angle for  $150^\circ$ ?
- (c) Unit-circle lab for «Unit circle fluency round 7»: sample three angles; record (cos,sin).

**T.V.9.045 Desmos trig graph challenge 1**

- (a) Unit circle: coordinates at  $180^\circ$ ?
- (b) Reference angle for  $180^\circ$ ?
- (c) Desmos lab for «Desmos trig graph challenge 1»: vary one parameter, capture before/after, write one observation.

**T.V.9.046 Desmos trig graph challenge 2**

- (a) Unit circle: coordinates at  $180^\circ$ ?
- (b) Reference angle for  $180^\circ$ ?
- (c) Desmos lab for «Desmos trig graph challenge 2»: vary one parameter, capture before/after, write one observation.

**T.V.9.047 Desmos trig graph challenge 3**

- (a) Unit circle: coordinates at  $45^\circ$ ?
- (b) Reference angle for  $45^\circ$ ?
- (c) Desmos lab for «Desmos trig graph challenge 3»: vary one parameter, capture before/after, write one observation.

**T.V.9.048 Desmos trig graph challenge 4**

- (a) Unit circle: coordinates at  $45^\circ$ ?
- (b) Reference angle for  $45^\circ$ ?
- (c) Desmos lab for «Desmos trig graph challenge 4»: vary one parameter, capture before/after, write one observation.

**T.V.9.049 Desmos trig graph challenge 5**

- (a) Unit circle: coordinates at  $0^\circ$ ?
- (b) Reference angle for  $0^\circ$ ?
- (c) Desmos lab for «Desmos trig graph challenge 5»: vary one parameter, capture before/after, write one observation.

#### **T.V.9.050 Desmos trig graph challenge 6**

- (a) Unit circle: coordinates at  $60^\circ$ ?
- (b) Reference angle for  $60^\circ$ ?
- (c) Desmos lab for «Desmos trig graph challenge 6»: vary one parameter, capture before/after, write one observation.

#### **T.V.9.051 Desmos trig graph challenge 7**

- (a) Unit circle: coordinates at  $120^\circ$ ?
- (b) Reference angle for  $120^\circ$ ?
- (c) Desmos lab for «Desmos trig graph challenge 7»: vary one parameter, capture before/after, write one observation.

#### **T.V.10.001 Prove identities structured exam write-ups**

- (a) AP/IB style: Which theorem or definition is central to «Prove identities structured exam write-ups»?
- (b) Write a justification sentence frame for «Prove identities structured exam write-ups»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.
- (c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Prove identities structured exam write-ups», then interpret.

#### **T.V.10.002 Pythagorean identities rearrange toolkit**

- (a) Legs 9 cm, 12 cm. Hypotenuse?
- (b) Hypotenuse 15 m, leg 9 m. Other leg?
- (c) Ladder 15 m; base 9 m from wall. Reach height?

#### **T.V.10.003 Compound angle applications: expand $\sin(A \pm B)$**

- (a) Exact value of  $\sin 30^\circ$ .
- (b) Exact value of  $\cos 60^\circ$ .
- (c) Without calculator, evaluate  $2 \sin 30^\circ \cos 30^\circ$ .

#### **T.V.10.004 Compound angle applications: find exact values**

- (a) Exact value of  $\sin 45^\circ$ .
- (b) Exact value of  $\cos 60^\circ$ .
- (c) Without calculator, evaluate  $2 \sin 45^\circ \cos 45^\circ$ .

#### **T.V.10.005 Double-angle in equation solving**

- (a) Exact value of  $\sin 60^\circ$ .
- (b) Exact value of  $\cos 30^\circ$ .
- (c) Without calculator, evaluate  $2 \sin 60^\circ \cos 60^\circ$ .

#### **T.V.10.006 Half-angle and t-formulae (where on syllabus)**

- (a) Exact value of  $\sin 60^\circ$ .
- (b) Exact value of  $\cos 60^\circ$ .
- (c) Without calculator, evaluate  $2 \sin 60^\circ \cos 60^\circ$ .

#### **T.V.10.007 Harmonic form $R \sin(\theta \pm \alpha)$ exam pack**

- (a) Exact value of  $\sin 45^\circ$ .
- (b) Exact value of  $\cos 45^\circ$ .
- (c) Without calculator, evaluate  $2 \sin 45^\circ \cos 45^\circ$ .

#### **T.V.10.008 Graphical solutions of trig equations**

- (a) Exact value of  $\sin 60^\circ$ .
- (b) Exact value of  $\cos 45^\circ$ .
- (c) Desmos lab for «Graphical solutions of trig equations»: vary one parameter, capture before/after, write one observation.

#### **T.V.10.009 General solutions in degrees and radians**

- (a) Exact value of  $\sin 45^\circ$ .
- (b) Exact value of  $\cos 45^\circ$ .
- (c) Without calculator, evaluate  $2 \sin 45^\circ \cos 45^\circ$ .

#### **T.V.10.010 Trig equations with multiple angles**

- (a) Exact value of  $\sin 30^\circ$ .

- (b) Exact value of  $\cos 30^\circ$ .
- (c) Without calculator, evaluate  $2 \sin 30^\circ \cos 30^\circ$ .

**T.V.10.011 Factor formulae applications (HL)**

- (a) Exact value of  $\sin 60^\circ$ .
- (b) Exact value of  $\cos 30^\circ$ .
- (c) Without calculator, evaluate  $2 \sin 60^\circ \cos 60^\circ$ .

**T.V.10.012 Desmos: verify identity numerically then prove**

- (a) Exact value of  $\sin 30^\circ$ .
- (b) Exact value of  $\cos 30^\circ$ .
- (c) Desmos lab for «Desmos: verify identity numerically then prove»: vary one parameter, capture before/after, write one observation.

**T.V.10.013 Exam mixed identity + equation mega-set**

- (a) Exact value of  $\sin 60^\circ$ .
- (b) Exact value of  $\cos 45^\circ$ .
- (c) Without calculator, evaluate  $2 \sin 60^\circ \cos 60^\circ$ .

**T.V.10.014 Linear combinations to single sinusoid modelling**

- (a) Exact value of  $\sin 45^\circ$ .
- (b) Exact value of  $\cos 45^\circ$ .
- (c) Desmos lab for «Linear combinations to single sinusoid modelling»: vary one parameter, capture before/after, write one observation.

**T.V.10.015 cos(A–B) expansion applications**

- (a) Exact value of  $\sin 60^\circ$ .
- (b) Exact value of  $\cos 45^\circ$ .
- (c) Without calculator, evaluate  $2 \sin 60^\circ \cos 60^\circ$ .

**T.V.10.016 sin(A+B) in wave interference light model**

- (a) Exact value of  $\sin 30^\circ$ .
- (b) Exact value of  $\cos 60^\circ$ .
- (c) Desmos lab for «sin(A+B) in wave interference light model»: vary one parameter, capture before/after, write one observation.

**T.V.10.017 Product-to-sum formulas light use**

- (a) Exact value of  $\sin 30^\circ$ .
- (b) Exact value of  $\cos 60^\circ$ .
- (c) Without calculator, evaluate  $2 \sin 30^\circ \cos 30^\circ$ .

**T.V.10.018 Solve trig equations on restricted intervals**

- (a) Exact value of  $\sin 45^\circ$ .
- (b) Exact value of  $\cos 45^\circ$ .
- (c) Desmos lab for «Solve trig equations on restricted intervals»: vary one parameter, capture before/after, write one observation.

**T.V.10.019 Identity proof: split into cases strategy**

- (a) Exact value of  $\sin 45^\circ$ .
- (b) Exact value of  $\cos 30^\circ$ .
- (c) Without calculator, evaluate  $2 \sin 45^\circ \cos 45^\circ$ .

**T.V.10.020 Equations: quadratic in sin/cos**

- (a) Exact value of  $\sin 45^\circ$ .
- (b) Exact value of  $\cos 45^\circ$ .
- (c) Without calculator, evaluate  $2 \sin 45^\circ \cos 45^\circ$ .

**T.V.10.021 Compound angle: expand cos(A+B) then simplify**

- (a) Exact value of  $\sin 60^\circ$ .
- (b) Exact value of  $\cos 30^\circ$ .
- (c) Without calculator, evaluate  $2 \sin 60^\circ \cos 60^\circ$ .

**T.V.10.022 Prove sin(A+B) from area or Ptolemy light**

- (a) Exact value of  $\sin 30^\circ$ .
- (b) Exact value of  $\cos 30^\circ$ .

(c) Without calculator, evaluate  $2 \sin 30^\circ \cos 30^\circ$ .

**T.V.10.023 Exam: which identity first? decision drills**

(a) Exact value of  $\sin 30^\circ$ .

(b) Exact value of  $\cos 60^\circ$ .

(c) Without calculator, evaluate  $2 \sin 30^\circ \cos 30^\circ$ .

**T.V.10.024 Hard identity: multi-formula chain**

(a) Exact value of  $\sin 30^\circ$ .

(b) Exact value of  $\cos 45^\circ$ .

(c) Without calculator, evaluate  $2 \sin 30^\circ \cos 30^\circ$ .

**T.V.10.025 Identities exam pack 1**

(a) Exact value of  $\sin 30^\circ$ .

(b) Exact value of  $\cos 45^\circ$ .

(c) Without calculator, evaluate  $2 \sin 30^\circ \cos 30^\circ$ .

**T.V.10.026 Identities exam pack 2**

(a) Exact value of  $\sin 30^\circ$ .

(b) Exact value of  $\cos 30^\circ$ .

(c) Without calculator, evaluate  $2 \sin 30^\circ \cos 30^\circ$ .

**T.V.10.027 Identities exam pack 3**

(a) Exact value of  $\sin 30^\circ$ .

(b) Exact value of  $\cos 60^\circ$ .

(c) Without calculator, evaluate  $2 \sin 30^\circ \cos 30^\circ$ .

**T.V.10.028 Identities exam pack 4**

(a) Exact value of  $\sin 60^\circ$ .

(b) Exact value of  $\cos 30^\circ$ .

(c) Without calculator, evaluate  $2 \sin 60^\circ \cos 60^\circ$ .

**T.V.10.029 Identities exam pack 5**

(a) Exact value of  $\sin 60^\circ$ .

(b) Exact value of  $\cos 30^\circ$ .

(c) Without calculator, evaluate  $2 \sin 60^\circ \cos 60^\circ$ .

**T.V.10.030 Trig equations graphical+algebraic 1**

(a) Exact value of  $\sin 45^\circ$ .

(b) Exact value of  $\cos 45^\circ$ .

(c) Desmos lab for «Trig equations graphical+algebraic 1»: vary one parameter, capture before/after, write one observation.

**T.V.10.031 Trig equations graphical+algebraic 2**

(a) Exact value of  $\sin 60^\circ$ .

(b) Exact value of  $\cos 45^\circ$ .

(c) Desmos lab for «Trig equations graphical+algebraic 2»: vary one parameter, capture before/after, write one observation.

**T.V.10.032 Trig equations graphical+algebraic 3**

(a) Exact value of  $\sin 30^\circ$ .

(b) Exact value of  $\cos 45^\circ$ .

(c) Desmos lab for «Trig equations graphical+algebraic 3»: vary one parameter, capture before/after, write one observation.

**T.V.10.033 Trig equations graphical+algebraic 4**

(a) Exact value of  $\sin 60^\circ$ .

(b) Exact value of  $\cos 30^\circ$ .

(c) Desmos lab for «Trig equations graphical+algebraic 4»: vary one parameter, capture before/after, write one observation.

**T.V.11.001 Coordinate proofs of parallelograms/rhombus**

(a) Notation/definition check for «Coordinate proofs of parallelograms/rhombus»: state the key idea in  $\leq 20$  words.

(b) Standard exam-style: Apply «Coordinate proofs of parallelograms/rhombus» in a worked example that uses the numbers 22 and 9. Show steps.

(c) US dialect challenge: Write and solve one markable question targeting «Coordinate proofs of parallelograms/rhombus» (include answer).

#### **T.V.11.002 Equation of perpendicular bisector loci link**

(a) Notation/definition check for «Equation of perpendicular bisector loci link»: state the key idea in  $\leq 20$  words.

(b) Standard exam-style: Apply «Equation of perpendicular bisector loci link» in a worked example that uses the numbers 12 and 7. Show steps.

(c) UK dialect challenge: Write and solve one markable question targeting «Equation of perpendicular bisector loci link» (include answer).

#### **T.V.11.003 G-GPE: circle and parabola algebraically**

(a) Notation/definition check for «G-GPE: circle and parabola algebraically»: state the key idea in  $\leq 20$  words.

(b) Standard exam-style: Apply «G-GPE: circle and parabola algebraically» in a worked example that uses the numbers 20 and 16. Show steps.

(c) US dialect challenge: Write and solve one markable question targeting «G-GPE: circle and parabola algebraically» (include answer).

#### **T.V.11.004 Partition section formula internal/external**

(a) Notation/definition check for «Partition section formula internal/external»: state the key idea in  $\leq 20$  words.

(b) Standard exam-style: Apply «Partition section formula internal/external» in a worked example that uses the numbers 13 and 12. Show steps.

(c) UK dialect challenge: Write and solve one markable question targeting «Partition section formula internal/external» (include answer).

#### **T.V.12.001 Complete square to identify conics**

(a) Notation/definition check for «Complete square to identify conics»: state the key idea in  $\leq 20$  words.

(b) Standard exam-style: Apply «Complete square to identify conics» in a worked example that uses the numbers 17 and 5. Show steps.

(c) US dialect challenge: Write and solve one markable question targeting «Complete square to identify conics» (include answer).

#### **T.V.12.002 Eccentricity and focus-directrix exam (light)**

(a) Notation/definition check for «Eccentricity and focus-directrix exam (light)»: state the key idea in  $\leq 20$  words.

(b) Standard exam-style: Apply «Eccentricity and focus-directrix exam (light)» in a worked example that uses the numbers 4 and 15. Show steps.

(c) IB dialect challenge: Write and solve one markable question targeting «Eccentricity and focus-directrix exam (light)» (include answer).

#### **T.V.13.001 Describe fully: translation reflection rotation enlargement**

(a) Notation/definition check for «Describe fully: translation reflection rotation enlargement»: state the key idea in  $\leq 20$  words.

(b) Standard exam-style: Apply «Describe fully: translation reflection rotation enlargement» in a worked example that uses the numbers 22 and 14. Show steps.

(c) UK dialect challenge: Write and solve one markable question targeting «Describe fully: translation reflection rotation enlargement» (include answer).

#### **T.V.13.002 Invariant points and lines under transforms**

(a) Notation/definition check for «Invariant points and lines under transforms»: state the key idea in  $\leq 20$  words.

(b) Standard exam-style: Apply «Invariant points and lines under transforms» in a worked example that uses the numbers 14 and 15. Show steps.

(c) UK dialect challenge: Write and solve one markable question targeting «Invariant points and lines under transforms» (include answer).

#### **T.V.13.003 Matrix representation of 2D transformations**

(a)  $\det\left(\begin{bmatrix} 1 & 5 \\ 2 & 4 \end{bmatrix}\right)$ ?

(b)  $\begin{bmatrix} 1 & 5 \\ 2 & 4 \end{bmatrix} \times \mathbf{I}$ ?

(c) Geometry construction lab for «Matrix representation of 2D transformations»: build the figure, measure, verify the claimed relation.

#### **T.V.13.004 Composition of transformations exam order**

(a) Notation/definition check for «Composition of transformations exam order»: state the key idea in  $\leq 20$  words.

(b) Standard exam-style: Apply «Composition of transformations exam order» in a worked example that uses the numbers 10 and 11. Show steps.

(c) UK dialect challenge: Write and solve one markable question targeting «Composition of transformations exam order» (include answer).

#### **T.V.13.005 Enlargement negative SF with centre**

(a) Notation/definition check for «Enlargement negative SF with centre»: state the key idea in  $\leq 20$  words.

(b) Standard exam-style: Apply «Enlargement negative SF with centre» in a worked example that uses the numbers 13 and 5. Show steps.

(c) Geometry construction lab for «Enlargement negative SF with centre»: build the figure, measure, verify the claimed relation.

#### **T.V.14.001 Plans elevations link to volume estimates**

(a) [Plans elevations link to volume estimates] Which view is the plan — front, side, or top?

- (b) Cuboid  $4 \times 2 \times 3$ : how many rectangles in a correct net?
- (c) Describe the front elevation of a square-based pyramid (base 4) standing on its base.

#### **T.V.14.002 G-GMD Cavalieri justification language**

- (a) AP/IB style: Which theorem or definition is central to «G-GMD Cavalieri justification language»?
- (b) Write a justification sentence frame for «G-GMD Cavalieri justification language»: Because \_\_\_, by \_\_\_, therefore \_\_\_.
- (c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «G-GMD Cavalieri justification language», then interpret.

#### **T.V.14.003 Density design problems (mass = density×volume)**

- (a) Notation/definition check for «Density design problems (mass = density×volume)»: state the key idea in  $\leq 20$  words.
- (b) Standard exam-style: Apply «Density design problems (mass = density×volume)» in a worked example that uses the numbers 20 and 3. Show steps.
- (c) UK dialect challenge: Write and solve one markable question targeting «Density design problems (mass = density×volume)» (include answer).

#### **T.V.14.004 Frustum volume (where assessed)**

- (a) Notation/definition check for «Frustum volume (where assessed)»: state the key idea in  $\leq 20$  words.
- (b) Standard exam-style: Apply «Frustum volume (where assessed)» in a worked example that uses the numbers 14 and 14. Show steps.
- (c) UK dialect challenge: Write and solve one markable question targeting «Frustum volume (where assessed)» (include answer).

#### **T.V.14.005 Cross-sections of cubes/pyramids exam**

- (a) Notation/definition check for «Cross-sections of cubes/pyramids exam»: state the key idea in  $\leq 20$  words.
- (b) Standard exam-style: Apply «Cross-sections of cubes/pyramids exam» in a worked example that uses the numbers 18 and 8. Show steps.
- (c) Geometry construction lab for «Cross-sections of cubes/pyramids exam»: build the figure, measure, verify the claimed relation.

#### **T.V.15.001 Exam construction: perpendicular bisector**

- (a) Notation/definition check for «Exam construction: perpendicular bisector»: state the key idea in  $\leq 20$  words.
- (b) Standard exam-style: Apply «Exam construction: perpendicular bisector» in a worked example that uses the numbers 18 and 16. Show steps.
- (c) Geometry construction lab for «Exam construction: perpendicular bisector»: build the figure, measure, verify the claimed relation.

#### **T.V.15.002 Exam construction: angle bisector**

- (a) Notation/definition check for «Exam construction: angle bisector»: state the key idea in  $\leq 20$  words.
- (b) Standard exam-style: Apply «Exam construction: angle bisector» in a worked example that uses the numbers 3 and 8. Show steps.
- (c) Geometry construction lab for «Exam construction: angle bisector»: build the figure, measure, verify the claimed relation.

#### **T.V.15.003 Exam construction: equilateral / $60^\circ$**

- (a) Notation/definition check for «Exam construction: equilateral /  $60^\circ$ »: state the key idea in  $\leq 20$  words.
- (b) Standard exam-style: Apply «Exam construction: equilateral /  $60^\circ$ » in a worked example that uses the numbers 17 and 15. Show steps.
- (c) Geometry construction lab for «Exam construction: equilateral /  $60^\circ$ »: build the figure, measure, verify the claimed relation.

#### **T.V.15.004 Exam construction: regular hexagon in circle**

- (a) Notation/definition check for «Exam construction: regular hexagon in circle»: state the key idea in  $\leq 20$  words.
- (b) Standard exam-style: Apply «Exam construction: regular hexagon in circle» in a worked example that uses the numbers 17 and 11. Show steps.
- (c) Geometry construction lab for «Exam construction: regular hexagon in circle»: build the figure, measure, verify the claimed relation.

#### **T.V.15.005 Exam construction: parallel through point**

- (a) Notation/definition check for «Exam construction: parallel through point»: state the key idea in  $\leq 20$  words.
- (b) Standard exam-style: Apply «Exam construction: parallel through point» in a worked example that uses the numbers 21 and 11. Show steps.
- (c) Geometry construction lab for «Exam construction: parallel through point»: build the figure, measure, verify the claimed relation.

#### **T.V.15.006 Loci constructions combined with regions**

- (a) Notation/definition check for «Loci constructions combined with regions»: state the key idea in  $\leq 20$  words.
- (b) Standard exam-style: Apply «Loci constructions combined with regions» in a worked example that uses the numbers 11 and 3. Show steps.
- (c) Geometry construction lab for «Loci constructions combined with regions»: build the figure, measure, verify the claimed relation.

#### **T.V.15.007 Construct $45^\circ$ and $90^\circ$ exam techniques**

- (a) Notation/definition check for «Construct  $45^\circ$  and  $90^\circ$  exam techniques»: state the key idea in  $\leq 20$  words.
- (b) Standard exam-style: Apply «Construct  $45^\circ$  and  $90^\circ$  exam techniques» in a worked example that uses the numbers 8 and 4. Show steps.

(c) Geometry construction lab for «Construct  $45^\circ$  and  $90^\circ$  exam techniques»: build the figure, measure, verify the claimed relation.

#### **T.V.15.008 Construct triangle given ASA accurately**

(a) Notation/definition check for «Construct triangle given ASA accurately»: state the key idea in  $\leq 20$  words.

(b) Standard exam-style: Apply «Construct triangle given ASA accurately» in a worked example that uses the numbers 20 and 5. Show steps.

(c) Geometry construction lab for «Construct triangle given ASA accurately»: build the figure, measure, verify the claimed relation.

#### **T.V.15.009 Construction accuracy drill 1**

(a) Notation/definition check for «Construction accuracy drill 1»: state the key idea in  $\leq 20$  words.

(b) Standard exam-style: Apply «Construction accuracy drill 1» in a worked example that uses the numbers 7 and 14. Show steps.

(c) Geometry construction lab for «Construction accuracy drill 1»: build the figure, measure, verify the claimed relation.

#### **T.V.15.010 Construction accuracy drill 2**

(a) Notation/definition check for «Construction accuracy drill 2»: state the key idea in  $\leq 20$  words.

(b) Standard exam-style: Apply «Construction accuracy drill 2» in a worked example that uses the numbers 10 and 6. Show steps.

(c) Geometry construction lab for «Construction accuracy drill 2»: build the figure, measure, verify the claimed relation.

#### **T.V.15.011 Construction accuracy drill 3**

(a) Notation/definition check for «Construction accuracy drill 3»: state the key idea in  $\leq 20$  words.

(b) Standard exam-style: Apply «Construction accuracy drill 3» in a worked example that uses the numbers 6 and 10. Show steps.

(c) Geometry construction lab for «Construction accuracy drill 3»: build the figure, measure, verify the claimed relation.

#### **T.V.9.052 Unit circle: verticality drill with readout**

(a) Unit circle: coordinates at  $120^\circ$ ?

(b) Reference angle for  $120^\circ$ ?

(c) Unit-circle lab for «Unit circle: verticality drill with readout»: sample three angles; record (cos,sin).

#### **T.V.9.053 Unit circle: horizontality drill with readout**

(a) Unit circle: coordinates at  $150^\circ$ ?

(b) Reference angle for  $150^\circ$ ?

(c) Unit-circle lab for «Unit circle: horizontality drill with readout»: sample three angles; record (cos,sin).

#### **T.V.9.054 Desmos: rebuild tide model from data points**

(a) Unit circle: coordinates at  $180^\circ$ ?

(b) Reference angle for  $180^\circ$ ?

(c) Desmos lab for «Desmos: rebuild tide model from data points»: vary one parameter, capture before/after, write one observation.

#### **T.V.10.034 Compound angle application: $\sin(75^\circ)$ exact**

(a) Exact value of  $\sin 60^\circ$ .

(b) Exact value of  $\cos 45^\circ$ .

(c) Without calculator, evaluate  $2 \sin 60^\circ \cos 60^\circ$ .

#### **T.V.10.035 Compound angle application: $\cos(15^\circ)$ exact**

(a) Exact value of  $\sin 30^\circ$ .

(b) Exact value of  $\cos 45^\circ$ .

(c) Without calculator, evaluate  $2 \sin 30^\circ \cos 30^\circ$ .

#### **T.V.6.032 Exact values memory palace exam warm-up**

(a) Exact value of  $\sin 30^\circ$ .

(b) Exact value of  $\cos 45^\circ$ .

(c) Without calculator, evaluate  $2 \sin 30^\circ \cos 30^\circ$ .

#### **T.V.7.023 Bearings reverse and multi-leg combo challenge**

(a) Write  $110^\circ$  from north as a three-figure bearing.

(b) Bearing of B from A is  $110$ . Bearing of A from B?

(c) Geometry construction lab for «Bearings reverse and multi-leg combo challenge»: build the figure, measure, verify the claimed relation.

# Era VI — Change

Dark green lab notebook

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## T.VI.1.001 AP FRQ: limit justification from table/graph

- (a) AP/IB style: Which theorem or definition is central to «AP FRQ: limit justification from table/graph»?
- (b) Write a justification sentence frame for «AP FRQ: limit justification from table/graph»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.
- (c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «AP FRQ: limit justification from table/graph», then interpret.

## T.VI.1.002 Formal limit language where AP demands precision

- (a) Evaluate  $\lim_{x \rightarrow 6} (x^2 - 36)/(x - 6)$ .
- (b) IVT:  $f$  continuous on  $[6, 9]$ ,  $f(6) = -2$ ,  $f(9) = 5$ . Justify a root of  $f(x) = 0$ .
- (c) Distinguish a vertical asymptote from a removable hole in one sentence.

## T.VI.1.003 IVT justification FRQ stems

- (a) AP/IB style: Which theorem or definition is central to «IVT justification FRQ stems»?
- (b) Write a justification sentence frame for «IVT justification FRQ stems»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.
- (c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «IVT justification FRQ stems», then interpret.

## T.VI.1.004 Continuity checklist FRQ

- (a) AP/IB style: Which theorem or definition is central to «Continuity checklist FRQ»?
- (b) Write a justification sentence frame for «Continuity checklist FRQ»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.
- (c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Continuity checklist FRQ», then interpret.

## T.VI.1.005 Limits analytically vs numerically GDC

- (a) Evaluate  $\lim_{x \rightarrow 3} (x^2 - 9)/(x - 3)$ .
- (b) IVT:  $f$  continuous on  $[3, 6]$ ,  $f(3) = -2$ ,  $f(6) = 5$ . Justify a root of  $f(x) = 0$ .
- (c) GDC lab for «Limits analytically vs numerically GDC»: record menu path and numerical output.

## T.VI.1.006 Squeeze theorem justification write-ups

- (a) AP/IB style: Which theorem or definition is central to «Squeeze theorem justification write-ups»?
- (b) Write a justification sentence frame for «Squeeze theorem justification write-ups»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.
- (c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Squeeze theorem justification write-ups», then interpret.

## T.VI.1.007 Horizontal asymptotes via limits at infinity FRQ

- (a) AP/IB style: Which theorem or definition is central to «Horizontal asymptotes via limits at infinity FRQ»?
- (b) Write a justification sentence frame for «Horizontal asymptotes via limits at infinity FRQ»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.
- (c) Desmos lab for «Horizontal asymptotes via limits at infinity FRQ»: vary one parameter, capture before/after, write one observation.

## T.VI.1.008 Vertical asymptotes vs holes FRQ language

- (a) AP/IB style: Which theorem or definition is central to «Vertical asymptotes vs holes FRQ language»?
- (b) Write a justification sentence frame for «Vertical asymptotes vs holes FRQ language»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.
- (c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Vertical asymptotes vs holes FRQ language», then interpret.

## T.VI.1.009 Epsilon-N / epsilon-delta light AP+

- (a) Evaluate  $\lim_{x \rightarrow 4} (x^2 - 16)/(x - 4)$ .
- (b) IVT:  $f$  continuous on  $[4, 7]$ ,  $f(4) = -2$ ,  $f(7) = 5$ . Justify a root of  $f(x) = 0$ .
- (c) Distinguish a vertical asymptote from a removable hole in one sentence.

## T.VI.1.010 One-sided limits for piecewise continuity

- (a) Evaluate  $\lim_{x \rightarrow 6} (x^2 - 36)/(x - 6)$ .
- (b) IVT:  $f$  continuous on  $[6, 9]$ ,  $f(6) = -2$ ,  $f(9) = 5$ . Justify a root of  $f(x) = 0$ .
- (c) Distinguish a vertical asymptote from a removable hole in one sentence.

## T.VI.2.001 AP FRQ: definition of derivative as limit

- (a) AP/IB style: Which theorem or definition is central to «AP FRQ: definition of derivative as limit»?
- (b) Write a justification sentence frame for «AP FRQ: definition of derivative as limit»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.
- (c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «AP FRQ: definition of derivative as limit», then interpret.

## T.VI.2.002 Differentiability vs continuity FRQ traps

- (a) AP/IB style: Which theorem or definition is central to «Differentiability vs continuity FRQ traps»?

- (b) Write a justification sentence frame for «Differentiability vs continuity FRQ traps»: Because \_\_\_, by \_\_\_, therefore \_\_\_.
- (c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Differentiability vs continuity FRQ traps», then interpret.

#### **T.VI.2.003 L'Hôpital packaging as named AP topic**

- (a) Differentiate  $y=4x^4$ .
- (b)  $f(x)=4x^4$ . Find  $f'(2)$ .
- (c)  $s(t)=4t^2$  metres. Velocity at  $t=4$  s?

#### **T.VI.2.004 Implicit differentiation AP FRQ**

- (a) AP/IB style: Which theorem or definition is central to «Implicit differentiation AP FRQ»?
- (b) Write a justification sentence frame for «Implicit differentiation AP FRQ»: Because \_\_\_, by \_\_\_, therefore \_\_\_.
- (c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Implicit differentiation AP FRQ», then interpret.

#### **T.VI.2.005 Related rates setup sentence frames**

- (a) Differentiate  $y=5x^2$ .
- (b)  $f(x)=5x^2$ . Find  $f'(2)$ .
- (c)  $s(t)=5t^2$  metres. Velocity at  $t=2$  s?

#### **T.VI.2.006 Calculator-active derivative estimates**

- (a) Differentiate  $y=6x^2$ .
- (b)  $f(x)=6x^2$ . Find  $f'(2)$ .
- (c) GDC lab for «Calculator-active derivative estimates»: record menu path and numerical output.

#### **T.VI.2.007 Chain rule layered AP items**

- (a) AP/IB style: Which theorem or definition is central to «Chain rule layered AP items»?
- (b) Write a justification sentence frame for «Chain rule layered AP items»: Because \_\_\_, by \_\_\_, therefore \_\_\_.
- (c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Chain rule layered AP items», then interpret.

#### **T.VI.2.008 Second derivative from parametric (BC)**

- (a) Differentiate  $y=7x^4$ .
- (b)  $f(x)=7x^4$ . Find  $f'(2)$ .
- (c)  $s(t)=7t^2$  metres. Velocity at  $t=4$  s?

#### **T.VI.2.009 Logarithmic differentiation AP items**

- (a) Differentiate  $y=7x^3$ .
- (b)  $f(x)=7x^3$ . Find  $f'(2)$ .
- (c)  $s(t)=7t^2$  metres. Velocity at  $t=3$  s?

#### **T.VI.2.010 Motion along a line: when particle at rest**

- (a) Differentiate  $y=7x^2$ .
- (b)  $f(x)=7x^2$ . Find  $f'(2)$ .
- (c)  $s(t)=7t^2$  metres. Velocity at  $t=2$  s?

#### **T.VI.2.011 Approximate derivative from table FRQ**

- (a) AP/IB style: Which theorem or definition is central to «Approximate derivative from table FRQ»?
- (b) Write a justification sentence frame for «Approximate derivative from table FRQ»: Because \_\_\_, by \_\_\_, therefore \_\_\_.
- (c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Approximate derivative from table FRQ», then interpret.

#### **T.VI.3.001 MVT justification FRQ stems**

- (a) AP/IB style: Which theorem or definition is central to «MVT justification FRQ stems»?
- (b) Write a justification sentence frame for «MVT justification FRQ stems»: Because \_\_\_, by \_\_\_, therefore \_\_\_.
- (c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «MVT justification FRQ stems», then interpret.

#### **T.VI.3.002 EVT / Extreme Value Theorem justification**

- (a) AP/IB style: Which theorem or definition is central to «EVT / Extreme Value Theorem justification»?
- (b) Write a justification sentence frame for «EVT / Extreme Value Theorem justification»: Because \_\_\_, by \_\_\_, therefore \_\_\_.
- (c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «EVT / Extreme Value Theorem justification», then interpret.

#### **T.VI.3.003 Candidates test write-up discipline**

- (a) Differentiate  $y=4x^2$ .
- (b)  $f(x)=4x^2$ . Find  $f'(2)$ .

(c)  $s(t)=4t^2$  metres. Velocity at  $t=2$  s?

#### **T.VI.3.004 $f, f', f''$ sign chart FRQ classic**

(a) AP/IB style: Which theorem or definition is central to « $f, f', f''$  sign chart FRQ classic»?

(b) Write a justification sentence frame for « $f, f', f''$  sign chart FRQ classic»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.

(c) Desmos lab for « $f, f', f''$  sign chart FRQ classic»: vary one parameter, capture before/after, write one observation.

#### **T.VI.3.005 Optimization AP contextual FRQ**

(a) AP/IB style: Which theorem or definition is central to «Optimization AP contextual FRQ»?

(b) Write a justification sentence frame for «Optimization AP contextual FRQ»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.

(c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Optimization AP contextual FRQ», then interpret.

#### **T.VI.3.006 Linearisation / tangent approx error talk**

(a) Differentiate  $y=5x^4$ .

(b)  $f(x)=5x^4$ . Find  $f'(2)$ .

(c)  $s(t)=5t^2$  metres. Velocity at  $t=4$  s?

#### **T.VI.3.007 Particle motion: velocity/acceleration FRQ**

(a) AP/IB style: Which theorem or definition is central to «Particle motion: velocity/acceleration FRQ»?

(b) Write a justification sentence frame for «Particle motion: velocity/acceleration FRQ»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.

(c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Particle motion: velocity/acceleration FRQ», then interpret.

#### **T.VI.3.008 Newton's method AP/IB numerical**

(a) Differentiate  $y=6x^5$ .

(b)  $f(x)=6x^5$ . Find  $f'(2)$ .

(c) GDC lab for «Newton's method AP/IB numerical»: record menu path and numerical output.

#### **T.VI.3.009 Related rates classic cone/ladder/shadow**

(a) Differentiate  $y=2x^2$ .

(b)  $f(x)=2x^2$ . Find  $f'(2)$ .

(c)  $s(t)=2t^2$  metres. Velocity at  $t=2$  s?

#### **T.VI.3.010 Concavity and inflection FRQ write-ups**

(a) AP/IB style: Which theorem or definition is central to «Concavity and inflection FRQ write-ups»?

(b) Write a justification sentence frame for «Concavity and inflection FRQ write-ups»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.

(c) Desmos lab for «Concavity and inflection FRQ write-ups»: vary one parameter, capture before/after, write one observation.

#### **T.VI.3.011 L'Hôpital repeated applications**

(a) Differentiate  $y=3x^4$ .

(b)  $f(x)=3x^4$ . Find  $f'(2)$ .

(c)  $s(t)=3t^2$  metres. Velocity at  $t=4$  s?

#### **T.VI.3.012 Graphing calculator: find zeros of $f'$**

(a) Differentiate  $y=4x^3$ .

(b)  $f(x)=4x^3$ . Find  $f'(2)$ .

(c) GDC lab for «Graphing calculator: find zeros of  $f'$ »: record menu path and numerical output.

#### **T.VI.3.013 AP calculator-active vs inactive practice modes**

(a) Differentiate  $y=5x^3$ .

(b)  $f(x)=5x^3$ . Find  $f'(2)$ .

(c) GDC lab for «AP calculator-active vs inactive practice modes»: record menu path and numerical output.

#### **T.VI.3.014 Absolute extrema on closed interval FRQ**

(a) AP/IB style: Which theorem or definition is central to «Absolute extrema on closed interval FRQ»?

(b) Write a justification sentence frame for «Absolute extrema on closed interval FRQ»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.

(c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Absolute extrema on closed interval FRQ», then interpret.

#### **T.VI.4.001 Riemann sum notation left/right/mid FRQ**

(a) AP/IB style: Which theorem or definition is central to «Riemann sum notation left/right/mid FRQ»?

(b) Write a justification sentence frame for «Riemann sum notation left/right/mid FRQ»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.

(c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Riemann sum notation left/right/mid FRQ», then interpret.

#### T.VI.4.002 FTC graphical accumulation FRQ

- (a) AP/IB style: Which theorem or definition is central to «FTC graphical accumulation FRQ»?
- (b) Write a justification sentence frame for «FTC graphical accumulation FRQ»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.
- (c) Graph lab for «FTC graphical accumulation FRQ»: plot, mark key points, state the trend.

#### T.VI.4.003 Average value theorem applications

- (a)  $\int 5x^4 dx$ .
- (b) Evaluate  $\int_{-1}^1 5x dx$ .
- (c) If  $v(t)$  is velocity, what does  $\int_{-1}^1 v(t) dt$  represent?

#### T.VI.4.004 Definite integral properties FRQ

- (a) AP/IB style: Which theorem or definition is central to «Definite integral properties FRQ»?
- (b) Write a justification sentence frame for «Definite integral properties FRQ»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.
- (c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Definite integral properties FRQ», then interpret.

#### T.VI.4.005 Trapezoidal rule approximation FRQ

- (a) AP/IB style: Which theorem or definition is central to «Trapezoidal rule approximation FRQ»?
- (b) Write a justification sentence frame for «Trapezoidal rule approximation FRQ»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.
- (c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Trapezoidal rule approximation FRQ», then interpret.

#### T.VI.4.006 FTC derivative of integral with variable limits

- (a)  $\int 6x^4 dx$ .
- (b) Evaluate  $\int_{-1}^1 6x dx$ .
- (c) If  $v(t)$  is velocity, what does  $\int_{-1}^1 v(t) dt$  represent?

#### T.VI.4.007 AP numerical integration on GDC

- (a)  $\int 5x^3 dx$ .
- (b) Evaluate  $\int_{-1}^1 5x dx$ .
- (c) GDC lab for «AP numerical integration on GDC»: record menu path and numerical output.

#### T.VI.4.008 Accumulation from rate-in rate-out FRQ

- (a) AP/IB style: Which theorem or definition is central to «Accumulation from rate-in rate-out FRQ»?
- (b) Write a justification sentence frame for «Accumulation from rate-in rate-out FRQ»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.
- (c) Graph lab for «Accumulation from rate-in rate-out FRQ»: plot, mark key points, state the trend.

#### T.VI.4.009 Signed area vs total area language

- (a)  $\int 6x^2 dx$ .
- (b) Evaluate  $\int_{-1}^1 6x dx$ .
- (c) Graph lab for «Signed area vs total area language»: plot, mark key points, state the trend.

#### T.VI.5.001 Integration by parts IB AA / AP BC

- (a)  $\int 5x^3 dx$ .
- (b) Evaluate  $\int_{-1}^1 5x dx$ .
- (c) If  $v(t)$  is velocity, what does  $\int_{-1}^1 v(t) dt$  represent?

#### T.VI.5.002 Partial fractions integration exam

- (a)  $\int 2x^4 dx$ .
- (b) Evaluate  $\int_{-1}^1 2x dx$ .
- (c) If  $v(t)$  is velocity, what does  $\int_{-1}^1 v(t) dt$  represent?

#### T.VI.5.003 Improper integrals convergence tests light

- (a)  $\int 4x^4 dx$ .
- (b) Evaluate  $\int_{-1}^1 4x dx$ .
- (c) If  $v(t)$  is velocity, what does  $\int_{-1}^1 v(t) dt$  represent?

#### T.VI.5.004 Trigonometric integrals powers of sin/cos

- (a)  $\int 2x^1 dx$ .
- (b) Evaluate  $\int_{-1}^1 2x dx$ .
- (c) If  $v(t)$  is velocity, what does  $\int_{-1}^1 v(t) dt$  represent?

#### T.VI.5.005 Trigonometric substitution light (BC/HL)

- (a)  $\int 4x^4 dx$ .
- (b) Evaluate  $\int_{-1}^1 4x dx$ .
- (c) If  $v(t)$  is velocity, what does  $\int_{-1}^1 v(t) dt$  represent?

#### T.VI.6.001 Area between curves AP FRQ classic

- (a) 150 cm to nearest 10 cm. Write the error interval for length  $L$ .
- (b)  $w=6.0$  cm,  $l=9.5$  cm each to nearest 0.1 cm. Upper bound of perimeter?
- (c) Maximise speed = distance/time given rounded measurements. Which bounds?

#### T.VI.6.002 Disk/washer volume FRQ

- (a) AP/IB style: Which theorem or definition is central to «Disk/washer volume FRQ»?
- (b) Write a justification sentence frame for «Disk/washer volume FRQ»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.
- (c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Disk/washer volume FRQ», then interpret.

#### T.VI.6.003 Known cross-section volume FRQ

- (a) AP/IB style: Which theorem or definition is central to «Known cross-section volume FRQ»?
- (b) Write a justification sentence frame for «Known cross-section volume FRQ»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.
- (c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Known cross-section volume FRQ», then interpret.

#### T.VI.6.004 Net change theorem worded FRQ

- (a) AP/IB style: Which theorem or definition is central to «Net change theorem worded FRQ»?
- (b) Write a justification sentence frame for «Net change theorem worded FRQ»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.
- (c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Net change theorem worded FRQ», then interpret.

#### T.VI.6.005 Arc length setup (BC)

- (a) Differentiate  $y=5x^5$ .
- (b)  $f(x)=5x^5$ . Find  $f'(2)$ .
- (c)  $s(t)=5t^2$  metres. Velocity at  $t=5$  s?

#### T.VI.6.006 Shell method when washer awkward

- (a) Differentiate  $y=6x^4$ .
- (b)  $f(x)=6x^4$ . Find  $f'(2)$ .
- (c)  $s(t)=6t^2$  metres. Velocity at  $t=4$  s?

#### T.VI.6.007 Work pumping liquid FRQ style

- (a) AP/IB style: Which theorem or definition is central to «Work pumping liquid FRQ style»?
- (b) Write a justification sentence frame for «Work pumping liquid FRQ style»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.
- (c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Work pumping liquid FRQ style», then interpret.

#### T.VI.6.008 Probability density integral applications

- (a) Differentiate  $y=5x^5$ .
- (b)  $f(x)=5x^5$ . Find  $f'(2)$ .
- (c)  $s(t)=5t^2$  metres. Velocity at  $t=5$  s?

#### T.VI.6.009 Displacement from velocity graph areas

- (a) Differentiate  $y=2x^4$ .
- (b)  $f(x)=2x^4$ . Find  $f'(2)$ .
- (c) Graph lab for «Displacement from velocity graph areas»: plot, mark key points, state the trend.

#### T.VI.7.001 Slope fields sketch and match FRQ

- (a) AP/IB style: Which theorem or definition is central to «Slope fields sketch and match FRQ»?
- (b) Write a justification sentence frame for «Slope fields sketch and match FRQ»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.
- (c) Desmos lab for «Slope fields sketch and match FRQ»: vary one parameter, capture before/after, write one observation.

#### T.VI.7.002 Euler's method table FRQ

- (a) AP/IB style: Which theorem or definition is central to «Euler's method table FRQ»?
- (b) Write a justification sentence frame for «Euler's method table FRQ»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.
- (c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Euler's method table FRQ», then interpret.

#### T.VI.7.003 Logistic DE: $dP/dt=kP(M-P)$ exam pack

- (a) Differentiate  $y=6x^4$ .

- (b)  $f(x)=6x^4$ . Find  $f'(2)$ .
- (c)  $s(t)=6t^2$  metres. Velocity at  $t=4$  s?

#### **T.VI.7.004 Separable DE with initial condition FRQ**

- (a) AP/IB style: Which theorem or definition is central to «Separable DE with initial condition FRQ»?
- (b) Write a justification sentence frame for «Separable DE with initial condition FRQ»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.
- (c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Separable DE with initial condition FRQ», then interpret.

#### **T.VI.7.005 Growth/decay DE contextual FRQ**

- (a) AP/IB style: Which theorem or definition is central to «Growth/decay DE contextual FRQ»?
- (b) Write a justification sentence frame for «Growth/decay DE contextual FRQ»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.
- (c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Growth/decay DE contextual FRQ», then interpret.

#### **T.VI.7.006 Newton cooling DE exam**

- (a) Differentiate  $y=7x^3$ .
- (b)  $f(x)=7x^3$ . Find  $f'(2)$ .
- (c)  $s(t)=7t^2$  metres. Velocity at  $t=3$  s?

#### **T.VI.7.007 Mixing problems DE setup**

- (a) Differentiate  $y=7x^3$ .
- (b)  $f(x)=7x^3$ . Find  $f'(2)$ .
- (c)  $s(t)=7t^2$  metres. Velocity at  $t=3$  s?

#### **T.VI.7.008 Equilibrium solutions stability talk**

- (a) Differentiate  $y=4x^5$ .
- (b)  $f(x)=4x^5$ . Find  $f'(2)$ .
- (c)  $s(t)=4t^2$  metres. Velocity at  $t=5$  s?

#### **T.VI.7.009 IB DE modelling emphasis pack**

- (a) Differentiate  $y=7x^4$ .
- (b)  $f(x)=7x^4$ . Find  $f'(2)$ .
- (c)  $s(t)=7t^2$  metres. Velocity at  $t=4$  s?

#### **T.VI.7.010 Particular vs general solution language**

- (a) Differentiate  $y=2x^4$ .
- (b)  $f(x)=2x^4$ . Find  $f'(2)$ .
- (c)  $s(t)=2t^2$  metres. Velocity at  $t=4$  s?

#### **T.VI.7.011 Euler vs analytic solution comparison**

- (a) Differentiate  $y=5x^5$ .
- (b)  $f(x)=5x^5$ . Find  $f'(2)$ .
- (c) GDC lab for «Euler vs analytic solution comparison»: record menu path and numerical output.

#### **T.VI.7.012 Logistic carrying capacity interpretation**

- (a) Differentiate  $y=3x^4$ .
- (b)  $f(x)=3x^4$ . Find  $f'(2)$ .
- (c) Desmos lab for «Logistic carrying capacity interpretation»: vary one parameter, capture before/after, write one observation.

#### **T.VI.7.013 Slope field: draw solution through a point**

- (a) Differentiate  $y=4x^2$ .
- (b)  $f(x)=4x^2$ . Find  $f'(2)$ .
- (c) Desmos lab for «Slope field: draw solution through a point»: vary one parameter, capture before/after, write one observation.

#### **T.VI.7.014 Euler method stepped table 1**

- (a) Differentiate  $y=5x^2$ .
- (b)  $f(x)=5x^2$ . Find  $f'(2)$ .
- (c)  $s(t)=5t^2$  metres. Velocity at  $t=2$  s?

#### **T.VI.7.015 Logistic model interpretation 1**

- (a) Differentiate  $y=3x^3$ .
- (b)  $f(x)=3x^3$ . Find  $f'(2)$ .

(c) Desmos lab for «Logistic model interpretation 1»: vary one parameter, capture before/after, write one observation.

#### T.VI.7.016 Euler method stepped table 2

(a) Differentiate  $y=4x^2$ .

(b)  $f(x)=4x^2$ . Find  $f'(2)$ .

(c)  $s(t)=4t^2$  metres. Velocity at  $t=2$  s?

#### T.VI.7.017 Logistic model interpretation 2

(a) Differentiate  $y=2x^3$ .

(b)  $f(x)=2x^3$ . Find  $f'(2)$ .

(c) Desmos lab for «Logistic model interpretation 2»: vary one parameter, capture before/after, write one observation.

#### T.VI.7.018 Euler method stepped table 3

(a) Differentiate  $y=7x^3$ .

(b)  $f(x)=7x^3$ . Find  $f'(2)$ .

(c)  $s(t)=7t^2$  metres. Velocity at  $t=3$  s?

#### T.VI.7.019 Logistic model interpretation 3

(a) Differentiate  $y=6x^4$ .

(b)  $f(x)=6x^4$ . Find  $f'(2)$ .

(c) Desmos lab for «Logistic model interpretation 3»: vary one parameter, capture before/after, write one observation.

#### T.VI.8.001 Parametric derivatives $dy/dx$ FRQ

(a) AP/IB style: Which theorem or definition is central to «Parametric derivatives  $dy/dx$  FRQ»?

(b) Write a justification sentence frame for «Parametric derivatives  $dy/dx$  FRQ»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.

(c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Parametric derivatives  $dy/dx$  FRQ», then interpret.

#### T.VI.8.002 Parametric arc length FRQ

(a) AP/IB style: Which theorem or definition is central to «Parametric arc length FRQ»?

(b) Write a justification sentence frame for «Parametric arc length FRQ»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.

(c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Parametric arc length FRQ», then interpret.

#### T.VI.8.003 Particle path parametric motion FRQ

(a) AP/IB style: Which theorem or definition is central to «Particle path parametric motion FRQ»?

(b) Write a justification sentence frame for «Particle path parametric motion FRQ»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.

(c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Particle path parametric motion FRQ», then interpret.

#### T.VI.8.004 Polar area $\frac{1}{2}\int r^2 d\theta$ FRQ

(a) 170 cm to nearest 10 cm. Write the error interval for length L.

(b)  $w=4.8$  cm,  $l=11.2$  cm each to nearest 0.1 cm. Upper bound of perimeter?

(c) Maximise speed = distance/time given rounded measurements. Which bounds?

#### T.VI.8.005 Polar slope $dy/dx$ in polar form

(a)  $\int 2x^3 dx$ .

(b) Evaluate  $\int_{-1}^1 2x dx$ .

(c) If  $v(t)$  is velocity, what does  $\int_{-1}^1 v(t) dt$  represent?

#### T.VI.8.006 Common polar curves recognition pack

(a)  $\int 6x^1 dx$ .

(b) Evaluate  $\int_{-1}^1 6x dx$ .

(c) Desmos lab for «Common polar curves recognition pack»: vary one parameter, capture before/after, write one observation.

#### T.VI.8.007 Convert polar $\leftrightarrow$ cartesian exam

(a)  $\int 2x^1 dx$ .

(b) Evaluate  $\int_{-1}^1 2x dx$ .

(c) If  $v(t)$  is velocity, what does  $\int_{-1}^1 v(t) dt$  represent?

#### T.VI.8.008 Area between polar curves FRQ

(a) AP/IB style: Which theorem or definition is central to «Area between polar curves FRQ»?

(b) Write a justification sentence frame for «Area between polar curves FRQ»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.

(c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Area between polar curves FRQ», then interpret.

**T.VI.8.009 Desmos polar graphing exploration**

- (a)  $\int 5x^1 dx$ .
- (b) Evaluate  $\int_{-1}^1 5x dx$ .
- (c) Desmos lab for «Desmos polar graphing exploration»: vary one parameter, capture before/after, write one observation.

**T.VI.8.010 Vector-valued position/velocity/acceleration**

- (a)  $a=(1, 4)$ ,  $b=(4, 5)$ . Find  $a+b$  as a column vector.
- (b)  $|a|$  for  $a=(1, 4)$ ?
- (c) Position  $A=(1,4)$ ,  $B=(5,9)$ . Vector  $AB$ ?

**T.VI.8.011 Polar to area of roses petals count**

- (a)  $\int 5x^2 dx$ .
- (b) Evaluate  $\int_{-1}^1 5x dx$ .
- (c) Desmos lab for «Polar to area of roses petals count»: vary one parameter, capture before/after, write one observation.

**T.VI.8.012 Eliminate parameter then differentiate check**

- (a)  $\int 2x^1 dx$ .
- (b) Evaluate  $\int_{-1}^1 2x dx$ .
- (c) If  $v(t)$  is velocity, what does  $\int_{-1}^1 v(t) dt$  represent?

**T.VI.8.013 Polar area shared regions careful bounds**

- (a) 150 cm to nearest 10 cm. Write the error interval for length  $L$ .
- (b)  $w=5.8$  cm,  $l=10.5$  cm each to nearest 0.1 cm. Upper bound of perimeter?
- (c) Desmos lab for «Polar area shared regions careful bounds»: vary one parameter, capture before/after, write one observation.

**T.VI.8.014 Polar/parametric FRQ segment 1**

- (a) AP/IB style: Which theorem or definition is central to «Polar/parametric FRQ segment 1»?
- (b) Write a justification sentence frame for «Polar/parametric FRQ segment 1»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.
- (c) Desmos lab for «Polar/parametric FRQ segment 1»: vary one parameter, capture before/after, write one observation.

**T.VI.8.015 Polar/parametric FRQ segment 2**

- (a) AP/IB style: Which theorem or definition is central to «Polar/parametric FRQ segment 2»?
- (b) Write a justification sentence frame for «Polar/parametric FRQ segment 2»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.
- (c) Desmos lab for «Polar/parametric FRQ segment 2»: vary one parameter, capture before/after, write one observation.

**T.VI.8.016 Polar/parametric FRQ segment 3**

- (a) AP/IB style: Which theorem or definition is central to «Polar/parametric FRQ segment 3»?
- (b) Write a justification sentence frame for «Polar/parametric FRQ segment 3»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.
- (c) Desmos lab for «Polar/parametric FRQ segment 3»: vary one parameter, capture before/after, write one observation.

**T.VI.8.017 Polar/parametric FRQ segment 4**

- (a) AP/IB style: Which theorem or definition is central to «Polar/parametric FRQ segment 4»?
- (b) Write a justification sentence frame for «Polar/parametric FRQ segment 4»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.
- (c) Desmos lab for «Polar/parametric FRQ segment 4»: vary one parameter, capture before/after, write one observation.

**T.VI.9.001 Series tests battery: nth-term first**

- (a) nth term of 2, 4, 6, 8, ...
- (b) 20th term:  $a=2$ ,  $d=2$ .
- (c)  $S_n$  for  $a=2$ ,  $d=2$ .

**T.VI.9.002 Integral test with remainder estimate**

- (a) 120 cm to nearest 10 cm. Write the error interval for length  $L$ .
- (b)  $w=6.2$  cm,  $l=9.8$  cm each to nearest 0.1 cm. Upper bound of perimeter?
- (c) Maximise speed = distance/time given rounded measurements. Which bounds?

**T.VI.9.003 Comparison and limit comparison drills**

- (a) Evaluate  $\lim_{x \rightarrow 6} (x^2-36)/(x-6)$ .
- (b) IVT:  $f$  continuous on  $[6,9]$ ,  $f(6)=-2$ ,  $f(9)=5$ . Justify a root of  $f(x)=0$ .
- (c) Distinguish a vertical asymptote from a removable hole in one sentence.

**T.VI.9.004 Alternating series test + error bound**

- (a) 170 cm to nearest 10 cm. Write the error interval for length L.
- (b)  $w=4.5$  cm,  $l=9.3$  cm each to nearest 0.1 cm. Upper bound of perimeter?
- (c) Maximise speed = distance/time given rounded measurements. Which bounds?

#### T.VI.9.005 Ratio test radius of convergence workflow

- (a) nth term of 8, 15, 22, 29, ...
- (b) 20th term:  $a=8$ ,  $d=7$ .
- (c)  $S_{\infty}$  for  $a=8$ ,  $d=7$ .

#### T.VI.9.006 Absolute vs conditional convergence FRQ

- (a) AP/IB style: Which theorem or definition is central to «Absolute vs conditional convergence FRQ»?
- (b) Write a justification sentence frame for «Absolute vs conditional convergence FRQ»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.
- (c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Absolute vs conditional convergence FRQ», then interpret.

#### T.VI.9.007 Which test? decision tree mega-drill

- (a) nth term of 7, 11, 15, 19, ...
- (b) 20th term:  $a=7$ ,  $d=4$ .
- (c)  $S_{\infty}$  for  $a=7$ ,  $d=4$ .

#### T.VI.9.008 p-series and geometric recognition speed

- (a) nth term of 2, 6, 10, 14, ...
- (b) 20th term:  $a=2$ ,  $d=4$ .
- (c)  $S_{\infty}$  for  $a=2$ ,  $d=4$ .

#### T.VI.9.009 Root test when ratio inconclusive

- (a) nth term of 8, 12, 16, 20, ...
- (b) 20th term:  $a=8$ ,  $d=4$ .
- (c)  $S_{\infty}$  for  $a=8$ ,  $d=4$ .

#### T.VI.9.010 Telescoping series recognition

- (a) nth term of 3, 7, 11, 15, ...
- (b) 20th term:  $a=3$ ,  $d=4$ .
- (c)  $S_{\infty}$  for  $a=3$ ,  $d=4$ .

#### T.VI.9.011 Limit comparison with asymptotic equivalence

- (a) Evaluate  $\lim_{x \rightarrow 5} (x^2-25)/(x-5)$ .
- (b) IVT:  $f$  continuous on  $[5,8]$ ,  $f(5)=-2$ ,  $f(8)=5$ . Justify a root of  $f(x)=0$ .
- (c) Distinguish a vertical asymptote from a removable hole in one sentence.

#### T.VI.9.012 Series tests mixed battery 1

- (a) nth term of 4, 9, 14, 19, ...
- (b) 20th term:  $a=4$ ,  $d=5$ .
- (c)  $S_{\infty}$  for  $a=4$ ,  $d=5$ .

#### T.VI.9.013 Series tests mixed battery 2

- (a) nth term of 2, 5, 8, 11, ...
- (b) 20th term:  $a=2$ ,  $d=3$ .
- (c)  $S_{\infty}$  for  $a=2$ ,  $d=3$ .

#### T.VI.9.014 Series tests mixed battery 3

- (a) nth term of 5, 11, 17, 23, ...
- (b) 20th term:  $a=5$ ,  $d=6$ .
- (c)  $S_{\infty}$  for  $a=5$ ,  $d=6$ .

#### T.VI.9.015 Series tests mixed battery 4

- (a) nth term of 7, 14, 21, 28, ...
- (b) 20th term:  $a=7$ ,  $d=7$ .
- (c)  $S_{\infty}$  for  $a=7$ ,  $d=7$ .

#### T.VI.9.016 Series tests mixed battery 5

- (a) nth term of 8, 13, 18, 23, ...

(b) 20th term:  $a=8, d=5$ .

(c)  $S_n$  for  $a=8, d=5$ .

#### T.VI.10.001 Taylor/Maclaurin polynomial construction FRQ

(a) AP/IB style: Which theorem or definition is central to «Taylor/Maclaurin polynomial construction FRQ»?

(b) Write a justification sentence frame for «Taylor/Maclaurin polynomial construction FRQ»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.

(c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Taylor/Maclaurin polynomial construction FRQ», then interpret.

#### T.VI.10.002 Lagrange error bound FRQ

(a) 80 cm to nearest 10 cm. Write the error interval for length L.

(b)  $w=2.6$  cm,  $l=9.6$  cm each to nearest 0.1 cm. Upper bound of perimeter?

(c) Maximise speed = distance/time given rounded measurements. Which bounds?

#### T.VI.10.003 Alternating series error for Taylor

(a) nth term of 3, 10, 17, 24, ...

(b) 20th term:  $a=3, d=7$ .

(c)  $S_n$  for  $a=3, d=7$ .

#### T.VI.10.004 Manipulate known Maclaurin series

(a) nth term of 7, 10, 13, 16, ...

(b) 20th term:  $a=7, d=3$ .

(c)  $S_n$  for  $a=7, d=3$ .

#### T.VI.10.005 Interval of convergence endpoint checks

(a) nth term of 6, 10, 14, 18, ...

(b) 20th term:  $a=6, d=4$ .

(c)  $S_n$  for  $a=6, d=4$ .

#### T.VI.10.006 Series for $e^x$ , $\sin x$ , $\cos x$ , $1/(1-x)$ fluency

(a) nth term of 3, 5, 7, 9, ...

(b) 20th term:  $a=3, d=2$ .

(c)  $S_n$  for  $a=3, d=2$ .

#### T.VI.10.007 Approximate integrals via series FRQ

(a) AP/IB style: Which theorem or definition is central to «Approximate integrals via series FRQ»?

(b) Write a justification sentence frame for «Approximate integrals via series FRQ»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.

(c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Approximate integrals via series FRQ», then interpret.

#### T.VI.10.008 IB AA HL Maclaurin exam packaging

(a) nth term of 5, 7, 9, 11, ...

(b) 20th term:  $a=5, d=2$ .

(c)  $S_n$  for  $a=5, d=2$ .

#### T.VI.10.009 Binomial series expansion $(1+x)^k$

(a) nth term of 2, 7, 12, 17, ...

(b) 20th term:  $a=2, d=5$ .

(c)  $S_n$  for  $a=2, d=5$ .

#### T.VI.10.010 Taylor series for $\ln(1+x)$ and $\arctan$

(a) nth term of 4, 7, 10, 13, ...

(b) 20th term:  $a=4, d=3$ .

(c)  $S_n$  for  $a=4, d=3$ .

#### T.VI.10.011 Error bounds choose Lagrange vs alternating

(a) 120 cm to nearest 10 cm. Write the error interval for length L.

(b)  $w=4.2$  cm,  $l=9.1$  cm each to nearest 0.1 cm. Upper bound of perimeter?

(c) Maximise speed = distance/time given rounded measurements. Which bounds?

#### T.VI.10.012 Replace $x$ by $x^2$ etc. in Maclaurin

(a) nth term of 4, 6, 8, 10, ...

(b) 20th term:  $a=4, d=2$ .

(c)  $S_{\text{max}}$  for  $a=4$ ,  $d=2$ .

#### **T.VI.10.013 Taylor error bound practice 1**

- (a) 170 cm to nearest 10 cm. Write the error interval for length  $L$ .
- (b)  $w=4.6$  cm,  $l=6.8$  cm each to nearest 0.1 cm. Upper bound of perimeter?
- (c) Maximise speed = distance/time given rounded measurements. Which bounds?

#### **T.VI.10.014 Taylor error bound practice 2**

- (a) 40 cm to nearest 10 cm. Write the error interval for length  $L$ .
- (b)  $w=4.2$  cm,  $l=6.9$  cm each to nearest 0.1 cm. Upper bound of perimeter?
- (c) Maximise speed = distance/time given rounded measurements. Which bounds?

#### **T.VI.10.015 Taylor error bound practice 3**

- (a) 150 cm to nearest 10 cm. Write the error interval for length  $L$ .
- (b)  $w=6.0$  cm,  $l=9.3$  cm each to nearest 0.1 cm. Upper bound of perimeter?
- (c) Maximise speed = distance/time given rounded measurements. Which bounds?

#### **T.VI.11.001 Study design: experiment vs observational**

- (a)  $X \sim N(50, 5^2)$ . Approx  $P(45 < X < 55)$  (empirical rule)?
- (b) z-score of  $x=60$  ( $\mu=50$ ,  $\sigma=5$ )?
- (c) GDC: outline finding  $P(X > 55)$  for  $X \sim N(50, 5^2)$ .

#### **T.VI.11.002 Sampling methods named pack**

- (a)  $X \sim N(70, 10^2)$ . Approx  $P(60 < X < 80)$  (empirical rule)?
- (b) z-score of  $x=90$  ( $\mu=70$ ,  $\sigma=10$ )?
- (c) GDC: outline finding  $P(X > 80)$  for  $X \sim N(70, 10^2)$ .

#### **T.VI.11.003 Bias types: undercoverage, nonresponse, response**

- (a)  $X \sim N(70, 10^2)$ . Approx  $P(60 < X < 80)$  (empirical rule)?
- (b) z-score of  $x=90$  ( $\mu=70$ ,  $\sigma=10$ )?
- (c) GDC: outline finding  $P(X > 80)$  for  $X \sim N(70, 10^2)$ .

#### **T.VI.11.004 Simulation-based inference pedagogy**

- (a)  $X \sim N(50, 5^2)$ . Approx  $P(45 < X < 55)$  (empirical rule)?
- (b) z-score of  $x=60$  ( $\mu=50$ ,  $\sigma=5$ )?
- (c) GDC: outline finding  $P(X > 55)$  for  $X \sim N(50, 5^2)$ .

#### **T.VI.11.005 One-proportion z-interval procedure**

- (a)  $X \sim N(50, 5^2)$ . Approx  $P(45 < X < 55)$  (empirical rule)?
- (b) z-score of  $x=60$  ( $\mu=50$ ,  $\sigma=5$ )?
- (c) GDC: outline finding  $P(X > 55)$  for  $X \sim N(50, 5^2)$ .

#### **T.VI.11.006 One-proportion z-test procedure**

- (a)  $X \sim N(70, 10^2)$ . Approx  $P(60 < X < 80)$  (empirical rule)?
- (b) z-score of  $x=90$  ( $\mu=70$ ,  $\sigma=10$ )?
- (c) GDC: outline finding  $P(X > 80)$  for  $X \sim N(70, 10^2)$ .

#### **T.VI.11.007 Two-proportion z procedures**

- (a)  $X \sim N(100, 15^2)$ . Approx  $P(85 < X < 115)$  (empirical rule)?
- (b) z-score of  $x=130$  ( $\mu=100$ ,  $\sigma=15$ )?
- (c) GDC: outline finding  $P(X > 115)$  for  $X \sim N(100, 15^2)$ .

#### **T.VI.11.008 One-sample t-interval for mean**

- (a)  $X \sim N(100, 15^2)$ . Approx  $P(85 < X < 115)$  (empirical rule)?
- (b) z-score of  $x=130$  ( $\mu=100$ ,  $\sigma=15$ )?
- (c) GDC: outline finding  $P(X > 115)$  for  $X \sim N(100, 15^2)$ .

#### **T.VI.11.009 One-sample t-test for mean**

- (a)  $X \sim N(70, 10^2)$ . Approx  $P(60 < X < 80)$  (empirical rule)?
- (b) z-score of  $x=90$  ( $\mu=70$ ,  $\sigma=10$ )?
- (c) GDC: outline finding  $P(X > 80)$  for  $X \sim N(70, 10^2)$ .

**T.VI.11.010 Two-sample t procedures / paired data**

- (a)  $X \sim N(70, 10^2)$ . Approx  $P(60 < X < 80)$  (empirical rule)?
- (b) z-score of  $x=90$  ( $\mu=70$ ,  $\sigma=10$ )?
- (c) GDC: outline finding  $P(X > 80)$  for  $X \sim N(70, 10^2)$ .

**T.VI.11.011 Chi-square GOF test**

- (a)  $X \sim N(100, 15^2)$ . Approx  $P(85 < X < 115)$  (empirical rule)?
- (b) z-score of  $x=130$  ( $\mu=100$ ,  $\sigma=15$ )?
- (c) GDC: outline finding  $P(X > 115)$  for  $X \sim N(100, 15^2)$ .

**T.VI.11.012 Chi-square independence / homogeneity**

- (a)  $X \sim N(70, 10^2)$ . Approx  $P(60 < X < 80)$  (empirical rule)?
- (b) z-score of  $x=90$  ( $\mu=70$ ,  $\sigma=10$ )?
- (c) GDC: outline finding  $P(X > 80)$  for  $X \sim N(70, 10^2)$ .

**T.VI.11.013 Slope inference / CI for regression slope**

- (a)  $X \sim N(70, 10^2)$ . Approx  $P(60 < X < 80)$  (empirical rule)?
- (b) z-score of  $x=90$  ( $\mu=70$ ,  $\sigma=10$ )?
- (c) GDC: outline finding  $P(X > 80)$  for  $X \sim N(70, 10^2)$ .

**T.VI.11.014 AP Stats FRQ investigative communication**

- (a) AP/IB style: Which theorem or definition is central to «AP Stats FRQ investigative communication»?
- (b) Write a justification sentence frame for «AP Stats FRQ investigative communication»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.
- (c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «AP Stats FRQ investigative communication», then interpret.

**T.VI.11.015 Formula sheet / table fluency ( $z, t, \chi^2$ )**

- (a)  $X \sim N(70, 10^2)$ . Approx  $P(60 < X < 80)$  (empirical rule)?
- (b) z-score of  $x=90$  ( $\mu=70$ ,  $\sigma=10$ )?
- (c) GDC: outline finding  $P(X > 80)$  for  $X \sim N(70, 10^2)$ .

**T.VI.11.016 IB AI inference GDC workflows**

- (a)  $X \sim N(50, 5^2)$ . Approx  $P(45 < X < 55)$  (empirical rule)?
- (b) z-score of  $x=60$  ( $\mu=50$ ,  $\sigma=5$ )?
- (c) GDC lab for «IB AI inference GDC workflows»: record menu path and numerical output.

**T.VI.11.017 Type I / Type II errors in context**

- (a)  $X \sim N(70, 10^2)$ . Approx  $P(60 < X < 80)$  (empirical rule)?
- (b) z-score of  $x=90$  ( $\mu=70$ ,  $\sigma=10$ )?
- (c) GDC: outline finding  $P(X > 80)$  for  $X \sim N(70, 10^2)$ .

**T.VI.11.018 Conditions check mantra before every procedure**

- (a)  $X \sim N(50, 5^2)$ . Approx  $P(45 < X < 55)$  (empirical rule)?
- (b) z-score of  $x=60$  ( $\mu=50$ ,  $\sigma=5$ )?
- (c) GDC: outline finding  $P(X > 55)$  for  $X \sim N(50, 5^2)$ .

**T.VI.11.019 Pooling for two-proportion z-test**

- (a)  $X \sim N(50, 5^2)$ . Approx  $P(45 < X < 55)$  (empirical rule)?
- (b) z-score of  $x=60$  ( $\mu=50$ ,  $\sigma=5$ )?
- (c) GDC: outline finding  $P(X > 55)$  for  $X \sim N(50, 5^2)$ .

**T.VI.11.020 df for t and  $\chi^2$  fluency**

- (a)  $X \sim N(100, 15^2)$ . Approx  $P(85 < X < 115)$  (empirical rule)?
- (b) z-score of  $x=130$  ( $\mu=100$ ,  $\sigma=15$ )?
- (c) GDC: outline finding  $P(X > 115)$  for  $X \sim N(100, 15^2)$ .

**T.VI.11.021 Confidence level vs interval width tradeoff**

- (a)  $X \sim N(100, 15^2)$ . Approx  $P(85 < X < 115)$  (empirical rule)?
- (b) z-score of  $x=130$  ( $\mu=100$ ,  $\sigma=15$ )?
- (c) GDC: outline finding  $P(X > 115)$  for  $X \sim N(100, 15^2)$ .

**T.VI.11.022 Writing hypotheses  $H_0$   $H_a$  templates**

- (a)  $X \sim N(50, 5^2)$ . Approx  $P(45 < X < 55)$  (empirical rule)?
- (b) z-score of  $x=60$  ( $\mu=50$ ,  $\sigma=5$ )?
- (c) GDC: outline finding  $P(X > 55)$  for  $X \sim N(50, 5^2)$ .

#### **T.VI.11.023 Conclusion sentence frames AP style**

- (a)  $X \sim N(100, 15^2)$ . Approx  $P(85 < X < 115)$  (empirical rule)?
- (b) z-score of  $x=130$  ( $\mu=100$ ,  $\sigma=15$ )?
- (c) GDC: outline finding  $P(X > 115)$  for  $X \sim N(100, 15^2)$ .

#### **T.VI.11.024 IB AI bivariate: Pearson r interpretation**

- (a)  $X \sim N(50, 5^2)$ . Approx  $P(45 < X < 55)$  (empirical rule)?
- (b) z-score of  $x=60$  ( $\mu=50$ ,  $\sigma=5$ )?
- (c) GDC lab for «IB AI bivariate: Pearson r interpretation»: record menu path and numerical output.

#### **T.VI.11.025 IB AI: Spearman rank correlation (if in guide)**

- (a)  $X \sim N(50, 5^2)$ . Approx  $P(45 < X < 55)$  (empirical rule)?
- (b) z-score of  $x=60$  ( $\mu=50$ ,  $\sigma=5$ )?
- (c) GDC: outline finding  $P(X > 55)$  for  $X \sim N(50, 5^2)$ .

#### **T.VI.11.026 Margin of error from CI half-width**

- (a)  $X \sim N(70, 10^2)$ . Approx  $P(60 < X < 80)$  (empirical rule)?
- (b) z-score of  $x=90$  ( $\mu=70$ ,  $\sigma=10$ )?
- (c) GDC: outline finding  $P(X > 80)$  for  $X \sim N(70, 10^2)$ .

#### **T.VI.11.027 AP Stats 2026 CED Unit checklist overlay**

- (a)  $X \sim N(70, 10^2)$ . Approx  $P(60 < X < 80)$  (empirical rule)?
- (b) z-score of  $x=90$  ( $\mu=70$ ,  $\sigma=10$ )?
- (c) GDC: outline finding  $P(X > 80)$  for  $X \sim N(70, 10^2)$ .

#### **T.VI.11.028 Chi-square expected counts computation**

- (a)  $X \sim N(100, 15^2)$ . Approx  $P(85 < X < 115)$  (empirical rule)?
- (b) z-score of  $x=130$  ( $\mu=100$ ,  $\sigma=15$ )?
- (c) GDC: outline finding  $P(X > 115)$  for  $X \sim N(100, 15^2)$ .

#### **T.VI.11.029 Blocking in experiments**

- (a)  $X \sim N(50, 5^2)$ . Approx  $P(45 < X < 55)$  (empirical rule)?
- (b) z-score of  $x=60$  ( $\mu=50$ ,  $\sigma=5$ )?
- (c) GDC: outline finding  $P(X > 55)$  for  $X \sim N(50, 5^2)$ .

#### **T.VI.11.030 Interpreting slope and intercept in context**

- (a)  $X \sim N(70, 10^2)$ . Approx  $P(60 < X < 80)$  (empirical rule)?
- (b) z-score of  $x=90$  ( $\mu=70$ ,  $\sigma=10$ )?
- (c) GDC: outline finding  $P(X > 80)$  for  $X \sim N(70, 10^2)$ .

#### **T.VI.11.031 Standard error vs standard deviation language**

- (a)  $X \sim N(50, 5^2)$ . Approx  $P(45 < X < 55)$  (empirical rule)?
- (b) z-score of  $x=60$  ( $\mu=50$ ,  $\sigma=5$ )?
- (c) GDC: outline finding  $P(X > 55)$  for  $X \sim N(50, 5^2)$ .

#### **T.VI.11.032 Full inference write-up timed drills**

- (a)  $X \sim N(50, 5^2)$ . Approx  $P(45 < X < 55)$  (empirical rule)?
- (b) z-score of  $x=60$  ( $\mu=50$ ,  $\sigma=5$ )?
- (c) GDC: outline finding  $P(X > 55)$  for  $X \sim N(50, 5^2)$ .

#### **T.VI.11.033 Power of a test conceptually**

- (a)  $X \sim N(100, 15^2)$ . Approx  $P(85 < X < 115)$  (empirical rule)?
- (b) z-score of  $x=130$  ( $\mu=100$ ,  $\sigma=15$ )?
- (c) GDC: outline finding  $P(X > 115)$  for  $X \sim N(100, 15^2)$ .

#### **T.VI.11.034 Interpreting P-value correctly (language drills)**

- (a)  $X \sim N(100, 15^2)$ . Approx  $P(85 < X < 115)$  (empirical rule)?

- (b) z-score of  $x=130$  ( $\mu=100$ ,  $\sigma=15$ )?
- (c) GDC: outline finding  $P(X>115)$  for  $X\sim N(100,15^2)$ .

#### **T.VI.11.035 IB AI $\chi^2$ GDC and critical value methods**

- (a)  $X\sim N(100,15^2)$ . Approx  $P(85<X<115)$  (empirical rule)?
- (b) z-score of  $x=130$  ( $\mu=100$ ,  $\sigma=15$ )?
- (c) GDC lab for «IB AI  $\chi^2$  GDC and critical value methods»: record menu path and numerical output.

#### **T.VI.11.036 Matched pairs design advantages**

- (a)  $X\sim N(70,10^2)$ . Approx  $P(60<X<80)$  (empirical rule)?
- (b) z-score of  $x=90$  ( $\mu=70$ ,  $\sigma=10$ )?
- (c) GDC: outline finding  $P(X>80)$  for  $X\sim N(70,10^2)$ .

#### **T.VI.11.037 AP Stats: interpreting $r^2$ in context**

- (a)  $X\sim N(100,15^2)$ . Approx  $P(85<X<115)$  (empirical rule)?
- (b) z-score of  $x=130$  ( $\mu=100$ ,  $\sigma=15$ )?
- (c) GDC: outline finding  $P(X>115)$  for  $X\sim N(100,15^2)$ .

#### **T.VI.11.038 AP Stats procedure write-up 1**

- (a)  $X\sim N(50,5^2)$ . Approx  $P(45<X<55)$  (empirical rule)?
- (b) z-score of  $x=60$  ( $\mu=50$ ,  $\sigma=5$ )?
- (c) GDC: outline finding  $P(X>55)$  for  $X\sim N(50,5^2)$ .

#### **T.VI.11.039 AP Stats procedure write-up 2**

- (a)  $X\sim N(50,5^2)$ . Approx  $P(45<X<55)$  (empirical rule)?
- (b) z-score of  $x=60$  ( $\mu=50$ ,  $\sigma=5$ )?
- (c) GDC: outline finding  $P(X>55)$  for  $X\sim N(50,5^2)$ .

#### **T.VI.11.040 AP Stats procedure write-up 3**

- (a)  $X\sim N(50,5^2)$ . Approx  $P(45<X<55)$  (empirical rule)?
- (b) z-score of  $x=60$  ( $\mu=50$ ,  $\sigma=5$ )?
- (c) GDC: outline finding  $P(X>55)$  for  $X\sim N(50,5^2)$ .

#### **T.VI.11.041 AP Stats procedure write-up 4**

- (a)  $X\sim N(50,5^2)$ . Approx  $P(45<X<55)$  (empirical rule)?
- (b) z-score of  $x=60$  ( $\mu=50$ ,  $\sigma=5$ )?
- (c) GDC: outline finding  $P(X>55)$  for  $X\sim N(50,5^2)$ .

#### **T.VI.11.042 AP Stats procedure write-up 5**

- (a)  $X\sim N(70,10^2)$ . Approx  $P(60<X<80)$  (empirical rule)?
- (b) z-score of  $x=90$  ( $\mu=70$ ,  $\sigma=10$ )?
- (c) GDC: outline finding  $P(X>80)$  for  $X\sim N(70,10^2)$ .

#### **T.VI.11.043 IB AI inference GDC task 1**

- (a)  $X\sim N(50,5^2)$ . Approx  $P(45<X<55)$  (empirical rule)?
- (b) z-score of  $x=60$  ( $\mu=50$ ,  $\sigma=5$ )?
- (c) GDC lab for «IB AI inference GDC task 1»: record menu path and numerical output.

#### **T.VI.11.044 IB AI inference GDC task 2**

- (a)  $X\sim N(100,15^2)$ . Approx  $P(85<X<115)$  (empirical rule)?
- (b) z-score of  $x=130$  ( $\mu=100$ ,  $\sigma=15$ )?
- (c) GDC lab for «IB AI inference GDC task 2»: record menu path and numerical output.

#### **T.VI.11.045 IB AI inference GDC task 3**

- (a)  $X\sim N(100,15^2)$ . Approx  $P(85<X<115)$  (empirical rule)?
- (b) z-score of  $x=130$  ( $\mu=100$ ,  $\sigma=15$ )?
- (c) GDC lab for «IB AI inference GDC task 3»: record menu path and numerical output.

#### **T.VI.12.001 SUVAT links to calculus motion FRQ**

- (a) AP/IB style: Which theorem or definition is central to «SUVAT links to calculus motion FRQ»?
- (b) Write a justification sentence frame for «SUVAT links to calculus motion FRQ»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.

(c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «SUVAT links to calculus motion FRQ», then interpret.

#### **T.VI.12.002 Variable acceleration from $a(t)$**

(a) AP/IB style: Which theorem or definition is central to «Variable acceleration from  $a(t)$ »?

(b) Write a justification sentence frame for «Variable acceleration from  $a(t)$ »: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.

(c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Variable acceleration from  $a(t)$ », then interpret.

#### **T.VI.3.015 AP FRQ justification practice 1**

(a) AP/IB style: Which theorem or definition is central to «AP FRQ justification practice 1»?

(b) Write a justification sentence frame for «AP FRQ justification practice 1»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.

(c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «AP FRQ justification practice 1», then interpret.

#### **T.VI.3.016 AP FRQ justification practice 2**

(a) AP/IB style: Which theorem or definition is central to «AP FRQ justification practice 2»?

(b) Write a justification sentence frame for «AP FRQ justification practice 2»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.

(c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «AP FRQ justification practice 2», then interpret.

#### **T.VI.3.017 AP FRQ justification practice 3**

(a) AP/IB style: Which theorem or definition is central to «AP FRQ justification practice 3»?

(b) Write a justification sentence frame for «AP FRQ justification practice 3»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.

(c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «AP FRQ justification practice 3», then interpret.

#### **T.VI.3.018 AP FRQ justification practice 4**

(a) AP/IB style: Which theorem or definition is central to «AP FRQ justification practice 4»?

(b) Write a justification sentence frame for «AP FRQ justification practice 4»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.

(c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «AP FRQ justification practice 4», then interpret.

#### **T.VI.3.019 AP FRQ justification practice 5**

(a) AP/IB style: Which theorem or definition is central to «AP FRQ justification practice 5»?

(b) Write a justification sentence frame for «AP FRQ justification practice 5»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.

(c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «AP FRQ justification practice 5», then interpret.

#### **T.VI.3.020 AP FRQ justification practice 6**

(a) AP/IB style: Which theorem or definition is central to «AP FRQ justification practice 6»?

(b) Write a justification sentence frame for «AP FRQ justification practice 6»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.

(c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «AP FRQ justification practice 6», then interpret.

#### **T.VI.3.021 AP FRQ justification practice 7**

(a) AP/IB style: Which theorem or definition is central to «AP FRQ justification practice 7»?

(b) Write a justification sentence frame for «AP FRQ justification practice 7»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.

(c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «AP FRQ justification practice 7», then interpret.

#### **T.VI.4.010 FTC accumulation FRQ variant 1**

(a) AP/IB style: Which theorem or definition is central to «FTC accumulation FRQ variant 1»?

(b) Write a justification sentence frame for «FTC accumulation FRQ variant 1»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.

(c) Graph lab for «FTC accumulation FRQ variant 1»: plot, mark key points, state the trend.

#### **T.VI.1.011 Limit & continuity FRQ variant 1**

(a) AP/IB style: Which theorem or definition is central to «Limit & continuity FRQ variant 1»?

(b) Write a justification sentence frame for «Limit & continuity FRQ variant 1»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.

(c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Limit & continuity FRQ variant 1», then interpret.

#### **T.VI.4.011 FTC accumulation FRQ variant 2**

(a) AP/IB style: Which theorem or definition is central to «FTC accumulation FRQ variant 2»?

(b) Write a justification sentence frame for «FTC accumulation FRQ variant 2»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.

(c) Graph lab for «FTC accumulation FRQ variant 2»: plot, mark key points, state the trend.

#### **T.VI.1.012 Limit & continuity FRQ variant 2**

(a) AP/IB style: Which theorem or definition is central to «Limit & continuity FRQ variant 2»?

(b) Write a justification sentence frame for «Limit & continuity FRQ variant 2»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.

(c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Limit & continuity FRQ variant 2», then interpret.

### T.VI.4.012 FTC accumulation FRQ variant 3

- (a) AP/IB style: Which theorem or definition is central to «FTC accumulation FRQ variant 3»?
- (b) Write a justification sentence frame for «FTC accumulation FRQ variant 3»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.
- (c) Graph lab for «FTC accumulation FRQ variant 3»: plot, mark key points, state the trend.

### T.VI.1.013 Limit & continuity FRQ variant 3

- (a) AP/IB style: Which theorem or definition is central to «Limit & continuity FRQ variant 3»?
- (b) Write a justification sentence frame for «Limit & continuity FRQ variant 3»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.
- (c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Limit & continuity FRQ variant 3», then interpret.

### T.VI.2.012 Product and quotient rule mixed AP set

- (a) Differentiate  $y=2x^5$ .
- (b)  $f(x)=2x^5$ . Find  $f'(2)$ .
- (c)  $s(t)=2t^2$  metres. Velocity at  $t=5$  s?

### T.VI.3.022 Related rates with similar triangles

- (a) Differentiate  $y=5x^3$ .
- (b)  $f(x)=5x^3$ . Find  $f'(2)$ .
- (c)  $s(t)=5t^2$  metres. Velocity at  $t=3$  s?

### T.VI.3.023 Curve sketching from calculus summary table

- (a) Differentiate  $y=5x^5$ .
- (b)  $f(x)=5x^5$ . Find  $f'(2)$ .
- (c) Desmos lab for «Curve sketching from calculus summary table»: vary one parameter, capture before/after, write one observation.

### T.VI.4.013 Midpoint Riemann sum vs trapezoid compare

- (a)  $\int 3x^1 dx$ .
- (b) Evaluate  $\int_{-1}^1 3x dx$ .
- (c) If  $v(t)$  is velocity, what does  $\int_{-1}^1 v(t) dt$  represent?

### T.VI.6.010 Washer method horizontal axis vs vertical

- (a) Differentiate  $y=2x^5$ .
- (b)  $f(x)=2x^5$ . Find  $f'(2)$ .
- (c)  $s(t)=2t^2$  metres. Velocity at  $t=5$  s?

### T.VI.7.020 Verify a proposed solution to a DE

- (a) Differentiate  $y=5x^3$ .
- (b)  $f(x)=5x^3$ . Find  $f'(2)$ .
- (c)  $s(t)=5t^2$  metres. Velocity at  $t=3$  s?

### T.VI.8.018 Speed vs velocity parametric distinction

- (a)  $\int 3x^1 dx$ .
- (b) Evaluate  $\int_{-1}^1 3x dx$ .
- (c) If  $v(t)$  is velocity, what does  $\int_{-1}^1 v(t) dt$  represent?

### T.VI.9.017 Geometric series of functions radius

- (a) nth term of 3, 10, 17, 24, ...
- (b) 20th term:  $a=3$ ,  $d=7$ .
- (c)  $S_{20}$  for  $a=3$ ,  $d=7$ .

### T.VI.10.016 Center of Taylor series not zero

- (a) nth term of 8, 10, 12, 14, ...
- (b) 20th term:  $a=8$ ,  $d=2$ .
- (c)  $S_{20}$  for  $a=8$ ,  $d=2$ .

### T.VI.11.046 Interpreting confidence interval correctly

- (a)  $X \sim N(100, 15^2)$ . Approx  $P(85 < X < 115)$  (empirical rule)?
- (b) z-score of  $x=130$  ( $\mu=100$ ,  $\sigma=15$ )?
- (c) GDC: outline finding  $P(X > 115)$  for  $X \sim N(100, 15^2)$ .

### T.VI.11.047 Randomization in experiments vs sampling

- (a)  $X \sim N(70, 10^2)$ . Approx  $P(60 < X < 80)$  (empirical rule)?
- (b) z-score of  $x=90$  ( $\mu=70$ ,  $\sigma=10$ )?
- (c) GDC: outline finding  $P(X > 80)$  for  $X \sim N(70, 10^2)$ .

#### T.VI.1.014 Limit of piecewise defined at junction

- (a) Evaluate  $\lim_{x \rightarrow 5} (x^2 - 25)/(x - 5)$ .
- (b) IVT:  $f$  continuous on  $[5, 8]$ ,  $f(5) = -2$ ,  $f(8) = 5$ . Justify a root of  $f(x) = 0$ .
- (c) Distinguish a vertical asymptote from a removable hole in one sentence.

#### T.VI.4.014 Fundamental theorem verbal interpretations

- (a) AP/IB style: Which theorem or definition is central to «Fundamental theorem verbal interpretations»?
- (b) Write a justification sentence frame for «Fundamental theorem verbal interpretations»: Because \_\_\_\_, by \_\_\_\_, therefore \_\_\_\_.
- (c) Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Fundamental theorem verbal interpretations», then interpret.

#### T.VI.8.019 Polar intercepts and symmetry tests

- (a)  $\int 2x^1 dx$ .
- (b) Evaluate  $\int_{-1}^1 2x dx$ .
- (c) Desmos lab for «Polar intercepts and symmetry tests»: vary one parameter, capture before/after, write one observation.

#### T.VI.10.017 Series solution approximation of DE light

- (a) nth term of 2, 9, 16, 23, ...
- (b) 20th term:  $a = 2$ ,  $d = 7$ .
- (c)  $S_n$  for  $a = 2$ ,  $d = 7$ .

#### T.VI.11.048 Simulation of sampling distribution of phat

- (a)  $X \sim N(50, 5^2)$ . Approx  $P(45 < X < 55)$  (empirical rule)?
- (b) z-score of  $x=60$  ( $\mu=50$ ,  $\sigma=5$ )?
- (c) GDC: outline finding  $P(X > 55)$  for  $X \sim N(50, 5^2)$ .

# Era VII — Space

Muted purple wireframe journal

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## T.VII.1.001 Column vectors GCSE notation

(a)  $a=(6, 1)$ ,  $b=(3, 2)$ . Find  $a+b$  as a column vector.

(b)  $|a|$  for  $a=(6, 1)$ ?

(c) Position  $A=(6,1)$ ,  $B=(9,3)$ . Vector  $AB$ ?

## T.VII.1.002 Vector geometry: midpoint and section formula

(a)  $a=(5, 4)$ ,  $b=(1, 5)$ . Find  $a+b$  as a column vector.

(b)  $|a|$  for  $a=(5, 4)$ ?

(c) Position  $A=(5,4)$ ,  $B=(6,9)$ . Vector  $AB$ ?

## T.VII.1.003 Vector proof: parallelogram / midpoint theorem

(a)  $a=(5, 3)$ ,  $b=(0, 2)$ . Find  $a+b$  as a column vector.

(b)  $|a|$  for  $a=(5, 3)$ ?

(c) Position  $A=(5,3)$ ,  $B=(5,5)$ . Vector  $AB$ ?

## T.VII.1.004 Vector proof: collinearity and ratio

(a)  $a=(2, 5)$ ,  $b=(1, 2)$ . Find  $a+b$  as a column vector.

(b)  $|a|$  for  $a=(2, 5)$ ?

(c) Position  $A=(2,5)$ ,  $B=(3,7)$ . Vector  $AB$ ?

## T.VII.1.005 Position vectors in plane exam pack

(a)  $a=(3, 1)$ ,  $b=(3, 4)$ . Find  $a+b$  as a column vector.

(b)  $|a|$  for  $a=(3, 1)$ ?

(c) Position  $A=(3,1)$ ,  $B=(6,5)$ . Vector  $AB$ ?

## T.VII.1.006 Magnitude and unit vectors GCSE/IB

(a)  $a=(1, -3)$ ,  $b=(-2, 6)$ . Find  $a+b$  as a column vector.

(b)  $|a|$  for  $a=(1, -3)$ ?

(c) Position  $A=(1,-3)$ ,  $B=(-1,3)$ . Vector  $AB$ ?

## T.VII.1.007 Vector geometric proof exercise 1

(a)  $a=(4, 3)$ ,  $b=(-1, 6)$ . Find  $a+b$  as a column vector.

(b)  $|a|$  for  $a=(4, 3)$ ?

(c) Position  $A=(4,3)$ ,  $B=(3,9)$ . Vector  $AB$ ?

## T.VII.1.008 Vector geometric proof exercise 2

(a)  $a=(1, 2)$ ,  $b=(0, 4)$ . Find  $a+b$  as a column vector.

(b)  $|a|$  for  $a=(1, 2)$ ?

(c) Position  $A=(1,2)$ ,  $B=(1,6)$ . Vector  $AB$ ?

## T.VII.1.009 Vector geometric proof exercise 3

(a)  $a=(1, 2)$ ,  $b=(2, 3)$ . Find  $a+b$  as a column vector.

(b)  $|a|$  for  $a=(1, 2)$ ?

(c) Position  $A=(1,2)$ ,  $B=(3,5)$ . Vector  $AB$ ?

## T.VII.3.001 IB HL matrices for 2D transformations pack

(a)  $\det([[1,3],[5,3]])$ ?

(b)  $[[1,3],[5,3]] \times I$ ?

(c) Geometry construction lab for «IB HL matrices for 2D transformations pack»: build the figure, measure, verify the claimed relation.

## T.VII.3.002 Solve 2×2 systems with inverse matrices

(a)  $\det([[4,1],[5,4]])$ ?

(b)  $[[4,1],[5,4]] \times I$ ?

(c) If  $\det \neq 0$ , inverse of  $[[4,1],[5,4]]$ ?

## T.VII.3.003 Determinant 2×2 geometric area link

(a)  $\det([[3,3],[5,2]])$ ?

- (b)  $\begin{bmatrix} 3 & 3 \\ 5 & 2 \end{bmatrix} \times \mathbf{I}$ ?
- (c) If  $\det \neq 0$ , inverse of  $\begin{bmatrix} 3 & 3 \\ 5 & 2 \end{bmatrix}$ ?

#### T.VII.3.004 Matrix multiplication non-commutativity demos

- (a)  $\det(\begin{bmatrix} 2 & 3 \\ 2 & 3 \end{bmatrix})$ ?
- (b)  $\begin{bmatrix} 2 & 3 \\ 2 & 3 \end{bmatrix} \times \mathbf{I}$ ?
- (c) If  $\det \neq 0$ , inverse of  $\begin{bmatrix} 2 & 3 \\ 2 & 3 \end{bmatrix}$ ?

#### T.VII.3.005 AP Precalc / (+) matrices systems bridge

- (a)  $\det(\begin{bmatrix} 1 & 5 \\ 5 & 1 \end{bmatrix})$ ?
- (b)  $\begin{bmatrix} 1 & 5 \\ 5 & 1 \end{bmatrix} \times \mathbf{I}$ ?
- (c) If  $\det \neq 0$ , inverse of  $\begin{bmatrix} 1 & 5 \\ 5 & 1 \end{bmatrix}$ ?

#### T.VII.3.006 Inverse of 2x2 formula fluency

- (a)  $\det(\begin{bmatrix} 4 & 4 \\ 2 & 5 \end{bmatrix})$ ?
- (b)  $\begin{bmatrix} 4 & 4 \\ 2 & 5 \end{bmatrix} \times \mathbf{I}$ ?
- (c) If  $\det \neq 0$ , inverse of  $\begin{bmatrix} 4 & 4 \\ 2 & 5 \end{bmatrix}$ ?

#### T.VII.3.007 Identity and zero matrix roles

- (a)  $\det(\begin{bmatrix} 2 & 4 \\ 2 & 5 \end{bmatrix})$ ?
- (b)  $\begin{bmatrix} 2 & 4 \\ 2 & 5 \end{bmatrix} \times \mathbf{I}$ ?
- (c) If  $\det \neq 0$ , inverse of  $\begin{bmatrix} 2 & 4 \\ 2 & 5 \end{bmatrix}$ ?

#### T.VII.3.008 Transpose properties $(A+B)^T$ etc.

- (a)  $\det(\begin{bmatrix} 5 & 3 \\ 5 & 3 \end{bmatrix})$ ?
- (b)  $\begin{bmatrix} 5 & 3 \\ 5 & 3 \end{bmatrix} \times \mathbf{I}$ ?
- (c) If  $\det \neq 0$ , inverse of  $\begin{bmatrix} 5 & 3 \\ 5 & 3 \end{bmatrix}$ ?

#### T.VII.3.009 Solving matrix equations $AX=B$

- (a)  $\det(\begin{bmatrix} 4 & 3 \\ 1 & 1 \end{bmatrix})$ ?
- (b)  $\begin{bmatrix} 4 & 3 \\ 1 & 1 \end{bmatrix} \times \mathbf{I}$ ?
- (c) If  $\det \neq 0$ , inverse of  $\begin{bmatrix} 4 & 3 \\ 1 & 1 \end{bmatrix}$ ?

#### T.VII.3.010 Coded messages / Hill cipher light (fun)

- (a)  $\det(\begin{bmatrix} 5 & 3 \\ 2 & 4 \end{bmatrix})$ ?
- (b)  $\begin{bmatrix} 5 & 3 \\ 2 & 4 \end{bmatrix} \times \mathbf{I}$ ?
- (c) If  $\det \neq 0$ , inverse of  $\begin{bmatrix} 5 & 3 \\ 2 & 4 \end{bmatrix}$ ?

#### T.VII.3.011 US Precalculus matrix applications pack

- (a)  $\det(\begin{bmatrix} 2 & 4 \\ 1 & 4 \end{bmatrix})$ ?
- (b)  $\begin{bmatrix} 2 & 4 \\ 1 & 4 \end{bmatrix} \times \mathbf{I}$ ?
- (c) If  $\det \neq 0$ , inverse of  $\begin{bmatrix} 2 & 4 \\ 1 & 4 \end{bmatrix}$ ?

#### T.VII.3.012 Exam mixed 2x2 matrix mega-set

- (a)  $\det(\begin{bmatrix} 4 & 5 \\ 2 & 5 \end{bmatrix})$ ?
- (b)  $\begin{bmatrix} 4 & 5 \\ 2 & 5 \end{bmatrix} \times \mathbf{I}$ ?
- (c) If  $\det \neq 0$ , inverse of  $\begin{bmatrix} 4 & 5 \\ 2 & 5 \end{bmatrix}$ ?

#### T.VII.3.013 Matrix encoding of simultaneous equations

- (a)  $\det(\begin{bmatrix} 4 & 5 \\ 4 & 4 \end{bmatrix})$ ?
- (b)  $\begin{bmatrix} 4 & 5 \\ 4 & 4 \end{bmatrix} \times \mathbf{I}$ ?
- (c) If  $\det \neq 0$ , inverse of  $\begin{bmatrix} 4 & 5 \\ 4 & 4 \end{bmatrix}$ ?

#### T.VII.3.014 Matrix representation of complex multiply light

- (a)  $\det(\begin{bmatrix} 3 & 5 \\ 3 & 4 \end{bmatrix})$ ?
- (b)  $\begin{bmatrix} 3 & 5 \\ 3 & 4 \end{bmatrix} \times \mathbf{I}$ ?
- (c) If  $\det \neq 0$ , inverse of  $\begin{bmatrix} 3 & 5 \\ 3 & 4 \end{bmatrix}$ ?

#### T.VII.5.001 Geometric linear maps: stretch shear rotate

- (a)  $\det(\begin{bmatrix} 2 & 1 \\ 5 & 5 \end{bmatrix})$ ?
- (b)  $\begin{bmatrix} 2 & 1 \\ 5 & 5 \end{bmatrix} \times \mathbf{I}$ ?

(c) Geometry construction lab for «Geometric linear maps: stretch shear rotate»: build the figure, measure, verify the claimed relation.

#### **T.VII.5.002 Composition of matrix transformations order**

(a)  $\det\left(\begin{bmatrix} 1 & 5 \\ 2 & 5 \end{bmatrix}\right)$ ?

(b)  $\begin{bmatrix} 1 & 5 \\ 2 & 5 \end{bmatrix} \times \mathbf{I}$ ?

(c) If  $\det \neq 0$ , inverse of  $\begin{bmatrix} 1 & 5 \\ 2 & 5 \end{bmatrix}$ ?

#### **T.VII.5.003 Kernel and image geometric meaning (light)**

(a)  $\det\left(\begin{bmatrix} 3 & 5 \\ 2 & 5 \end{bmatrix}\right)$ ?

(b)  $\begin{bmatrix} 3 & 5 \\ 2 & 5 \end{bmatrix} \times \mathbf{I}$ ?

(c) If  $\det \neq 0$ , inverse of  $\begin{bmatrix} 3 & 5 \\ 2 & 5 \end{bmatrix}$ ?

#### **T.VII.5.004 Reflection matrices in x/y axes and y=x**

(a)  $\det\left(\begin{bmatrix} 3 & 2 \\ 2 & 2 \end{bmatrix}\right)$ ?

(b)  $\begin{bmatrix} 3 & 2 \\ 2 & 2 \end{bmatrix} \times \mathbf{I}$ ?

(c) Geometry construction lab for «Reflection matrices in x/y axes and y=x»: build the figure, measure, verify the claimed relation.

#### **T.VII.5.005 Rotation matrices by $\theta$ standard form**

(a) Write 85000 in standard form.

(b) Write  $4.1 \times 10^4$  as an ordinary number.

(c) Geometry construction lab for «Rotation matrices by  $\theta$  standard form»: build the figure, measure, verify the claimed relation.

#### **T.VII.5.006 Enlargement matrix kI**

(a)  $\det\left(\begin{bmatrix} 5 & 3 \\ 2 & 3 \end{bmatrix}\right)$ ?

(b)  $\begin{bmatrix} 5 & 3 \\ 2 & 3 \end{bmatrix} \times \mathbf{I}$ ?

(c) Geometry construction lab for «Enlargement matrix kI»: build the figure, measure, verify the claimed relation.

#### **T.VII.5.007 Shear matrices and invariant lines**

(a)  $\det\left(\begin{bmatrix} 5 & 5 \\ 4 & 2 \end{bmatrix}\right)$ ?

(b)  $\begin{bmatrix} 5 & 5 \\ 4 & 2 \end{bmatrix} \times \mathbf{I}$ ?

(c) Geometry construction lab for «Shear matrices and invariant lines»: build the figure, measure, verify the claimed relation.

#### **T.VII.5.008 Inverse transformation matrix meaning**

(a)  $\det\left(\begin{bmatrix} 3 & 4 \\ 3 & 4 \end{bmatrix}\right)$ ?

(b)  $\begin{bmatrix} 3 & 4 \\ 3 & 4 \end{bmatrix} \times \mathbf{I}$ ?

(c) If  $\det \neq 0$ , inverse of  $\begin{bmatrix} 3 & 4 \\ 3 & 4 \end{bmatrix}$ ?

#### **T.VII.5.009 Determinant as oriented area scale factor**

(a)  $\det\left(\begin{bmatrix} 1 & 3 \\ 3 & 3 \end{bmatrix}\right)$ ?

(b)  $\begin{bmatrix} 1 & 3 \\ 3 & 3 \end{bmatrix} \times \mathbf{I}$ ?

(c) Geometry construction lab for «Determinant as oriented area scale factor»: build the figure, measure, verify the claimed relation.

#### **T.VII.1.010 N-VM (+) vectors applications bridge**

(a)  $a=(6, -1)$ ,  $b=(3, 5)$ . Find  $a+b$  as a column vector.

(b)  $|a|$  for  $a=(6, -1)$ ?

(c) Position  $A=(6,-1)$ ,  $B=(9,4)$ . Vector  $AB$ ?

#### **T.VII.1.011 Dot product: perpendicular test exam**

(a)  $a=(4, -2)$ ,  $b=(1, 4)$ . Find  $a+b$  as a column vector.

(b)  $|a|$  for  $a=(4, -2)$ ?

(c) Position  $A=(4,-2)$ ,  $B=(5,2)$ . Vector  $AB$ ?

#### **T.VII.1.012 Vector equation of a line (IB)**

(a)  $a=(6, 4)$ ,  $b=(4, 6)$ . Find  $a+b$  as a column vector.

(b)  $|a|$  for  $a=(6, 4)$ ?

(c) Position  $A=(6,4)$ ,  $B=(10,10)$ . Vector  $AB$ ?

#### **T.VII.1.013 Angle between vectors exam**

(a)  $a=(5, 0)$ ,  $b=(1, 3)$ . Find  $a+b$  as a column vector.

(b)  $|a|$  for  $a=(5, 0)$ ?

(c) Position  $A=(5,0)$ ,  $B=(6,3)$ . Vector  $AB$ ?

**T.VII.1.014 IB AI/AA vector applications in geometry**

(a)  $a=(6, 2)$ ,  $b=(-2, 4)$ . Find  $a+b$  as a column vector.

(b)  $|a|$  for  $a=(6, 2)$ ?

(c) Position  $A=(6,2)$ ,  $B=(4,6)$ . Vector  $AB$ ?

**T.VII.1.015 Unit vector in direction of motion**

(a)  $a=(5, 0)$ ,  $b=(4, 3)$ . Find  $a+b$  as a column vector.

(b)  $|a|$  for  $a=(5, 0)$ ?

(c) Position  $A=(5,0)$ ,  $B=(9,3)$ . Vector  $AB$ ?

**T.VII.1.016 Scalar projection applications**

(a)  $a=(5, -1)$ ,  $b=(0, 1)$ . Find  $a+b$  as a column vector.

(b)  $|a|$  for  $a=(5, -1)$ ?

(c) Position  $A=(5,-1)$ ,  $B=(5,0)$ . Vector  $AB$ ?

**T.VII.1.017 3D coordinates distance and midpoint**

(a)  $a=(5, -1)$ ,  $b=(4, 4)$ . Find  $a+b$  as a column vector.

(b)  $|a|$  for  $a=(5, -1)$ ?

(c) Position  $A=(5,-1)$ ,  $B=(9,3)$ . Vector  $AB$ ?

**T.VII.1.018 Vector proof of median concurrency light**

(a)  $a=(6, 2)$ ,  $b=(3, 3)$ . Find  $a+b$  as a column vector.

(b)  $|a|$  for  $a=(6, 2)$ ?

(c) Position  $A=(6,2)$ ,  $B=(9,5)$ . Vector  $AB$ ?

**T.VII.1.019 Displacement vs position vector clarity**

(a)  $a=(6, 2)$ ,  $b=(4, 3)$ . Find  $a+b$  as a column vector.

(b)  $|a|$  for  $a=(6, 2)$ ?

(c) Position  $A=(6,2)$ ,  $B=(10,5)$ . Vector  $AB$ ?

**T.VII.1.020 IB AA HL vector product magnitude area**

(a)  $a=(2, -1)$ ,  $b=(3, 2)$ . Find  $a+b$  as a column vector.

(b)  $|a|$  for  $a=(2, -1)$ ?

(c) Position  $A=(2,-1)$ ,  $B=(5,1)$ . Vector  $AB$ ?

**T.VII.1.021 Resultant forces as vector sums**

(a)  $a=(6, 5)$ ,  $b=(1, 2)$ . Find  $a+b$  as a column vector.

(b)  $|a|$  for  $a=(6, 5)$ ?

(c) Position  $A=(6,5)$ ,  $B=(7,7)$ . Vector  $AB$ ?

**T.VII.1.022 Unit vector  $i,j,k$  notation fluency**

(a)  $a=(3, 5)$ ,  $b=(1, 1)$ . Find  $a+b$  as a column vector.

(b)  $|a|$  for  $a=(3, 5)$ ?

(c) Position  $A=(3,5)$ ,  $B=(4,6)$ . Vector  $AB$ ?

**T.VII.1.023 Parallel vectors scalar multiple test**

(a)  $a=(6, -1)$ ,  $b=(0, 5)$ . Find  $a+b$  as a column vector.

(b)  $|a|$  for  $a=(6, -1)$ ?

(c) Position  $A=(6,-1)$ ,  $B=(6,4)$ . Vector  $AB$ ?

**T.VII.1.024 Vector journeys multi-stage collinearity**

(a)  $a=(1, 5)$ ,  $b=(1, 1)$ . Find  $a+b$  as a column vector.

(b)  $|a|$  for  $a=(1, 5)$ ?

(c) Position  $A=(1,5)$ ,  $B=(2,6)$ . Vector  $AB$ ?

**T.VII.1.025 Component form from magnitude and direction**

(a)  $a=(6, -2)$ ,  $b=(-2, 4)$ . Find  $a+b$  as a column vector.

(b)  $|a|$  for  $a=(6, -2)$ ?

(c) Position  $A=(6,-2)$ ,  $B=(4,2)$ . Vector  $AB$ ?

**T.VII.2.001 Gaussian elimination exam discipline**

- (a)  $\det([[5,2],[2,3]])$ ?
- (b)  $[[5,2],[2,3]] \times \mathbf{I}$ ?
- (c) If  $\det \neq 0$ , inverse of  $[[5,2],[2,3]]$ ?

#### T.VII.2.002 Consistent vs inconsistent systems language

- (a)  $\det([[4,5],[1,4]])$ ?
- (b)  $[[4,5],[1,4]] \times \mathbf{I}$ ?
- (c) If  $\det \neq 0$ , inverse of  $[[4,5],[1,4]]$ ?

#### T.VII.2.003 Augmented matrix notation fluency

- (a)  $\det([[4,2],[4,5]])$ ?
- (b)  $[[4,2],[4,5]] \times \mathbf{I}$ ?
- (c) If  $\det \neq 0$ , inverse of  $[[4,2],[4,5]]$ ?

#### T.VII.2.004 Parameter count for infinite solutions

- (a)  $\det([[5,5],[5,1]])$ ?
- (b)  $[[5,5],[5,1]] \times \mathbf{I}$ ?
- (c) If  $\det \neq 0$ , inverse of  $[[5,5],[5,1]]$ ?

#### T.VII.6.001 Determinant expand vs row-reduce choice

- (a)  $\det([[1,2],[3,3]])$ ?
- (b)  $[[1,2],[3,3]] \times \mathbf{I}$ ?
- (c) If  $\det \neq 0$ , inverse of  $[[1,2],[3,3]]$ ?

#### T.VII.6.002 Singular matrices and no inverse

- (a)  $\det([[3,4],[2,2]])$ ?
- (b)  $[[3,4],[2,2]] \times \mathbf{I}$ ?
- (c) If  $\det \neq 0$ , inverse of  $[[3,4],[2,2]]$ ?

#### T.VII.6.003 Cramer's rule 2x2 (optional method)

- (a) Notation/definition check for «Cramer's rule 2x2 (optional method)»: state the key idea in  $\leq 20$  words.
- (b) Standard exam-style: Apply «Cramer's rule 2x2 (optional method)» in a worked example that uses the numbers 5 and 5. Show steps.
- (c) IB dialect challenge: Write and solve one markable question targeting «Cramer's rule 2x2 (optional method)» (include answer).

#### T.VII.7.001 Eigenvalues 2x2 exam pack (light)

- (a)  $\det([[1,1],[4,3]])$ ?
- (b)  $[[1,1],[4,3]] \times \mathbf{I}$ ?
- (c) If  $\det \neq 0$ , inverse of  $[[1,1],[4,3]]$ ?

#### T.VII.7.002 Diagonalisation idea for matrix powers light

- (a)  $\det([[4,4],[5,2]])$ ?
- (b)  $[[4,4],[5,2]] \times \mathbf{I}$ ?
- (c) If  $\det \neq 0$ , inverse of  $[[4,4],[5,2]]$ ?

#### T.VII.4.001 Linear independence of 2–3 vectors in R2/R3

- (a)  $a=(5, 5)$ ,  $b=(4, 3)$ . Find  $a+b$  as a column vector.
- (b)  $|a|$  for  $a=(5, 5)$ ?
- (c) Position  $A=(5,5)$ ,  $B=(9,8)$ . Vector  $AB$ ?

#### T.VII.4.002 Basis of R<sup>2</sup> standard and rotated

- (a)  $a=(5, -2)$ ,  $b=(1, 3)$ . Find  $a+b$  as a column vector.
- (b)  $|a|$  for  $a=(5, -2)$ ?
- (c) Position  $A=(5,-2)$ ,  $B=(6,1)$ . Vector  $AB$ ?

#### T.VII.8.001 Orthogonal vectors in R<sup>2</sup> exam

- (a) Notation/definition check for «Orthogonal vectors in R<sup>2</sup> exam»: state the key idea in  $\leq 20$  words.
- (b) Standard exam-style: Apply «Orthogonal vectors in R<sup>2</sup> exam» in a worked example that uses the numbers 9 and 12. Show steps.
- (c) IB dialect challenge: Write and solve one markable question targeting «Orthogonal vectors in R<sup>2</sup> exam» (include answer).

#### T.VII.10.001 Partial derivatives notation exam light

- (a) Notation/definition check for «Partial derivatives notation exam light»: state the key idea in  $\leq 20$  words.

- (b) Standard exam-style: Apply «Partial derivatives notation exam light» in a worked example that uses the numbers 10 and 11. Show steps.
- (c) IB dialect challenge: Write and solve one markable question targeting «Partial derivatives notation exam light» (include answer).

#### **T.VII.14.001 Vector fields intuition AP+ bridge**

- (a)  $a=(4, -2)$ ,  $b=(1, 6)$ . Find  $a+b$  as a column vector.
- (b)  $|a|$  for  $a=(4, -2)$ ?
- (c) Desmos lab for «Vector fields intuition AP+ bridge»: vary one parameter, capture before/after, write one observation.

#### **T.VII.3.015 Matrix transform exam figure 1**

- (a)  $\det\left(\begin{bmatrix} 2 & 1 \\ 4 & 1 \end{bmatrix}\right)$ ?
- (b)  $\begin{bmatrix} 2 & 1 \\ 4 & 1 \end{bmatrix} \times I$ ?
- (c) Geometry construction lab for «Matrix transform exam figure 1»: build the figure, measure, verify the claimed relation.

#### **T.VII.1.026 Vector geometry GCSE Higher 1**

- (a)  $a=(3, 0)$ ,  $b=(4, 6)$ . Find  $a+b$  as a column vector.
- (b)  $|a|$  for  $a=(3, 0)$ ?
- (c) Position  $A=(3,0)$ ,  $B=(7,6)$ . Vector  $AB$ ?

#### **T.VII.3.016 Matrix transform exam figure 2**

- (a)  $\det\left(\begin{bmatrix} 2 & 4 \\ 2 & 4 \end{bmatrix}\right)$ ?
- (b)  $\begin{bmatrix} 2 & 4 \\ 2 & 4 \end{bmatrix} \times I$ ?
- (c) Geometry construction lab for «Matrix transform exam figure 2»: build the figure, measure, verify the claimed relation.

#### **T.VII.1.027 Vector geometry GCSE Higher 2**

- (a)  $a=(2, 2)$ ,  $b=(4, 5)$ . Find  $a+b$  as a column vector.
- (b)  $|a|$  for  $a=(2, 2)$ ?
- (c) Position  $A=(2,2)$ ,  $B=(6,7)$ . Vector  $AB$ ?

#### **T.VII.3.017 Matrix transform exam figure 3**

- (a)  $\det\left(\begin{bmatrix} 1 & 4 \\ 5 & 3 \end{bmatrix}\right)$ ?
- (b)  $\begin{bmatrix} 1 & 4 \\ 5 & 3 \end{bmatrix} \times I$ ?
- (c) Geometry construction lab for «Matrix transform exam figure 3»: build the figure, measure, verify the claimed relation.

#### **T.VII.1.028 Vector geometry GCSE Higher 3**

- (a)  $a=(1, 0)$ ,  $b=(-1, 2)$ . Find  $a+b$  as a column vector.
- (b)  $|a|$  for  $a=(1, 0)$ ?
- (c) Position  $A=(1,0)$ ,  $B=(0,2)$ . Vector  $AB$ ?

#### **T.VII.3.018 Matrix transform exam figure 4**

- (a)  $\det\left(\begin{bmatrix} 1 & 4 \\ 4 & 3 \end{bmatrix}\right)$ ?
- (b)  $\begin{bmatrix} 1 & 4 \\ 4 & 3 \end{bmatrix} \times I$ ?
- (c) Geometry construction lab for «Matrix transform exam figure 4»: build the figure, measure, verify the claimed relation.

#### **T.VII.1.029 Vector geometry GCSE Higher 4**

- (a)  $a=(2, 2)$ ,  $b=(1, 2)$ . Find  $a+b$  as a column vector.
- (b)  $|a|$  for  $a=(2, 2)$ ?
- (c) Position  $A=(2,2)$ ,  $B=(3,4)$ . Vector  $AB$ ?

#### **T.VII.5.010 Standard matrix recognition 1**

- (a)  $\det\left(\begin{bmatrix} 3 & 1 \\ 1 & 4 \end{bmatrix}\right)$ ?
- (b)  $\begin{bmatrix} 3 & 1 \\ 1 & 4 \end{bmatrix} \times I$ ?
- (c) Geometry construction lab for «Standard matrix recognition 1»: build the figure, measure, verify the claimed relation.

#### **T.VII.5.011 Standard matrix recognition 2**

- (a)  $\det\left(\begin{bmatrix} 4 & 1 \\ 3 & 5 \end{bmatrix}\right)$ ?
- (b)  $\begin{bmatrix} 4 & 1 \\ 3 & 5 \end{bmatrix} \times I$ ?
- (c) Geometry construction lab for «Standard matrix recognition 2»: build the figure, measure, verify the claimed relation.

#### **T.VII.3.019 Matrix transform area scale $|\det|$ interpretation**

- (a)  $\det\left(\begin{bmatrix} 2 & 5 \\ 1 & 1 \end{bmatrix}\right)$ ?
- (b)  $\begin{bmatrix} 2 & 5 \\ 1 & 1 \end{bmatrix} \times I$ ?

**(c)** Geometry construction lab for «Matrix transform area scale  $|\det|$  interpretation»: build the figure, measure, verify the claimed relation.